INCH-POUND

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MILITARY SPECIFICATION

CONTROL UNIT, ALARM SET

C-9412/FSS-9(V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the alarm set control unit. It is used to process signals from intrusion detection sensors and to control the mode of operation of an area which has been secured. It is a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-up.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6350

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MILITARY

MIL-P-116	- Preservation, Methods of.

STANDARDS

FEDERAL

FED-STD-101	- Test procedures for Packaging Materials.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment, Subsystem and System.
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-810 1/	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.
MIL-STD-45662	- Calibration System Requirements.

1/ The "C" revision of MIL-STD-810 shall be used.

(Unless otherwise specified, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 191111-5094.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

TA13220E4085	- Control Unit, Alarm Set C-9412/FSS-9(V).
TA13220E4213	- Printed Wiring Board Assembly, Tamper/Alarm.
TA13220E4214	- Printed Wiring Board Assembly, Voltage Monitor.
TA13220E4215	- Printed Wiring Board Assembly, Audible Alarm.
TA13220E4216	- Printed Wiring Board Assembly, Instantaneous Alarm.
TA13220E4217	- Printed Wiring Board Assembly, Latched Alarm.
TA13220E4218	- Printed Wiring Board Assembly, Alarm I.D.
TA13220E4219	- Printed Wiring Board Assembly, L.E.D.
TA13220E4225	- Interconnecting Wiring Subassembly, Status Processor.
TA13226E1152	- Power Supply, Control Unit Assembly.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research,

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Development, and Engineering Center, ATTN: STRBE-JIE, Fort Belvoir, VA 22060-5606.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The alarm set control unit (here-after called "CU") shall be in accordance with Top Assembly TA13220E4085 and as specified herein.

3.1.1 Major components. The CU shall consist of the following major components:

a. Power supply assembly TA13226E1152.
b. Printed wiring board (PWB) assembly, tamper/alarm TA13220E4213.
c. PWB assembly, voltage monitor TA13220E4214.
d. PWB assembly, audible alarm TA 13220E4215.
e. PWB assembly, instantaneous alarm TA13220E4216.
f. PWB assembly, latched alarm TA13220E4217.
g. PWB assembly, alarm I.D. TA13220E4218.
h. PWB assembly, LED TA13220E4219.
i. Interconnecting wiring subassembly, TA13220E4225.

When specified (see 6.2), any of the major components listed above shall be furnished separately.

3.1.2 Function. The CU is the central control element of J-SIIDS. Its primary function is to process signals from intrusion detection sensors and to monitor the status of the controlled zone. This information is applied to output (data link) terminals for transmittal to peripheral monitor equipment. The CU also supplies primary and emergency power to its associated sensors; provides "secure", "access", and "test" modes of system operation within the controlled zone; and maintains control zone subsystem security by monitoring interconnecting sensor and audible alarm signal lines, and by providing anti-tamper protection for its own enclosure.

3.1.3 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the control unit or major components when furnished separately. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First Article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

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3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The CU shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the CU may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the CU may be newly fabricated from recovered materials to the maximum extent practicable, provided the CU produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the CU.

3.4 Environmental conditions. The CU and major components when furnished separately shall withstand individual environmental conditions specified herein, or combinations thereof, without damage (see 6.4.2), performance degradation, or malfunction when tested as specified in 4.5.2.6.

3.4.1 Operational.

a. Temperature range of -20 °F to +150 °F.
b. Humidity up to 95 percent relative.

3.4.2 Non-operational condition and storage.

a. Temperature range of -30 °F to +165 °F.
b. Humidity up to 95 percent relative.
c. Shock: bench handling; equipment not operating (without battery).

3.5 Detailed technical requirements. The CU shall perform as specified herein and shall not false alarm (see 6.4.1) or give other false indications (incorrect access or secure or ac fail indication) under any of the operating test conditions specified herein.

3.5.1 Tamper protection. The CU tamper switch shall be assembled into the CU enclosure such that the switch contacts open with not less than 0.125 inch nor more than 0.250 inch of door movement and before the door flanges are disengaged.

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3.5.2 Terminal board connections. All terminal boards and terminals shall be labeled. The required function of each terminal board and each terminal is as follows:

Terminal Bd. and Terminal No.	Function
TB1 1/	Alternating current (ac) power interface
1	ac high
2	ac neutral
3	Ground
4	Switched ac (high)
5	ac indicator lamp (negative)
6	ac indicator lamp (positive)
TB2 1/	CU subassembly interface
1	27.5 Volts, direct current (Vdc) (battery positive)
2	27.5 Vdc
3	20.0 Vdc
4	Ground (battery negative)
5	5 Vdc
TB3 1/	CU subassembly interface
1	ac presence output
2	ac presence output
3	Sonalert input (positive)
4	Sonalert input (negative)
TB4 1/	Audible Alarm signal interface
1	Alarm output
2	Signal common
3	Tamper input
4	5 Vdc
TB5 1/	Audible Alarm power interface
1	ac power (high)
2	ac power (neutral)
3	Ground

1/ Indicates TB's physically located on power supply assembly.

Terminal Bd.and Terminal No.	Function
TB6	Data link interface
S1-I	Instantaneous alarm signal
S1-IR	Instantaneous alarm signal return
S1-D	Delayed alarm signal
S1-DR	Delayed alarm signal return
S-2	Access-secure signal
S-2R	Access-secure signal return
S-3	ac power on/off signal
TB7	Data link interface
S-3R	ac power on/off signal return

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S-4	Ground
S-5	20 Vdc
S-6	dc return
S-7	Resynchronization (resync) initiation
S-8	Dialer tamper signal
S-9	Dialer tamper return
TB8-F	Duress sensor interface
1	Duress alarm signal
2	Signal common
3	Tamper alarm signal
4	20 Vdc
5	dc return
6	Ground
7	Audible Alarm inhibit signal
TB9-A thru TB13-E	Intrusion sensor interface
1	Intrusion alarm signal
2	Signal common
3	Tamper alarm signal
4	20 Vdc
5	dc return
6	Ground
7	Access command signal (27.5 Vdc)

3.5.3 External inputs to CU. The CU shall recognize and process alarm and non-alarm inputs from the duress sensor, intrusion sensors, and the Audible Alarm. Three types of alarm inputs shall be recognized; duress alarms, intrusion alarms, and tamper alarms. The criteria for recognizing the alarm or non-alarm state from duress and intrusion sensors at any specific input shall be as follows:

a. Alarm state: The CU shall recognize the alarm state when the resistance across the input terminals is greater than or equal to 100 kilohms (K ohms). An instantaneous change from 2 K ohms to 1 megohm and greater persisting for at least 225 milliseconds (ms) shall be processed as an alarm.
b. Non-alarm state: The CU shall recognize the non-alarm state when the resistance across the input terminals is less than or equal to 2K ohms. The temporary existence of a resistance greater than or equal to 1 megohm for less than 60 ms, when the resistance before and after is less than or equal to 2K ohms, shall be processed as the non-alarm state.

3.5.3.1 Intrusion alarms. Alarm states presented between the intrusion alarm signal and signal common input terminals on TB9-A through TB13-E shall be processed as intrusion alarms.

3.5.3.2 Duress alarms. Alarm states presented between the duress alarm signal and signal common input terminals on TB8-F shall be processed as duress alarms.

3.5.3.3 Tamper alarms. Alarm states presented between tamper alarm signal and signal common input terminal pairs on TB8-F and TB9-A through TB13-E and between TB7-S-8 and TB7-S-9 shall be processed as tamper alarms. Presence of a voltage

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less than 1 Vdc across the Audible Alarm input terminals TB4-3 and -2 shall also be processed as a tamper alarm. Presence of a voltage of 5 ±1 Vdc across these terminals shall be recognized as the non-alarm state. The CU shall also recognize tamper alarms from internal sources specified in 3.5.4.

3.5.4 Internal tamper alarm sources. In addition to the external sources, tamper alarms shall also be recognized when internally generated by the tamper switch and by the regulated voltage monitor circuit. The CU shall recognize the presence of 20 ±1 Vdc at terminal TB2-3 reference TB2-4 as a non-tamper alarm and the presence of less than or equal to 18.3 Vdc or greater than or equal to 21.7 Vdc as a tamper alarm.

3.5.5 CU outputs. The CU shall process the alarm and other status inputs as specified herein and provide impedance or voltage levels, as appropriate, on specified output terminal pairs on TB6, TB7, and TB4. The CU shall present either a high impedance of greater than 100K ohms or a low impedance of less than 2K ohms in accordance with the following:

Terminals (TB6 and TB7)	Status (Impedance State)
S1-I reference S1-IR	Instantaneous alarm (high), non-alarm (low)
S1-D reference S1-DR	Delayed alarm (high), non-alarm (low)
S-2 reference S-2R	Secure mode (high), test/reset or access mode (low)
S-3 reference S-3R	ac power off (high), on (low)

The CU shall also provide alarm/non-alarm outputs at output terminals TB4-1 reference TB4-2. In the non-alarm condition a 5 ±0.5 Vdc signal shall be provided on TB4-1. In the alarm condition a 0 to 1 Vdc signal shall be provided on TB4-1.

3.5.6 Alarm signal processing. The CU shall process input alarms from external and internal sources and provide outputs dependent on the type of input alarm, the settings of the mode switch (see 3.5.7), the setting of the latch/non-latch switch and timing relationships specified herein.

3.5. Instantaneous alarm output. Tamper and duress alarm inputs of sufficient duration shall immediately cause the alarm state to be presented at the instantaneous alarm output (TB6-S1-I and its return), regardless of the mode switch setting. Intrusion alarm inputs of sufficient duration shall immediately cause the alarm state to be presented to the instantaneous alarm output, except when the mode switch is in the ACCESS position, the intrusion alarm input shall not be processed. Once initiated, the alarm state at the instantaneous output shall be present until 10 ±2 seconds after removal of all input alarms. In the TEST/RESET mode switch setting, a sonalert (see 3.5.11.3.1) physically located in the power supply shall sound whenever an alarm is present at the instantaneous alarm output.

3.5.6.2 Delayed alarm output for tamper and duress alarm inputs. Tamper and duress alarm inputs of sufficient duration shall immediately cause the alarm state to be presented at the delayed alarm output (TB6-S1-D and its return), regardless of the mode switch setting. In the NON-LATCH switch position, the delayed output shall automatically reset to the non-alarm state 10 ±2 seconds after removal of the alarm input, in all three modes. In the LATCH position, when in ACCESS or SECURE mode, the delayed output shall be latched in the alarm

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state until reset by momentarily rotating the mode switch to the TEST/RESET position, at which time the delayed output shall return to the non-alarm state, provided the input alarm has been removed. The latch/non-latch switch has no effect on the delayed alarm output in the TEST/RESET mode. The delayed alarm output shall automatically reset 10 ±2 seconds after removal of the input alarm.

3.5.6.3 Delayed alarm output for intrusion alarms. In the ACCESS mode switch setting, intrusion alarms shall be inhibited. In the TEST/RESET setting, any intrusion alarm input shall immediately result in an alarm output at the delayed alarm output terminals. This alarm state shall continue for the duration of the alarm and 10 ±2 seconds after removal of the alarm input. In the SECURE mode setting, intrusion alarms shall be subject to an entrance and exit time delay as specified in 3.5.6.3.1. Once initiated, if the latch/non-latch switch is in the NON-LATCH position, the delayed output alarm shall be present for 10 ±2 seconds if the input alarm has been removed, or until 10 ±2 seconds after removal of the input alarm if the input alarm was still present after expiration of the entrance delay. If the latch/non-latch switch is in the LATCH position an alarm output once initiated shall remain until reset by rotating the mode switch to the TEST/RESET position.

3.5.6.3.1 Entrance and exit time delay functions. When the mode switch is rotated to the SECURE position, a time interval, the exit time delay, shall be initiated. During the exit time delay interval intrusion alarm inputs shall be inhibited from causing delayed alarm outputs. This time interval period shall be adjustable from a minimum of 10 ±2 seconds to a maximum of 115 ±35 seconds by adjusting potentiometer R6 on the A9 circuit card. Intrusion alarm inputs occurring after expiration of the exit delay shall not be presented at the delayed alarm output until after expiration of an additional time interval, the entrance time delay. This time interval shall be adjustable from a minimum of 10 ±2 seconds to a maximum of 115 ±35 seconds by adjusting potentiometer R8 on the A10 circuit card. The entrance delay shall only be able to be initiated after expiration of the exit delay, but once initiated shall result in a delayed alarm output after its expiration, unless reset by rotating the mode switch to the TEST/RESET position before expiration of the entrance delay.

3.5.6.4 Alarm outputs to Audible Alarm. Alarm outputs to the Audible Alarm presented at TB4-1 reference TB4-2, shall track delayed alarm outputs at TB6-S1-D for tamper and intrusion input alarms, except that alarm outputs to the Audible Alarm, once initiated, shall latch regardless of the position of the latch/non-latch switch and alarm outputs to the Audible Alarm shall be inhibited when the CU mode switch is in the test/reset position. Alarm outputs shall be reset by rotating the mode switch to the TEST/RESET position. Duress alarm inputs shall be inhibited from causing alarm outputs to the Audible Alarm.

3.5.6.5 Audible Alarm inhibit. With a low resistance (less than 100 ohms) applied from terminal 7 on terminal board TB8-F to ground, tamper alarm inputs shall be inhibited from producing alarms at the Audible Alarm output when the CU mode switch is in the secure or access mode.

3.5.7 Mode switch. The CU mode switch shall be a three position, key-operated, double-action, high security type switch. Two keys shall be furnished for the lock. The key shall be removable only in the SECURE position. The switch shall be capable of 50,000 switching operations opening a 30 volt, 100 milliampere (mA) direct current (dc) circuit (resistive load), and shall be

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sealed against dirt and dust. The CU operational requirements in each of the three positions shall be as specified in 3.5.7.1 through 3.5.7.3.

3.5.7.1 Access. In the ACCESS mode a low impedance state shall be presented between terminals TB6-S-2 and -S-2R within one second of the time the switch is rotated from the SECURE position. This low impedance state shall be maintained while the switch is in the ACCESS position and for 5.25 ±0.75 seconds after being rotated into the SECURE position. Alarm processing shall be as specified in 3.5.6. The 27.5 Vdc battery charging voltage shall be supplied to each of the terminal 7's reference terminal 2's of TB9-A through TB13-E.

3.5.7.2 Test/reset. In the TEST/RESET mode a low impedance state shall be presented between terminals TB6-S-2 and -S-2R within one second of the time the switch is rotated from the SECURE position. This low impedance state shall be maintained for as long as the switch is in TEST/RESET position and for an additional minimum period of 5.25 ±0.75 seconds after the switch is returned to the SECURE position. Alarm processing shall be as specified in 3.5.6. The sonalert shall be enabled in this mode and shall sound each time and for as long as an alarm output exists between TB6-S1-I and its return. The alarm output at TB6-S1-D and its return shall track the output at TB6-S1-I and its return during the TEST/RESET mode.

3.5.7.3 Secure. In the SECURE mode a high impedance state shall be presented between terminals TB6-S-2 and -S-2R 5.25 ±0.75 seconds after the rotation of the mode switch from either ACCESS or TEST/RESET to the SECURE position. Alarm processing shall be as specified in 3.5.6.

3.5.8 Alarm identification. After the exit time delay expires, the alarm identification circuit shall be armed to activate the light emitting diode (LED) associated with the first intrusion or duress alarm input to occur. The first intrusion or duress alarm input to occur after the circuit is armed shall light the LED associated with its alarm input circuit. If the first alarm input is followed within 800 ms or less by additional intrusion or duress alarms from different alarm input circuits, the LED's associated with these alarm inputs shall also light. However, after a maximum of 1.6 seconds has passed from the time of the first alarm input, this circuit shall be disarmed. Intrusion or duress alarms occurring after the circuit is disarmed shall not cause any LED to light. However, the LED(s) which were activated during the armed period shall be held on until the operating mode switch is next turned from the TEST/RESET position to the SECURE position. Turning the operating mode switch to the SECURE position shall extinguish all alarm identification LED's.

3.5.9 Resync switch. When resync switch S2 is depressed, terminal TB7-S-7 shall be placed at ground potential.

3.5.10 Alarm line supervision. The circuit card connector (J13) provided in the CU shall provide individual electrical access to all intrusion, duress, and tamper alarm input lines and their associated signal return lines. The connector and associated wiring shall be in accordance with the applicable drawings.

3.5.11 Power supply assembly. The power supply assembly shall be in accordance with Top Assembly TA13226E1152 and as specified herein.

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3.5.11.1 Input voltage. The power supply assembly shall operate on primary voltage of 110 to 125 Volts ac (Vac), 48 to 62 Hertz (Hz).

3.5.11.2 Primary dc outputs. The power supply shall be capable of producing the following outputs:

Output Number	Output Voltage	Line Regulation	Load Regulation	Ripple Content
1A (TB2-1 to TB2-4)	27.5 ±1.0 Vdc 1/ at 1.7 amperes (A)	±5.0%	±2.0% no load to rated load	2 Volts (V) peak to peak (p-p) at 1.7 A
1B (TB2-2 to TB2-4)	27.5 ±1.0 Vdc 2/ at 0.25 A	±1.0%	±2.0% no load to rated load	50 millivolt p-p at 250 mA
2 (TB2-3 to TB2-4)	20 ±1.0 Vdc at 0.6 A	±1.0%	±3.0% no load to rated load	50 millivolt p-p at 0.6 A
3 (TB2-5 to TB2-4)	5 ±0.35 Vdc at 50 mA	±1.0%	±2.0% no load to rated load	50 millivolt p-p at 50 mA
4 (TB1-6 TB1-5)	7.0 Vdc at 150 mA	±25%	Not applicable	Not applicable

1/ This set of characteristics simulates fast charging of the 24 V lead-calcium standby battery.
2/ This set of characteristics simulates trickle charging battery.

3.5.11.2.1 Battery charger. Output No. 1 shall be current limited to no more than 2.4 A when loaded with 5 ±0.25 ohms.

3.5.11.3 Auxiliary outputs. The power supply shall provide the following auxiliary outputs:

Output No.	Output Requirement
5 (TB5-1 to TB5-2)	110 - 125 Vac (switched)
6 (TB4-4 to TB4-2)	5 ±0.35 Vdc
7 (TB3-1 to TB3-2	≤ 10 ohms with ac input specified in 3.5.11.1 present between TB1-4 and TB1-2 OR 200K ±20K ohms with no ac input between TB1-4 and TB1-2

3.5.11.3.1 Sonalert. The sonalert shall be capable of producing a minimum signal of 60 decibels (db) reference 0.0002 microbars sound pressure level (SPL) at 10 feet on axis when powered by 20 ±1 Vdc at TB3-3 (+) to TB3-4 (-).

3.5.11.4 Standby power supply. Standby power shall be supplied from a 24 V lead-calcium battery. The battery shall be in accordance with drawing C13220E4208. Batteries are not deliverable items. Removal of the ac power input shall cause the battery (or other source of 24 Vdc at 0.75 A) to immediately begin providing power to sustain outputs 2 and 3 as specified. The current drain

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from the battery (or other source) shall not exceed 0.7 A when providing maximum rated currents at outputs 2 and 3.

3.5.12 Tamper/alarm PWB assembly. The tamper/alarm PWB assembly, A1 through A6, shall be in accordance with Top Assembly TA13220E4213 and as specified herein. The tamper/alarm PWB assembly shall have the following electrical characteristics with 19.9 ±0.1 Vdc applied to terminal pin 10 and with resistances of 20K ohms applied at each of terminal pins 6, 8, and 18, and at test points 1 and 2 (all voltages and resistances are referenced to ground, terminal pin 1):

a. With a step input resistance change from 2K ohms ±1 percent to 1 megohm and greater at terminal pin 4, the voltage at terminal pins 6 and 8 and at test point 1 shall transition from less than 1 Vdc to greater than 18 Vdc 100 ±50 ms after the input resistance change.
b. With a step input resistance change from 2K ohms ±1 percent to 1 megohm and greater at terminal pin 12, the voltage at terminal pin 18 and at test point 1 shall transition from less than 1 Vdc to greater than 18 Vdc 100 ±50 ms after the input resistance change.

3.5.13 Voltage monitor PWB assembly. The voltage monitor PWB assembly A7 shall be in accordance with Top Assembly TA13220E4214 and as specified herein. The voltage monitor PWB assembly shall have the following electrical characteristics with resistances of 20K ohms ±1 percent applied at terminal pin 18 and at test points 1 and 2 (all voltages and resistances are referenced to ground, terminal pin 1):

a. With the supply voltage at terminal pin 10 set to 19.9 ±0.1 Vdc, a step input resistance change from 2K ohms ±1 percent to 1 megohm at terminal pin 4 shall cause the voltage at terminal pin 18 and at test point 1 to increase from less than 1 Vdc to greater than 16 Vdc 100 ±50 ms after the input resistance change.
b. With the supply voltage at terminal pin 10 set to 19.9 ±0.1 Vdc, a step input resistance change from 2K ohms ±1 percent to 1 megohm at terminal pin 7 shall cause the voltage at terminal pin 18 and at test point 2 to increase from less than 1 Vdc to greater than 16 Vdc 100 ±50 ms after the input resistance change.
c. With the supply voltage at terminal pin 10 set to 19.9 ±0.1 Vdc and with an input voltage at terminal pin 14 of 4 to 5.5 Vdc, the voltage at terminal pin 18 shall be less than 1 Vdc.
d. With the supply voltage at terminal pin 10 set to 19.9 ±0.1 Vdc and with an input voltage at terminal pin 14 of less than 1 Vdc, the voltage at terminal pin 18 shall be greater than 18 Vdc.
e. With the supply voltage at terminal pin 10 set to any value equal to or greater than 19.0 and equal to or less than 21.0 Vdc, the voltage at terminal pin 18 shall be less than 1 Vdc.
f. With a supply voltage at terminal pin 10 set to any value equal to or less than 18.3 or equal to or greater than 21.7 Vdc, the voltage at terminal pin 18 shall be greater than 12 Vdc.

3.5.14 Audible Alarm PWB assembly. The Audible Alarm PWB assembly A8 shall be in accordance with Top Assembly TA13220E4215 and as specified herein. The Audible Alarm PWB assembly shall have the electrical characteristics specified in 3.5.14.1 through 3.5.14.6 with 19.9 ±0.1 Vdc applied to terminal pin 9, and 4.9

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±0.1 Vdc at terminal pin 10. Unless otherwise specified, all voltages and impedances are referenced to ground (terminal pin 1).

3.5.14.1 Access timer requirements. After an input pulse between 18 and 20 Vdc of 20 ±0.5 ms and greater duration on terminal pins 15 or 16, an impedance of between 0 to 1 ohm shall exist between terminal pins 2 and 3 for a duration of 5.25 ±0.75 seconds. An impedance greater than 100K ohms shall exist between terminal pins 2 and 3 at any time greater than 6 seconds subsequent to input pulse termination.

3.5.14.2 Nonlatch timer requirements. After an input pulse between 18 and 20 Vdc of 20 ±0.5 ms and greater duration on terminal pin 5, an impedance of between 0 and 100 ohms shall exist between terminal pins 1 and 6 for a duration of 10 ±2 seconds. An impedance greater than 100K ohms shall exist between terminal pins 1 and 6 at any time greater than 12 seconds subsequent to input pulse termination.

3.5.14.3 Audible Alarm inhibit requirements. With 12 Vdc and greater applied to terminal pin 11 and an impedance greater than 100K ohms between terminal pins 1 and 12, an output voltage of between 18 and 20 Vdc shall exist at terminal pin 8. With 18 to 20 Vdc supplied to terminal pin 11 and an impedance of between 0 and 100 ohms between terminal pins 2 and 12, an output voltage between 0 and 1 Vdc shall exist at terminal pin 8.

3.5.14.4 Tamper alarm requirements. With 12 Vdc and greater applied to terminal pin 11 and an impedance greater than 100K ohms between terminal pins 1 and 12, an output voltage between 18 and 20 Vdc shall exist at terminal pin 8. With 0 to 1 Vdc applied to terminal pin 11, an output voltage between 0 and 1 Vdc shall exist at terminal pin 8.

3.5.14.5 Sonalert driver requirements. With a voltage between 4 to 5 Vdc applied to terminal pin 14, an impedance between 0 and 100 ohms shall exist between terminal pins 1 and 13. With a voltage less than 1 Vdc applied to terminal pin 14, an impedance greater than 100K ohms shall exist between terminal pins 1 and 13.

3.5.14.6 Access command requirements. With a voltage between 18 and 20 Vdc applied to terminal pin 16, an impedance between 0 and 1 ohm shall exist between terminal pins 17 and 18. With less than 1 Vdc applied to terminal pin 16, an impedance greater than 100K ohms shall exist between terminal pins 17 and 18.

3.5.15 Instantaneous alarm PWB assembly. The instantaneous alarm PWB assembly A9 shall be in accordance with Top Assembly TA13220E4216 and as specified herein. The instantaneous alarm PWB assembly shall have the electrical characteristics specified in 3.5.15.1 through 3.5.15.4 with 4.9 ±0.1 Vdc applied to terminal pin 11, and 19.9 ±0.1 Vdc applied to terminal pins 3 and 17. Unless otherwise specified, all voltages and impedances are referenced to ground (terminal pin 1).

3.5.15.1 Instantaneous timer requirements. An alarm input pulse at either terminal pins 8, 9, or 10 of 12 Vdc and greater for a duration of 75 ±1 ms and greater shall cause the following impedance or voltage conditions to exist:

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Pin	to	Pin	Impedance or Voltage
R		1	4.5 ±0.5 Vdc
16		18	0 to 1 ohm
4		B	Greater than 100K ohms
14		U	Greater than 100K ohms

These conditions shall remain until the alarm input terminates and for a minimum additional duration of 10 ±2 seconds. Subsequent to the termination of the alarm input pulse and subsequent to a minimum additional duration of 12 seconds, the following impedance conditions shall exist:

Pin	to	Pin	Impedance or Voltage
R		1	less than 1 Vdc
4		N	0 to 1 ohm
14		U	0 to 1 ohm
15		18	0 to 1 ohm

3.5.15.2 Nonlatch alarm output requirements. With 18 to 20 Vdc applied to terminal pin D (reference terminal pin C), an impedance greater than 100K ohms shall exist between terminal pins 4 and B and between terminal pins 4 and N. With the voltage removed, an impedance between 0 and 1 ohm shall exist between terminal pins 4 and N, and an impedance greater than 100K ohms shall exist between terminal pins 4 and B.

3.5.15.3 Exit delay timer requirements. A voltage input at terminal pin 13 between 19 and 21 Vdc with a duration of 20 ±0.5 ms shall result in a delayed output voltage at terminal pin S between 4 and 5 Vdc with a minimum duration of 25 ±1 ms. The output voltage shall occur at 10 ±2 seconds for the minimum exit delay setting and at 80 to 150 seconds for the maximum exit delay setting. For voltage inputs at terminal pin 13 exceeding the exit delay time, the output voltage at terminal pin S shall remain for 20 ±5 ms after removing the voltage at pin 13.

3.5.15.4 Exit delay alarm inhibit. Application of 15 to 20 Vdc to terminal pin 9 during the exit delay time shall not cause a voltage at terminal pin 12 of greater than 1 Vdc. After the exit delay time has expired, application of 15 to 20 Vdc to terminal pin 9 shall cause the voltage at terminal pin 12 to be between 4 and 5 Vdc. When less than 1 Vdc is applied to terminal pin 9, the voltage at terminal pin 12 shall be less than 1 Vdc.

3.5.16 Latched alarm PWB assembly. The latched alarm PWB assembly A10 shall be in accordance with Top Assembly TA13220E4217 and as specified herein. The latched alarm PWB assembly shall have the electrical characteristics specified in 3.5.16.1 through 3.5.16.6 with 4.9 ±0.1 Vdc applied to terminal pin 5. Unless otherwise specified, all voltages and impedances are referenced to ground (terminal pin 1).

3.5.16.1 Entrance timer requirements. With 20 ±1 Vdc applied at terminal pin H, an alarm input at terminal pin 3 between 4 and 5 Vdc with a duration of 20 ±0.5 ms shall result in a delayed output pulse at terminal pin 18 between 3 and 5 Vdc with a minimum duration of 25 ±1 ms. The output pulse shall occur at 10 ±2 seconds for the minimum entrance delay setting and at 80 to 150 seconds for the

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maximum entrance delay setting. For alarm inputs exceeding the entrance delay time, the duration of the delayed output at terminal pin 18 shall be equal to the time difference (±25 ms) between the input duration and the entrance delay time.

3.5.16.2 Audible alarm output requirements. With 18 to 20 Vdc applied to either of terminal pins P or H, and the entrance delay timer reset, the voltage at terminal pin 4 shall be between 4 and 5 Vdc. Once an alarm input is received at terminal pin 3 and a delayed output has occurred at terminal pin 18, the voltage at terminal pin 4 and test point 1 shall be latched at 0 to 1 Vdc. Momentary removal of the voltage at terminal pins P and H shall cause the latch to reset, the voltage at test point 1 to be greater than 18 Vdc, and the voltage at terminal pin 4 to be between 4 and 5 Vdc.

3.5.16.3 Duress alarm requirements. With 18 to 20 Vdc applied to terminal pin L, and K1 and K2 not energized, the impedance between terminal pins 8 and 10 and between terminal pins 8 and 9 shall be between 0 and 1 ohm. An alarm input at terminal pin 6 of between 4 and 5 Vdc with a duration of 20 ±0.5 ms, shall immediately cause the duress output relay K1 to latch ON and an impedance greater than 100K ohms to be present between terminal pins 8 and 9 and 8 and 10. Momentary removal of the voltage at terminal pin L shall cause the latch to reset and the impedance between terminal pins 8 and 10 and between terminal pins 8 and 9 to be between 0 and 1 ohm.

3.5.16.4 Intrusion alarm requirements. With 18 to 20 Vdc applied to terminal pin L, and the entrance delay timer and duress latch reset, and less than 1 Vdc applied to terminal pin 7, the impedance between terminal pins 8 and 10 shall be between 0 and 1 ohm. Once an intrusion alarm input is received at terminal pin 3 and a delayed output has occurred at terminal pin 18, the impedance between terminal pins 8 and 10 shall be latched at greater than 100K ohms. Momentary removal of the voltage at terminal pin L shall cause the latch to reset and the impedance between terminal pins 8 and 10 to be between 0 and 1 ohm.

3.5.16.5 Tamper alarm requirements. With 18 to 20 Vdc applied to terminal pin L, and the entrance delay timer and duress latch reset, and less than 1 Vdc applied to terminal pin 7, the impedance between terminal pins 8 and 10 shall be between 0 and 1 ohm. Application of greater than 18 Vdc at terminal pin 7 shall cause the impedance between terminal pins 8 and 10 to be latched at greater than 100K ohms. Momentary removal of the voltage at terminal pin L shall cause the latch to reset and the impedance between terminal pins 8 and 10 to be between 0 and 1 ohm provided the 18 Vdc is no longer present at terminal pin 7. When 18 Vdc or greater is present at terminal pin 7, the voltage at terminal pin 18 shall be greater than 16 Vdc.

3.5.16.6 Relay K4 requirements. With 18 to 20 Vdc applied to terminal pin 17, the impedance between terminal pins 12 and 13 and between terminal pins 10 and 15 shall be greater than 100K ohms, and the impedance between terminal pins 1 and R shall be between 0 and 1 ohm. With the voltage removed from terminal pin 17, the impedance between terminal pins 11 and R, terminal pins 12 and 14, and terminal pins 15 and 16, shall be between 0 and 1 ohm.

3.5.17 Alarm ID PWB assembly. The alarm ID PWB assembly, A11 shall be in accordance with Top Assembly TA13220E4218 and as specified herein. The alarm

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ID PWB assembly shall have the following electrical characteristics with supply voltages of 19.9 ±0.1 Vdc and 4.9 ±0.1 Vdc.

a. With terminal pin 8 having been grounded for at least 5 ms and with no alarm inputs, i.e. less than 1 Vdc at each of terminal pins 2, 4, 12, 14, 16, and 18, application of 20 Vdc to terminal pin 10 (reset pulse) shall cause a reset of all latches and shall cause all LED driver transistors to turn off.
b. Within 2 seconds after a reset pulse on terminal pin 10, an alarm input greater than 18 Vdc with a duration of 15 ±1 ms presented to any one of input terminal pins 2, 4, 12, 14, 16, or 18 shall cause the corresponding latch to latch and the corresponding LED driver transistor to turn on.
c. Any alarm input greater than 18 Vdc which occurs at any one of input terminal pins 2, 4, 12, 14, 16, or 18 within 0.9 ±0.1 seconds subsequent to the receipt of the leading edge of the first alarm input on any input terminal pin shall cause the corresponding latch to latch and the corresponding LED driver transistor to turn on.
d. Any alarm input greater than 18 Vdc which occurs at input terminal pins 2, 4, 12, 14, 16, or 18 after 1.3 ±0.2 seconds subsequent to receipt of the leading edge of an alarm input on any input terminal pin shall not affect the status of the latches or LED driver transistors. Latches which are set shall remain set. Latches which are reset shall remain reset.

3.5.18 LED PWB assembly. The LED PWB assembly A12 shall be in accordance with Top Assembly TA13220E4219 and as specified herein. The LED PWB assembly shall have all LED's mounted so that the bases of the LED's are parallel to the plane of the card. With 19.9 ±0.1 Vdc simultaneously applied to the anodes of all LED's and 2.2K ohm ±10 percent series load resistors, connected between each cathode and the 19.9 Vdc return, it shall be visually apparent that all LED's are lit with no angular misalignment.

3.5.19 Status processor interconnecting wiring subassembly. The status processor interconnecting wiring subassembly shall be in accordance with TA13220E4225 and as specified herein.

3.6 Burn-in. Each CU and major component when furnished separately shall be subjected to a 24 hour burn-in to reduce the number of infant mortality failures (see 4.5.2.4).

3.7 Electromagnetic susceptibility. The electromagnetic susceptibility characteristics of the CU and power supply when furnished separately shall conform to MIL-STD-461, methods RS03, CS01, CS02, and CS06, as modified in 3.7.1 and 3.7.2.

3.7.1 Radiated. Radiated susceptibility of the CU shall be conducted in an electric field of 5 volts per meter (V/m) for both steady-state and keyed conditions of the EMI field generating equipment. This field applies to frequencies up to 500 megahertz (MHz). Above 500 MHz, the electric field shall be 1 V/m.

3.7.2 Conducted. Conducted interference signals shall be applied to the ac power input lines as follows:

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a. Continuous wave, 30 Hz to 400 kilohertz (kHz).
b. Interfering spike voltage, 10 spikes 5 to 10 microseconds wide at a repitition rate of 10 pulses per second, on input power leads only. Random phasing of the spikes with respect to the power line frequency shall be employed.

3.8 Reliability. The CU shall demonstrate a Mean-Time-Between-Failure (MTBF) of 10,000 hours or greater when tested as specified in 4.5.2.7.

3.9 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.10 Identification marking. The CU and major components shall be identified in accordance with MIL-STD-130 and the drawings referenced herein.

3.11 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.8) for manufacturing and testing the CU and major components when furnished separately:

Item No.	Description	Identification	QTY
1	Stable base drawing of PWB, Tamper/Alarm	13220E4169, sh 2	1 (see notes a,b)
2	Stable base drawing of PWB, Voltage Monitor	13220E4174, sh 2	1 (see notes a,b)
3	Stable base drawing of PWB, Audible Alarm	13220E4175, sh 2	1 (see notes a,b)
4	Stable base drawing of PWB, Instantaneous Alarm	13220E4177, sh 2	1 (see notes a,b)
5	Stable base drawing of PWB, Latched Alarm	13220E4178, sh 2	1 (see notes a,b)
6	Stable base drawing of PWB, Alarm I.D.	13220E4179, sh 2	1 (see notes a,b)
7	Stable base drawing of PWB, LED	13220E4180, sh 2	1 (see notes a,b)
8	Stable base drawing of PWB, Interconnect	13220E4172, sh 2	1 (see notes a,b)
9	Stable base drawing of PWB, Power Supply	13226E1154, sh 2	1 (see notes a,b)
10	Control Unit, Alarm Set	C-9412/FSS-9(V) NSN 6350-00-228-2735	1 (see note c)

a. Provided when procuring Control Unit.
b. Provided when it's associated assembly procured as major component procured separately.
c. Provided when procuring power supply separately and EMI testing of first article is required.

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3.12 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9, except soldering. These requirements are applicable to wiring, welding, brazing, planting, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part (or equipment) unsuitable for the purposes intended.

3.12.1 Soldering. All soldering shall be performed in accordance with MIL-STD-454, requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The CU and major components when furnished separately shall be inspected as specified herein. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article CU and major components when furnished separately shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests.

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4.3.2.1 Complete CU's. Complete CU's shall be subjected to the tests marked "X" in column 1 of table I. Individual testing of major components is not required. Failure of any test shall be cause for rejection.

4.3.2.2 Major components when furnished separately. Major components when furnished separately shall be subjected to the tests marked "X" in column 1 of the applicable tables II through IV. Failure of any test shall be cause for rejection.

4.3.2.3 Measurement recording requirements. All measurements, including voltages, currents, resistances, times, and distances, observed during first article tests shall be recorded on the test data sheets and shall be to the precision necessary to demonstrate compliance with requirements in the Appendix test method being conducted.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II. Sampling for tests shall be in accordance with MIL-STD-105 and as specified herein.

4.4.2 Examination.

4.4.2.1 Individual. Each CU and major component when furnished separately, if applicable, shall be examined for the critical defect specified in 4.5.1. Presence of the defect shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major defects specified in 4.5.1. Sample size shall be determined by using MIL-STD-105, table I and table II-a. A lot shall be accepted when zero defects are found and rejected when 1 or more defects are found.

4.4.3 Tests.

4.4.3.1 Individual.

4.4.3.1.1 CU's. Each CU shall be subjected to the tests marked "X" in column 2 of table I. Individual testing of major components is not required. Failure of any test shall be cause for rejection.

4.4.3.1.2 Major components when furnished separately. Each major component when furnished separately shall be subjected to the tests marked "X" in column 2 of the applicable tables II through IV. Failure of any test shall be cause for rejection.

4.4.3.2 Samples.

4.4.3.2.1 CU's. For normal inspection, five units shall be randomly selected from the monthly production lot. Each sample shall be subjected to the tests marked "X" in column 3 of table I. When tightened inspection is required, sample size shall be eight units. Reduced inspection may be instituted, if approved by responsible Government official, when 10 consecutive monthly sample lots have been subjected to normal inspection, there have been zero defectives on original inspection and production is at a steady rate. Sample size during reduced

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inspection shall be 3 units. For each sample size, the lot shall be accepted when zero defects are found and rejected when 1 or more defects are found.

4.4.3.2.2 Major components when furnished separately. In normal inspection, five units shall be randomly selected from the monthly production lot. Each sample shall be subjected to the tests marked "X" in column 3 of tables II through IV, as applicable. When tightened inspection is required, sample size shall be eight units. Reduced inspection may be instituted, if approved by responsible Government official, when 10 consecutive monthly sample lots have been subjected to normal inspection, there have been zero defectives on original inspection and production is at a steady rate. Sample size during reduced inspection shall be 3 units. For each sample size, the lot shall be accepted when zero defects are found and rejected when 1 or more defects are found.

4.5 Inspection procedure.

4.5.1 Examination. The CU and major components when furnished separately shall be examined as specified herein for the following defects:

Critical

1. Welding burn through.

Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes, or treatment.
105. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the CU or major components when furnished separately.
106. Any dimension, other than those identified on the QAP requirements shown on the drawings, not as specified.
107. Any part (or component) not in accordance with the QAP requirements as shown on the drawings.
108. Electromagnetic interference suppression components and networks not as specified.
109. Components missing or not as specified.
110. Assembly not correct.
111. Identification or special marking missing, incomplete or illegible.
112. Interchangeability not as specified.
113. Workmanship including wiring and soldering not as specified.
114. Treatment and painting not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperature and humidity conditions. After successful completion of functional testing, each CU and major component when

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furnished separately shall be subjected to a 24-hour burn-in as specified in 4.5.2.4. Unless otherwise specified herein, the CU's and major components when furnished separately shall operate as specified herein without maintenance.

4.5.2.2 Test Schedule.

Table I. Test schedule for complete CU.

First Article	Quality Conformance		Test	Requirement Paragraph	Test Paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Burn-in	3.6	4.5.2.4
X	X	-	Functional	3.5.3 thru 3.5.11	4.5.2.5.1
X	X	-	Tamper switch tolerance	3.5.1	4.5.2.5.2
X	-	-	Mode switch life	3.5.7	4.5.2.5.3
X	-	X	Low Temp	3.4.1(a), 3.4.2(a)	4.5.2.6.1
X	-	X	High Temp	3.4.1(a), 3.4.2(a)	4.5.2.6.2
X	-	X	Humidity	3.4.1(b), 3.4.2(b)	4.5.2.6.3
X	-	-	Shock	3.4.2(c)	4.5.2.6.4
X	-	-	Electromagnetic susceptibility	3.7	4.5.2.7
X	-	-	Reliability	3.8	4.5.2.8

Table II. Test Schedule for power supply assembly.

First Article	Quality Conformance		Test	Requirement Paragraph	Test Paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Burn-in	3.6	4.5.2.4
X	X	-	Functional	3.5.11	4.5.2.5.4
X	-	X	Low Temp 1/	3.4.1(a), 3.4.2(a)	4.5.2.6.1
X	-	X	High Temp 1/	3.4.1(a), 3.4.2(a)	4.5.2.6.2
X	-	X	Humidity 1/	3.4.1(b), 3.4.2(b)	4.5.2.6.3

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X	-	-	Electromagnetic susceptibility 2/	3.7	4.5.2.7
X	-	X	Shock	3.4.2(c)	4.5.2.6.4

1/ Power supply shall be mounted in a vertical position (switch on top). No adjustments to the power supply are permissible during environmental test or at any time after the post burn-in alignment (see Appendix test method 7, step 2.3i).
2/ Place power supply assembly in Government-loaned CU to simulate normal environment when installed in CU.

Table III. Test Schedule for PWB assemblies.

First Article	Quality Conformance		Test	Requirement Paragraph	Test Paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Burn-in	3.6	4.5.2.4
X	X	-	Functional	1/	2/
X	-	X	Low Temp	3.4.1(a), 3.4.2(a)	4.5.2.6.1
X	-	X	High Temp	3.4.1(a), 3.4.2(a)	4.5.2.6.2
X	-	X	Humidity	3.4.1(b), 3.4.2(b)	4.5.2.6.3

1/ 3.5.12 through 3.5.18 as applicable.
2/ 4.5.2.4.5 through 4.5.2.4.11 as applicable.

Table IV. Test Schedule for status processor interconnecting wiring subassembly.

First Article	Quality Conformance		Test	Requirement Paragraph	Test Paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.5.19	4.5.2.5.12

4.5.2.3 Test methods. Test shall be conducted in accordance with the test methods in the appendix and as specified herein. Calibration of the test equipment shall be in accordance with MIL-STD-45662.

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4.5.2.4 Burn-in. Each CU and major component when furnished separately, except the status processor and interconnecting wiring board, shall be subjected to a 24-hour burn-in. Complete CU's shall be subjected to the burn-in test of Appendix test method 4. Power supplies when furnished separately shall be subjected to the burn-in test of Appendix test method 6. Other major components when furnished separately shall be continuously energized and subjected to the functional tests of Appendix test methods 7 through 13, as applicable, nine times, but no test shall be initiated less than 45 minutes after completion of the previous test period. A complete functional test shall be performed after completion of burn-in. Non compliance with the applicable portions of 3.5 shall constitute a failure. Failures during burn-in shall be analyzed, repaired, and corrective action taken as necessary. When specified in the contract or purchase order (see 6.2), burn-in test reports shall be forwarded to the contracting officer. Any unit failing during burn-in shall be subjected to an additional 24-hour burn-in after failure analysis and repair and successful completion of the functional test.

4.5.2.5 Operational tests. The CU and major components when furnished separately shall be tested in accordance with 4.5.2.5.1 through 4.5.2.5.12 as applicable.

4.5.2.5.1 CU functional test. The CU shall be tested in accordance with Appendix test method 1. Nonconformance to 3.5.3 through 3.5.11 shall constitute failure of this test.

4.5.2.5.2 Tamper switch tolerance. The CU shall be tested in accordance with Appendix test method 2. Nonconformance to 3.5.1 shall constitute failure of this test.

4.5.2.5.3 Mode switch life. The CU shall be tested in accordance with Appendix test method 3. Nonconformance to 3.5.7 shall constitute failure of the test.

4.5.2.5.4 Power supply assembly. The power supply assembly shall be tested in accordance with Appendix test method 6. Nonconformance to 3.5.11 shall constitute failure of this test.

4.5.2.5.5 Tamper/alarm PWB assembly. The tamper/alarm PWB assembly shall be tested in accordance with Appendix test method 7. Nonconformance to 3.5.12 shall constitute failure of this test.

4.5.2.5.6 Voltage monitor PWB assembly. The voltage monitor PWB assembly shall be tested in accordance with Appendix test method 8. Nonconformance to 3.5.13 shall constitute failure of this test.

4.5.2.5.7 Audible alarm PWB assembly. The audible alarm PWB assembly shall be tested in accordance with Appendix test method 9. Nonconformance to 3.5.14 shall constitute failure of the test.

4.5.2.5.8 Instantaneous alarm PWB assembly. The instantaneous alarm PWB assembly shall be tested in accordance with Appendix test method 10. Nonconformance to 3.5.15 shall constitute failure of this test.

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4.5.2.5.9 Latched alarm PWB assembly. The latched alarm PWB assembly shall be tested in accordance with Appendix test method 11. Nonconformance to 3.5.16 shall constitute failure of this test.

4.5.2.5.10 Alarm ID PWB assembly. The alarm ID PWB assembly shall be tested in accordance with Appendix test method. Nonconformance to 3.5.17 shall constitute failure of this test.

4.5.2.5.11 LED PWB assembly. The LED PWB assembly shall be tested in accordance with Appendix test method 12. Nonconformance to 3.5.18 shall constitute failure of this test.

4.5.2.5.12 Status processor interconnecting wiring subassembly. The status processor interconnecting wiring subassembly shall be tested in accordance with Appendix test method 13. Nonconformance to 3.5.19 shall constitute failure of this test.

4.5.2.6 Environmental. The CU and major components when furnished separately shall be subjected to the following series of tests specified in MIL-STD-810. During each test, conduct the functional test specified in 4.5.2.5.1 and 4.5.2.5.4 through 4.5.2.5.11 as applicable. At the completion of each environmental test, conduct the functional tests specified in 4.5.2.5.1 and 4.5.2.4.4 through 4.5.2.4.11 as applicable.

4.5.2.6.1 Low Temperature. Conduct low temperature test method 502.1, procedure I, storage temperature of -30 °F for not less than 48 hours, equipment operating temperature of -20 °F.

4.5.2.6.2 High Temperature. Conduct high temperature Test Method 501.1, Procedure II, except that the storage temperature during step 4 shall be +165 °F. The operating temperature of step 7 shall be +150 °F.

4.5.2.6.3 Humidity. Conduct Test Method 507.1, Procedure I with the following exceptions:

(a) Step 5 shall be for 5 cycles (not less than 120 hours). Perform the functional test at the completion of each cycle.
(b) During steps 2 and 3, the relative humidity shall be maintained at 95% or greater.
(c) During step 4, the relative humidity shall be maintained at 92% or greater.
(d) Measurement shall commence 1 hour and 30 minutes prior to the completion of each 24 hour cycle. The CU may be removed from the environmental chamber for measurements. However, unsuccessful completion of measurements or unsuccessful return of the CU to the test chamber within 1 hour and 30 minutes subsequent to the initiation of measurements shall constitute failure of the test.
(e) The major components may be removed from the environmental chamber for testing. Testing shall commence 1 hour and 30 minutes prior to the completion of each 24 hour cycle. Nonconformance to the appropriate functional requirements for the major components when tested in

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accordance with the applicable test methods in the Appendix or failure to return the major components to the test chamber within 1 hour and 30 minutes subsequent to the initiation of testing shall constitute failure of the test.

4.5.2.6.4 Shock. Bench handling: Conduct method 516.2, procedure V, equipment not operating (without battery).

4.5.2.6.5 Failure criteria. Nonconformance to 3.4 shall constitute failure of these tests.

4.5.2.7 Electromagnetic susceptibility. When required (see 6.2) the CU and power supply when furnished separately shall be subjected to the following electromagnetic susceptibility tests of MIL-STD-461. Functional testing required during electromagnetic testing shall be in accordance with Appendix test method 5 of this specification.

4.5.2.7.1 Radiated susceptibility. Test method RS03, except that the electric field requirements of RS03 shall be modified and limited as specified in 3.7.1.

4.5.2.7.2 Conducted susceptibility. Test methods CS01, CS02, and CS06, as modified in 3.7.2.

4.5.2.7.3 Failure criteria. Nonconformance to 3.7 as demonstrated by test method 5 shall constitute failure of this test.

4.5.2.8 Reliability. Using an upper test MTBF (theta sub naught) of 10,000 hours and a lower test MTBF (theta sub 1) of 5,000 hours, the CU shall be tested as specified herein, with "accept" and "reject" criteria in accordance with MIL-STD-781, test plan VIIIC. Continue testing until either an "accept" or "reject" decision is reached. All test incidents will be scored by the Government. A failure is defined as any malfunction which causes or may cause one or more of the following:

(a) Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
(b) Serious damage to the system.
(c) Serious personnel safety hazards.

4.5.2.8.1 Reliability Test procedure. The test shall be conducted under ambient laboratory conditions, without temperature or humidity cycling. Equipment shall be energized at the primary power input specified herein. The reliability test shall be conducted in accordance with Appendix test method 6. The test shall commence only after successful completion of the 24 hour burn-in, including functional test, specified in 4.5.2.4. A complete functional test in accordance with Appendix test method 1 shall be performed on each test item at completion of the reliability test period. Parts or components of known limited life, such as indicator lamps, shall have a life stipulated prior to test. Failures of these parts occurring after the end of the stipulated period are nonrelevant, but any dependant failures caused thereby are counted as relevant. Failures chargeable to workmanship deficiencies shall be counted as relevant. Alarms not deliberately excited as part of the equipment test routine shall be reported and cause of the alarm identified, if practicable. This accounting shall be maintained separately from the failure accounting. Preventive

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maintenance procedure specified for the equipment during normal operation shall be performed during the tests. No additional preventive maintenance shall be permitted during the test.

4.5.2.8.2 Failure criteria. Nonconformance to 3.7 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 First article pack inspection.

4.6.1.1 Examination. The first article pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Tests. The level A, B, or C first article pack of the control unit and power supply assembly when furnished separately shall be subjected to the free fall drop test as described in FED-STD-101, method 5007.1. Levels A and B packs shall be tested in accordance with Procedure E, drop heights conforming to those specified for the appropriate level and gross weight. Level C packs shall be tested using Procedure D with drop heights the same as specified for level B. Any damage to the contents of the pack or shifting of the contents shall constitute failure of the test and be cause for rejection of the first article pack.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. Sample size shall be determined by using MIL-STD-105, table I and table II-a. A lot shall be accepted when zero defects are found and rejected when 1 or more defects are found.

115. Complete CU's not preserved and packed as specified for levels A and B.
116. Keys not placed with the control unit as specified for level A, level B, or level C.
117. Major components furnished separately not preserved and packed as specified for level A or level B.
118. Components of unlike description consolidated or packed together for level A or level B.
119. Quantities packed together exceed the size or weight limitations of the box for level A or B.
120. Strapping not as specified for level A or B.
121. Marking missing, illegible, incorrect, or incomplete for levels A, B, or C.

5. Packaging.

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5.1 First article. The contractor shall furnish a first article pack for examination and testing within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied packaging and marking comply with the packaging requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3). The first article pack may be prepared utilizing either the first article or a production CU or major component when furnished separately. When the first article model is utilized, any packaging shall be removed by the supplier at no expense to the Government, to facilitate comparison between the first article and production CU's.

5.2 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Each CU and major components when furnished separately shall be preserved by one of the following procedures depending upon procurement of complete CU's or procurement of major components when furnished separately.

5.2.1.1 Procurement of complete CU's. Each CU shall be preserved in accordance with MIL-P-116, submethod IIb. The inner container shall conform to PPP-B-636, type CF, class Domestic, variety SW or DW, grade 200. The CU shall be cushioned and secured within the container to prevent movement or damage. The keys shall be placed in a cotton draw-string bag or heat-sealed bag and placed within the container. The outer container shall conform to PPP-B-636, W5c, style optional.

5.2.1.2 Procurement of major components when furnished separately. In addition to the following requirements, major components when furnished separately that are subject to electrostatic or electromagnetic damage shall be protected as specified in MIL-P-116 (see 5.4.3).

5.2.1.2.1 Major components when furnished separately. Major components when furnished separately shall be individually preserved in accordance with MIL-P-116, submethod IIb. The inner container shall conform to PPP-B-636, type CF, class Domestic, variety SW, grade 175. The major component shall be cushioned and secured within the container to prevent movement or damage. The outer container shall conform to PPP-B-636, W6c or W5c, style optional.

5.2.1.2.1.1 Consolidation. Major components of like description, preserved as specified in 5.2.1.2.1, shall be consolidated together in a close-fitting box conforming to PPP-B-636, W6c, or W5c, style optional, in quantities not to exceed the weight limitations of the box. The contents shall be cushioned within the box to prevent movement or damage.

5.2.2 Level B. Preservation shall be as specified in 5.2.1 for level A, except submethod IIc shall be used in place of submethod IIb. Each CU or major component when furnished separately shall be placed in a close-fitting box conforming to PPP-B-636, W6c, or W5c, style optional. The component shall be cushioned within the box to prevent movement or damage.

5.2.3 Level C. Each CU with its keys shall be preserved together. Major components when furnished separately shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first using

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activity. As a minimum, each CU or major component when furnished separately shall be placed in a fiberboard box.

5.3 Packing. Packing shall be level A, level B, or level C as specified (see 6.2).

5.3.1 Level A. The CU and major components when furnished separately, preserved as specified in 5.2, shall be packed by one of the following procedures depending upon procurement of complete CU's or procurement of major components when furnished separately.

5.3.1.1 Procurement of complete CU's. CU's preserved as specified in 5.2 shall be packed together in a close-fitting box confirming to PPP-B-601, overseas type, grade B, style optional, in quantities not to exceed the size and weight limitations of the box. Cushioning and blocking within the box used to prevent movement and isolate impact shock shall be in accordance with MIL-STD-1186. Closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

5.3.1.2 Procurement of major components when furnished separately. Components of like description preserved as specified in 5.2, shall be packed together as specified in 5.3.1.1.

5.3.2 Level B. Complete CU's and major components when furnished separately, preserved as specified in 5.2, shall be packed as specified in 5.3.1 as applicable, except boxes shall be domestic type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.3.3 Level C. Each CU and major components when furnished separately, preserved as specified in 5.2, shall be packed in shipping containers that will afford adequate protection against damage during shipment from the supply source to the first receiving activity.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129 and the special marking specified in 5.4.1 and 5.4.2.

5.4.2 Special Marking. Each shipping container shall be marked with project code labels in accordance with MIL-STD-129. The labels shall contain the Project Code "JID".

5.4.3 Sensitive electronic components. Major components when furnished separately, preserved as specified in 5.2.1.2 or 5.2.2 for components requiring electrostatic or electromagnetic protection, shall be additionally marked in accordance with MIL-STD-129 with a sensitive electronic device caution label and the following caution statement:

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CAUTION

OBSERVE PRECAUTIONS FOR HANDLING

ELECTROSTATIC SENSITIVE ITEMS

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The CU, along with associated sensors, data transmission system, monitor cabinet and audible alarm comprise an Intrusion Detection System known as Joint-Services Interior Intrusion Detection System (J-SIIDS).

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Nomenclature and quantity of item(s) to be furnished.
d. Technical Data Package List(s), including revision level, to which the item(s) are to be built.
e. When a first article is required for inspection and approval, the number of units required, and the allocation of units required to environmental and reliability tests respectively.
f. When EMI testing is required as part of first article tests (see 4.5.2.7).
g. When Government-loaned property is not required (see 3.11).
h. Level of preservation and level of packing required (see 5.2 and 5.3).
i. Schedule, content, and format of burn-in test reports (see 4.5.2.4).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 False alarm. An alarm generated by stimuli other than those for which the test item is designed to alarm.

6.4.2 Alarm signals.

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6.4.2.1 Intrusion alarm. Alarm received from the sensor subsystem in response to an intrusion or attempted intrusion of the control zone.

6.4.2.2 Tamper alarm. Alarm caused by an attack on the system itself (tamper or line supervisor alarms) and failures which cause an alarm.

6.4.2.3 Duress alarm. Alarm received from a latching alarm switch resulting from activation by security personnel.

6.4.3 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any components; physical change which impairs the mechanical integrity of the control unit; evidence of delamination or water penetration into integrated circuits, printed circuit boards, external components, or physical condition resulting in noncompliance of the item with the provisions of this specification.

6.4.4 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. A QAP indicates the minimum requirements which must be inspected on the product drawings to verify the objectives of the product and assure interchangeability of the repair parts.

6.5 Quality assurance provisions (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manual to be furnished with each CU.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of the end items.

6.8 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.11 when applicable (see 6.2).

6.9 Subject term (key word) listing.

Alarm system
Intrusion detection
Physical security

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:
Army - ME
Navy - SH
Air Force - 99

Review activity:
Air Force - 82
DLA-CS

User activity:
Army - ER

Preparing activity:
Army - ME

Project 6350-0114

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APPENDIX

TEST METHODS

FOR

CONTROL UNIT, ALARM SET

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether CU's and major components when furnished separately procured under this specification conform to the requirements specified. Figures 1 through 26 contain data for fabrication of test fixtures and preparation of test set-ups which are required to facilitate testing of the CU's and major components when furnished separately.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

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APPENDIX

TEST METHOD 1

CONTROL UNIT

FUNCTIONAL TEST

1. TESTS TO BE PERFORMED:

a. Power supply test.
b. Voltage monitor test.
c. Secure/access indication test.
d. Secure mode test.
e. Access mode test.
f. Test/reset mode test.
g. Intrusion alarm delay test.
h. Audible Alarm inhibit test.

2. TEST APPARATUS:

a. Digital voltmeter (DVM) - Hewlett-Packard 34740A display with 34702A multimeter or equivalent.
b. Power supply (2) - Harrison 6202B or equivalent.
c. Time interval meter - B&K Precision 1822 or equivalent.
d. Oscilloscope, Tektronix Model T922, or equivalent.
e. DVM, Hewlett-Packard Model 3466A, or equivalent (ammeter and ac voltmeter).
f. Autotransformer (VR1), GenRad Model W10-MT3A, or equivalent.
g. Ammeter, -5, 0, +5 A dc (M1), GE Model 250-LSIS, or equivalent.
h. Test fixture (see figure 1).
i. Test fixture schematic and parts list (see figures 2 and 3).
j. Control unit test set-up (see figure 4).

3. TEST FIXTURE PREPARATION:

a. Remove jumpers (terminal links) between terminals 1 and 2 and terminals 2 and 3 of TB8 through TB13 and between terminals S-8 and S-9 of TB7. Reinstall after completion of testing.
b. Connect CU to external test equipment as shown in figure 4. The +20 V power supply shall be adjusted to 20 ±0.5 Vdc and the +24 V power supply to 24 ±0.5 Vdc.
c. Pull out plunger on door tamper switch to maintanence position.
d. Place test fixture controls in the following positions:

Control	Position	Function
S1	1	Monitor
S2	2	DVM
S3	2 (INST)	Time function
S4	2	Start trigger
S5	1	Stop trigger
S6	OFF	Audible Alarm inhibit
S7	OFF	Manual

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APPENDIX

Control	Position	Function
S9	ON (CLOSED)	Audible Alarm tamper
S10	ON (CLOSED)	Dialer
S11	OFF	Timed alarm
S12	1	Alarm start
S13	1	Delayed input
S14	1	Alarm duration
S22-S27	OPEN	Loads
A1-F1	1	Sensor simulators
A3-F3	1	Sensor simulators

f. Set autotransformer (VR1) for output voltage of 115 Vac to power supply (TB1-1 to TB1-2).
g. Adjust potentiometers R6 on A9 PWB and R8 on A10 PWB fully clockwise.

3.1 Power supply alignment procedure. The following alignment procedure shall be performed on each power supply prior to its subjection to initial functional tests (before burn-in) and prior to the first functional test after completion of burn-in. It may not be performed at any time during environmental or reliability tests. This procedure shall be done at room ambient temperatures between 65 °F and 82 °F.

a. Disconnect leads in CU that are attached to TB2-3 and TB2-5. Two red leads are attached to TB2-3 and two brown leads to TB2-5. Protect lead ends so that they cannot touch any other terminals, components, or the chassis. Do not disconnect leads going to test fixture.
b. Open CU door and pull out tamper switch to the non-alarm position. Place mode switch in ACCESS position.
c. Close S22, S23, and S24. Place the CU power supply switch in the ON position. Allow unit to operate for 5 minutes prior to making the following adjustments:

NOTE: Voltage readings called for may be made with DVM provided for in figure 4 by placing DVM switch S2 on test fixture in the position appropriate to the output number to be measured. This correspondence is shown in table V.

1. Adjust R3 on card A1 of power supply to attain 27.5 ±0.1 Vdc at output #1B (DVM switch S2 on test fixture is set to position 2 for this reading).
2. Adjust R6 on card A1 to attain 20 ±0.1 Vdc at output #2.

d. Voltage at output #3 shall be 5 ±0.35 Vdc.
e. The ac power lamp on the CU door shall light when ac is present across TB1-1 and TB1-2. Voltage across lamp, measured by reading DVM with S2 in position 6, shall be 7 ±2 Vdc.
f. Open S22, S23, and S24 and turn off CU power supply switch.
g. Reconnect leads disconnected in step a if test procedure of paragraph 4 will not be performed immediately.

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TABLE V. Power supply acceptance data.

Serial No: _____

Temperature: _____

A	B	C	D	E	F	G	H	I	J	K	L
Output Number	DVM Switch Position *	Location	Voltage Rqmt @ -20° to 150°F	Output Voltage No Load @ 115 VAC	Output Voltage Loaded @ 115 Vac	Allowable Load Deviation From Column D	Ripple V p-p Loaded @ 115 Vac	Allowable Ripple V p-p	Output Voltage Loaded @ 110 Vac	Output Voltage Loaded @ 125 Vac	Allowable Line Deviation From Column D
1A	2	TB2-1 to TB2-4	27.5 Vdc ± 1.0 V			± 2.0%		2 V			± 5.0%
1B	2	TB2-2 to TB2-4	27.5 Vdc ± 1.0 V			± 2.0%		50 mV			± 1.0%
2	1	TB2-3 to TB2-4	20 Vdc ± 1.0 V			± 3.0%		50 mV			± 1.0%
3	5	TB2-5 to TB2-4	5 Vdc ± 0.35 V			± 2.0%		50 mV			± 1.0%
4	6	TB1-5 to TB1-6	7 Vdc ± 2.0 V	**		**	**	**			±25.0%

* Applicable to test method 1 only.
** Not applicable.

Current limit [para 4.1.5(c) and 2.4.5(c) for test methods 1 and 7 respectively]: _____ amps (1.70 ±0.15 A).
Heavy current limit [para 4.1.5(e) and 2.4.5(e)]: _____ amps (2.4 A maximum).

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4. TEST PROCEDURES

4.1 Primary power supply.

a. Perform step a. of 3.1 above.
b. Adjust the output of VR1 to 115 ±1 Vac.
c. Set CU mode switch to ACCESS position.
d. Place the CU power switch in the ON position. The ac power lamp shall light (located on CU door).
e. Observe and record the output voltage levels for outputs #1A, 1B, 2 and 3 in column E of table V.
f. The output voltages must be within the levels stated in column D of table V to be acceptable.

4.1.2 Load regulation.

a. Close S22, S23, S25, and S5 and open S24, S26, and S27.
b. Maintain the output on VR1 at 115 ±1 Vac.
c. Ammeter shall indicate 1.7 ±0.15 A.
d. Observe the voltages on outputs #1A, 2, 3, and 4, and record in column F of table V.
e. Open S25 and close S24.
f. Record the voltage on output #1B in column F of table V.
g. For acceptable load regulation, the output voltage measurements recorded in column F shall not differ from those recorded in column E by more than the amounts stated in column G. The voltages recorded in column F shall also meet the requirements specified in column D.

4.1.3 Ripple and noise (periodic and random deviations).

a. Close S22, S23, and S25 and open S24, S26, and S27.
b. Maintain the output of VR1 at 115 ±1 Vac.
c. With an oscilloscope substituted for the DVM, measure the ac p-p ripple voltages on outputs #1A, 2, and 3. Record this data in column H of table V.
d. Open S25 and close S24. Measure and record the ac p-p ripple voltage on output #1B in column H of table V.
e. For acceptable ripple content, the ac p-p voltages shall be less than those specified in column I of table V.

4.1.4 Line regulation.

a. Close S22, S23, and S25 and open S24, S26, and S27.
b. Adjust the output of VR1 to 110 ±1 Vac.
c. Measure and record, in column J of table V, the voltages on outputs #1A, 2, 3, and 4.
d. Open S25, close S24, and record the voltage on output #1B in column J of table V.
e. Open S24, close S25, and increase the output of VR1 to 125 ±1 Vac.
f. Measure and record, in column K of table V, the voltages on outputs #1A, 2, 3, and 4.
g. Open S25, close S24, and record the voltage on output #1B in column K.

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h. For acceptable line regulation, the output voltage measurements recorded in columns J and K shall not differ from those recorded in column F by more than the amounts stated in column L. The voltages recorded in columns J and K shall also meet the requirements specified in column D.

4.1.5 Current limit.

a. Close S22, S23, and S25 and open S24, S26, and S27.
b. Adjust the output of VR1 to 115 ±1 Vac.
c. Observe and record in Table V the ammeter reading. It shall be 1.7 ± 0.15 A.
d. Close S26. Record in table V the current indicated on the ammeter.
e. The current indicated on ammeter shall not exceed 2.4 A.
f. Open S26.
g. For acceptance, the current limit characteristics shall be within the above specified limits.

4.1.6 Standby power.

a. Close S22, S23, and S27 and open S24, S25, and S26.
b. Place the CU power supply switch in the ON position.
c. Turn on the +24 V power supply.
d. Remove ac power input from CU power supply. The ac power lamp on the CU door shall extinguish. The voltages at outputs #2 and #3 shall remain within allowable tolerances specified in column D of table V. The ammeter will indicate current is being supplied from the 24 volt power supply and shall indicate not more than 725 ma.
e. Restore ac power input. The ac power lamp shall illuminate. The voltages at outputs #2 and #3 shall be within allowable tolerances specified in column D of table V.
f. For acceptance, the standby power characteristics shall be within the above specified limits.

4.1.7 Auxiliary outputs.

a. Close S22, S23, and S24 and open S25, S26, and S27.
b. Place the power supply switch in the ON position and adjust the output of VR1 to 115 ±1 Vac.
c. The voltage from TB4-4 to TB4-2 shall be 5 ±0.35 Vdc.
d. The voltage across TB5-1 and TB5-2 shall be 115 ±1 Vac.
e. Turn CU power supply switch off. The voltage across TB5-1 and TB5-2 shall be 0 Vac.
f. For acceptance, the auxiliary output characteristics shall be within the above specified limits.

4.2 Voltage monitor test.

a. Place DVM switch S2 in position 1. Open S22, S23, S24, S25, S26, and S27, reconnect leads that were disconnected from CU TB2-3 and TB2-5 in 4.1.1, step a, and turn on CU power supply.

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b. Observing DVM, rotate monitor switch S1 from position 1 thru position 6. The DVM shall indicate 20 ±1 Vdc at each position.
c. Set CU power switch to OFF.
d. Remove fuse F2 and connect an external dc power supply to TB2-3 (+) and TB2-4 (ground). The 24V power supply shown in figure 3 may be removed from the test set-up and used for steps d through p.
e. Adjust external dc power supply to 20 ±0.5 Vdc.
f. Set CU power switch to ON.
g. Turn CU mode switch to TEST/RESET.
h. Turn mode switch to SECURE. (Allow for exit time delay before proceeding.)
i. Push reset switch S8 on test fixture. All indicator lights shall go out.
j. Decrease external power supply voltage slowly to less than 19 Vdc. Alarm indicator lights INST, DELAYED, and AA shall illuminate when the voltage is greater than 18.2 and less than 19.0 Vdc.
k. Readjust external dc power supply output to 20 Vdc.
l. Turn CU mode switch to TEST/RESET and back to SECURE. (Allow for exit time delay before proceeding.)
m. Push reset switch S8. All indicator lights shall go out.
n. Increase external power supply voltage to greater than 21 Vdc. Alarm indicator lights INST, DELAYED, and AA shall illuminate when the voltage is greater than 21.0 and less than 21.8 Vdc.
o. Set CU power switch to OFF.
p. Disconnect external power supply and replace fuse F2. If 24 V power supply was used (see step d above), restore as shown in figure 4. Its power shall be on and its voltage adjusted for 24 Vdc.
q. Set CU power switch to ON.
r. Turn mode switch to TEST/RESET then back to SECURE. Wait 6 seconds for expiration of secure/access delay.
s. Push reset switch S8. All indicator lights shall go out.
t. Set timer function switch S3 to AC FAIL (position 6), stop trigger switch S5 to position 2, DVM switch S2 to position 2, and monitor switch S1 to position 6, switch S27 to closed, and reset time interval meter. The DVM voltage shall be less than 0.5 Vdc.
u. Simultaneously unplug ac power from the CU and set manual switch S7 to ON. The timer interval meter shall begin running. When the timer stops,it shall read 3.5 ±1.5 seconds and indicator light AC FAIL shall illuminate. The DVM voltage shall be 10 ±2 Vdc.
v. Turn manual switch S7 to OFF and open switch S27.
w. Reapply ac power to the CU.
x. Push reset switch S8. Indicator light AC FAIL shall go out.

4.3 Secure/access indication test.

a. Set CU mode switch to SECURE. Set timer function switch S3 to S/A, start trigger switch S4 to 1, and stop trigger switch S5 to 2.
b. Push reset switch S8 until all indicator lights go off. Reset time interval meter to zero.
c. Turn CU mode switch to TEST/RESET and back to SECURE within 0.5 second. Indicator light A/S shall illuminate and the time interval meter shall begin running. When the timer stops, the meter shall indicate 4.5 to

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6.5 seconds.

d. Push reset switch S8. Indicator light A/S shall go out.
e. Reset time interval meter.
f. Rapidly turn mode switch to ACCESS and back to SECURE within 1 second. The conditions of step c. shall occur, except meter may indicate up to 7 seconds.
g. Push reset switch S8. Indicator light A/S shall go out.

4.4 Secure Mode Test.

a. Set latch/nonlatch switch on status processor to LATCH.
b. Set timer function switch S3 to INST, start trigger switch S4 to 2, and stop trigger switch S5 to 1.
c. Close S11 and set A1 to position 2 (in that order).
d. At least 12 seconds after performing step f. of 4.3, momentarily depress reset switch S8 and reset time interval meter.
e. Momentarily depress S12 to simulate a 250 ms intrusion alarm. Alarm indicator light INST shall illuminate, the proper LED on PWB A12 shall illuminate, and the time interval meter shall begin running.
f. When the timer stops, the meter shall indicate 10 ±2 seconds. Alarm indicator lights DELAYED and AA shall illuminate within ±4 seconds of the time at which the timer stops. It is not necessary to accurately measure the ±4 seconds.
g. At least 12 seconds after alarm indicator lights DELAYED and AA illuminate, momentarily depress S8. Alarm indicator light INST shall extinguish.

NOTE: All alarm indicator lights will be off while S8 is depressed. Throughout these procedures, it is understood that the condition described in a given step immediately after momentarily depressing S8 is the condition that shall exist after the release of S8.

h. Turn CU mode switch to TEST/RESET, then momentarily depress S8. Alarm indicator lights DELAYED and AA shall extinguish.
i. Turn CU mode switch to SECURE and reset time interval meter. LED A1 on PWB A12 shall extinguish.

NOTE: Throughout this and subsequent tests, disregard the status of indicator light A/S, unless otherwise specified.

j. Set S3 to position 3 (Egress), S5 to position 2, A1 to position 1, and reset the time interval meter if necessary.
k. Set B1 to position 2.
l. At least 12 seconds after performing step i., momentarily depress switch S12. Alarm indicator light INST and the proper LED on PWB A12 shall illuminate and the time interval meter shall begin running.
m. When the time interval meter stops, alarm indicator lights DELAYED and AA shall illuminate. The time interval meter shall indicate 10 ±2 seconds.
n. As soon as the time interval meter stops, turn the CU mode switch to TEST/RESET and back to SECURE, then momentarily depress S8 and reset time interval meter. Alarm indicator lights DELAYED and AA and the LED

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on PWB A12 shall extinguish. Alarm indicator light INST may extinguish. If it does not, wait 4 seconds and momentarily depress S8. Alarm indicator light INST shall extinguish. Reset time interval meter and set B1 to position 1.

o. Repeat steps k., l., m., and n. for sensors simulator switches C1, D1, and E1, except, in steps k. and n., the sensor simulator switch being tested shall be set to position 2 in step j. and back to position 1 in step n.
p. Set S3 to position 2 (INST) and S5 to position 1 and reset time interval meter.
q. Set switch A3 to position 2.
r. Momentarily depress switch S12 to simulate a 250 ms tamper alarm. Alarm indicator lights INST, DELAYED, and AA shall illuminate and the time interval meter shall begin running. None of the LED's on PWB A12 shall illuminate.
s. When the timer stops, it shall indicate 10 ±2 seconds.
t. Momentarily depress reset switch S8. Alarm indicator light INST shall extinguish. The DELAYED and AA alarms indicator lights shall be lit when the switch is released.
u. Turn mode switch to TEST/RESET and back to SECURE and reset time interval meter, and set A3 to position 1.
v. Momentarily depress switch S8. After release of S8, all alarm indicator lights shall be off.
w. Repeat steps q. through v., using sensor simulator switches B3, C3, D3, E3, F3, except, in steps q. and u., substitute the switch being used for A3. In one of these trials, allow 4 seconds to elapse after performing step s. and before step t.
x. Momentarily open AA tamper switch S9. Alarm indicator lights INST, DELAYED, and AA shall illuminate and the time interval meter shall begin running. None of the LED's on PWB A12 shall illuminate.
y. When the timer stops, it shall indicate 10 ±2 seconds.
z. Momentarily depress reset switch S8. Alarm indicator light INST shall extinguish. The DELAYED and AA alarm indicator lights shall be lit when the switch is released.
aa. Turn CU mode switch to TEST/RESET and back to SECURE and reset time interval meter.
bb. Momentarily depress reset switch S8. After release of S8, all alarm indicator lights shall be off.
cc. Repeat steps x. through bb. for dialer switch S10 and for the CU door tamper switch.
dd. At least 12 seconds after last turning mode switch to SECURE, set sensor simulator switch F1 to position 2 and momentarily depress S12 to simulate a duress alarm. Alarm indicator lights INST and DELAYED shall illuminate, the time interval meter shall begin running, and LED F on PWB A12 shall illuminate. The AA indicator light shall not light.
ee. When the timer stops, it shall read 10 ±2 seconds.
ff. Momentarily depress reset switch S8. Alarm indicator light INST shall extinguish.
gg. Turn CU mode switch to TEST/RESET and back to SECURE and momentarily depress S8. Alarm indicator light DELAYED and LED F on PWB A12 shall extinguish. Reset time interval meter and set F1 switch to position 1.
hh. Set switch S14 to position 2 (60 ms).

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ii. Set A1 to position 2 and momentarily depress S12. None of the alarm indicator lights or LED's on PWB A12 shall illuminate. After completion, set A1 to position 1.
jj. Repeat step ii. for switches B1 through F1 and switches A3 through F3. Results shall be same as for switch A1. After completion set switch S14 back to position 1.
kk. Set latch/nonlatch switch on status processor to NONLATCH.
ll. Set timer function switch S3 to LATCH/NL.
mm. At least 12 seconds after last turning mode switch to SECURE, set switch A1 to position 2 and momentarily operate S12. Alarm indicator light INST shall illuminate and LED A on PWB A12 shall illuminate. After 10 ±2 additional seconds (it is not necessary to measure this time accurately) alarm indicator lights DELAYED and AA shall illuminate and the time interval meter shall read 10 ±2 seconds.
nn. Momentarily depress reset switch S8. Alarm indicator lights INST and DELAYED shall extinguish.
oo. Turn CU mode switch to TEST/RESET and back to SECURE, momentarily depress S8, and reset time interval meter. Alarm indicator light AA and the proper LED on PWB A12 shall extinguish. Set A1 to position 1.
pp. Repeat steps q. through u. Responses shall be as before, except in step t., both alarm indicator lights INST and DELAYED shall extinguish.
qq. Repeat steps dd. through gg. Responses shall be as before except, in step ff., alarm indicator light DELAYED shall also extinguish.
rr. Set switch S13 to position 2, switch S14 to position 3 (1.3 seconds), sensor simulators switches A1 and B1 to position 2, then momentarily depress S12. LED A on board A12 shall illuminate and LED B shall not illuminate. Disregard alarm indicator lights.
ss. Turn mode switch to TEST/RESET and back to SECURE. LED A shall extinguish (Allow for exit time before proceeding).
tt. Set switch S14 to position 4, then momentarily depress switch S12. LED's A and B shall illuminate. Disregard alarm indicator lights.
uu. Turn mode switch to TEST/RESET and back to SECURE. LED's A and B shall extinguish (Allow for exit time before proceeding).
vv. Push transmitter resync switch on status processor. The test fixture RESYNC indicator light shall illuminate while switch is depressed.

4.5 Access mode test.

a. Set latch/nonlatch switch on status processor to LATCH and set CU mode switch to ACCESS position.
b. Set timer function switch S3 to INST, start trigger switch S4 to 2, stop trigger switch S5 to 1, S13 to position 1 and S14 to position 1.
c. Momentarily depress reset switch S8 and reset time interval meter.
d. Set sensor simulator switch A1 to position 2 and momentarily depress S12. Wait at least 12 seconds. None of the LED's on the test fixture shall illuminate. LED A on PWB A12 shall illuminate.
e. Turn CU mode switch to SECURE and return to ACCESS. LED A on PWB 12 shall extinguish when turned to SECURE mode.
f. Set sensor simulator switches A1 to position 1 and A3 to position 2 and momentarily depress S12. Alarm indicator lights INST, DELAYED, and AA shall illuminate and the time interval meter shall begin running. None of the LED's on the PWB A12 shall illuminate.

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g. When the timer stops, it shall indicate 10 ±2 seconds.
h. Momentarily depress reset switch S8. Alarm indicator light INST shall extinguish.
i. Turn mode switch to TEST/RESET and back to ACCESS. Reset time interval meter.
j. Momentarily depress reset switch S8. Alarm indicator lights DELAYED and AA shall extinguish.
k. Set sensor simulator switch A3 to position 1, F1 to position 2 and momentarily depress S12 to simulate a duress alarm. Alarm indicator lights INST and DELAYED shall illuminate, the time interval meter shall begin running, and LED F on PWB A12 shall illuminate. The AA indicator light shall not light. Set F1 to position 1.
l. When the timer stops, it shall indicate 10 ±2 seconds.
m. Momentarily depress reset switch S8. Alarm indicator light INST shall extinguish.
n. Turn mode switch to SECURE and back to ACCESS.
o. Momentarily depress reset switch S8, and reset time interval meter. Alarm indicator light DELAYED and LED F shall extinguish.
p. Set latch/nonlatch switch on status processor to NONLATCH.
q. Set timer function switch S3 to LATCH/NL.
r. Repeat steps f. through j. Responses shall be as before, except, in step h., when S8 is pushed, alarm indicator lights INST and DELAYED shall extinguish and AA shall remain on after S8 is released.
s. Repeat steps k. through o. Responses shall be as before, except, in step m., alarm indicator lights DELAYED and INST shall extinguish.
t. Turn DVM switch S2 to 2.
u. Turn monitor switch S1 from position 1 through 6. The DVM shall indicate 24 to 28.5 V at each position except position 6.
v. Set timer function switch S3 to EGRESS, start trigger switch S4 to 2, stop trigger switch S5 to 2, and set S11 to OPEN.
w. Push reset switch S8 and reset time interval meter.
x. Turn potentiometer A9R6 fully clockwise.
y. Turn mode switch from ACCESS to SECURE.
z. Set sensor simulators switch A1 to position 2 within 0.5 seconds of step y. Alarm indicator light INST shall illuminate, the DELAYED and 2AA indicator lights shall be off, and the time interval meter shall begin running.
aa. When the timer stops, alarm indicators DELAYED and AA shall illuminate, indicator light INST shall remain on, and the time interval meter shall indicate 20 ±4.5 seconds.
bb. Set sensor simulators switch A1 to position 1.
cc. Turn potentiometer A9R6 fully counterclockwise.
dd. Turn mode switch to ACCESS. Disregard sonalert if it sounds.
ee. Push reset switch S8 and reset time interval meter.
ff. Repeat steps y. through bb., except in step aa., the meter shall indicate 125 ±35 seconds.
hh. Turn potentiometer A9R6 fully clockwise.

4.6 Test/reset mode test.

a. Set timer function switch S3 to INST, start trigger switch S4 to 2, and stop trigger switch S5 to 1. Set S11 to CLOSED.

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b. Set latch/nonlatch switch on status processor to LATCH and set mode switch to TEST/RESET position.
c. Momentarily depress reset switch S8 and reset time interval meter.
d. Set sensor simulator switch A1 to position 2 and momentarily depress S12 to simulate an intrusion alarm. Alarm indicator lights INST and DELAYED shall illuminate, LED A on board A12 shall illuminate, the sonalert shall begin sounding, and the time interval meter shall begin running. The AA indicator light shall not illuminate.
e. When the timer stops, the sonalert shall stop sounding and the meter shall indicate 10 ±2 seconds.
f. Momentarily depress reset switch S8. All alarms indicator lights shall extinguish.
g. Turn mode switch to SECURE and back to TEST/RESET and reset time interval meter. LED A shall extinguish.
h. Set sensor simulator switch A1 to position 1 and A3 to position 2, then momentarily depress S12 (tamper alarm). Alarm indicator lights INST and DELAYED shall illuminate, the sonalert shall begin sounding, and the time interval meter shall begin running. The AA indicator light and LED A on PWB A12 shall not illuminate.
i. When the timer stops, it shall indicate 10 ±2 seconds. The sonalert shall stop sounding when the timer stops.
j. Momentarily depress reset switch S8 and reset time interval meter. All alarm indicator lights shall extinguish.
k. Set sensor simulator switch A3 to position 1 and F1 to position 2 and momentarily depress S12 to simulate a duress alarm. Alarm indicators INST and DELAYED shall illuminate, the sonalert shall begin sounding, the time interval meter shall begin running, and LED F on PWB A12 shall illuminate. The AA indicator light shall not light.
l. When the timer stops, it shall indicate 10 ±2 seconds. The sonalert shall stop sounding when the timer stops.
m. Momentarily depress reset switch S8. All alarm indicator lights shall extinguish.
n. Turn mode switch to SECURE and back to TEST/RESET. LED F shall extinguish. Reset time interval meter and set switch F1 to position 1.
o. Set latch/nonlatch switch on status processor to NONLATCH.
p. Set timer function switch S3 to LATCH/NL, start trigger switch S4 to 2, and stop trigger switch S5 to 1.
q. Repeat steps d. through n. Responses shall be identical to those obtained in corresponding steps.

4.7 Intrusion alarm delay test.

a. Set mode switch to SECURE. (Allow for exit time delay before proceeding.)
b. Push reset switch S8 and reset time interval meter.
c. Turn potentiometer A1OR8 fully clockwise.
d. Set timer function switch S3 to ACCESS, start trigger switch S4 to 2, and stop trigger switch S5 to 2. Set S11 to OPEN.
e. After the exit time delay has elapsed, simulate an intrusion by setting any sensor simulators load switch A1 through E1 to position 2. Alarm indicator light INST and the appropriate CU LED shall illuminate, and indicator lights DELAYED and AA shall be off. The time interval

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meter shall begin running.

f. When the timer stops, alarm indicator lights DELAYED and AA shall illuminate, indicator light INST and CU LED shall remain on, and the time interval meter shall indicate 10 ±2 seconds.
g. Set sensor simulators switch which was opened in step e. to position 1.
h. Turn potentiometer A1OR8 fully counterclockwise.
i. Turn mode switch to TEST/RESET then back to SECURE. Appropriate CU LED shall extinguish. Disregard sonalert if it sounds. (Allow for exit time before proceeding.)
j. Push reset switch S8 and reset time interval meter.
k. Repeat steps e. through g., except in step f., the time interval meter shall indicate 115 ±35 seconds.
l. Turn potentiometer A1OR8 fully clockwise.

4.8 AA inhibit test.

a. Turn mode switch to TEST/RESET then to SECURE. (Allow for exit time delay before proceeding.)
b. Push reset switch S8 and reset time interval meter.
c. Set AA inhibit switch S6 to ON.
d. Set sensor simulators switch F3 to position 2 to simulate a tamper alarm. Alarm indicator light INST shall illuminate and indicator lights DELAYED and AA shall be off.
e. Turn off AA inhibit switch S6. Alarm indicator light INST shall remain on and indicator lights DELAYED and AA shall illuminate.
f. Set sensor simulators switch F3 to position 1. Allow 12 seconds for alarm to stop before proceeding.
g. Push reset switch S8. LED's INST and DELAYED shall extinguish and LED AA shall remain lit.
h. Turn mode switch to TEST/RESET, then to SECURE.
i. Push reset switch S8. LED AA shall extinguish.

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APPENDIX

TEST METHOD 2

CONTROL UNIT

TAMPER SWITCH TOLERANCE TEST

1. TEST APPARATUS:

   a. Six-inch ruler.
   b. CU test set-up (see figure 4).

2. TEST PREPARATION:

   a. Connect CU to external test equipment as shown in figure 4.
   b. Set all sensor simulators switches on test fixture to CLOSED.
   c. Set CU switch to ON.
   d. Close and lock door.
   e. Set mode switch to TEST/RESET.

3. TEST PROCEDURE:

   a. Note the door flange position on the right side of the cabinet near the door lock. Make a small mark on the cabinet side panel to record the exact position of the door flange.
   b. Unlock and slowly open the cabinet door. Mark the door flange position on the side panel at its exact position when the internal alarm sounds. The door flange shall not open beyond the edge of the side panel before the CU sonalert sounds.
   c. Measure between marks. The measurement shall be not less than 0.125 inch nor more than 0.250 inch.

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APPENDIX

TEST METHOD 3

CONTROL UNIT

MODE SWITCH LIFE TEST

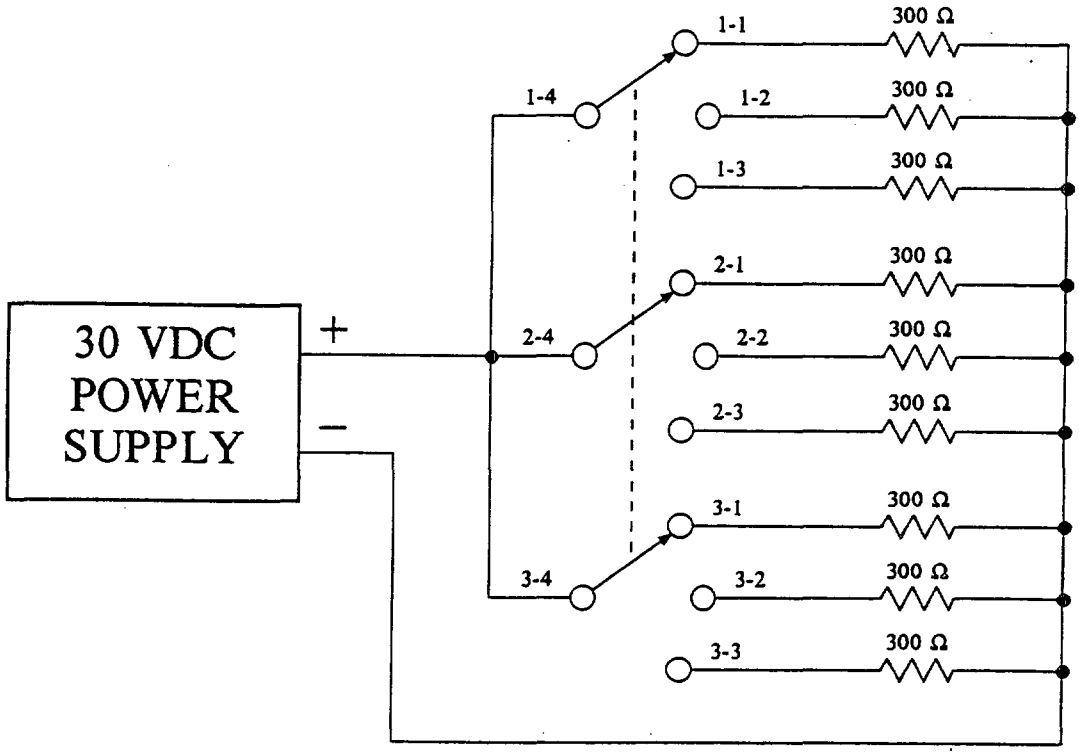
1. TEST APPARATUS:

   a. Power supply - B & K Precision Model 1610 or equivalent.
   b. Resistor (9) - 300 ohms, 3 watts minimum.
   c. Mode switch actuating device - As required.

2. TEST PROCEDURE:

   a. The following test shall be performed prior to installation of the switch in the CU:

      (1) Connect power supply set for 30 Vdc output across switch terminals as shown below.
      (2) Actuate switch 50,000 times.
      (3) After completion of life test, the mode switch shall show no visual sign of damage and shall function as specified after installation into the CU.



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APPENDIX

TEST METHOD 4

CONTROL UNIT BURN IN

1. TEST APPARATUS:

   a. DVM - Hewlett-Packard 34740A display with 34702A multimeter or equivalent.
   b. Power supply, Preston X-MOD-101 or equivalent (two required).
   c. Multimeter, Simpson Model 260 or equivalent.
   d. Resistor, 62 ohms, 10 watts (W).
   e. Time interval meter - B&K Precision model 1822 or equivalent.
   f. Test fixture (see figure 1).
   g. Test fixture schematic and parts list (see figures 2 and 3).
   h. CU test set-up (see figure 4).

2. TEST PREPARATION:

   a. Connect CU to external test equipment as shown in figure 4 or as required to perform test procedure. If substitute fixture is utilized perform steps similar to 2b. through 2f. below.
   b. Set all sensor simulators switches on test fixture to CLOSED and pull plunger out on door tamper switch.
   c. Set AA inhibit and manual switches S6 and S7 on test fixture to open on and dialer switch S10 to closed position.
   d. Set potentiometers on A9 and A10 PWB assemblies for minimum delay times.
   e. Set CU power switch to ON.
   f. Turn CU mode switch to TEST/RESET then to SECURE.
   g. After 6 seconds, momentarily depress reset switch S8.

3. TEST PROCEDURE:

   a. A functional test in accordance with test method 1 shall have been conducted prior to starting burn-in.
   b. Nine times each 24 hours during the test period, demonstrate the following:

      1. that a momentary intrusion or duress alarm input results in the correct output response at each of the 3 output terminal pairs and the alarm identification LED for each combination of latch/nonlatch switch and mode switch position (6 combinations).
      2. that a momentary tamper alarm input results in the correct output response at each of the 3 output terminal pairs for each combination of latch/nonlatch switch and mode switch position (6 combinations).
      3. that the entrance and exit delay functions are operating properly.
      4. that the secure/access timer is operating properly when switched to the TEST/RESET position and back to the SECURE position.

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APPENDIX

NOTE: Only one intrusion or duress alarm input and one tamper alarm input need be exercised at any one of the 9 times in the 24 hours. However, in the 24-hour period, each such input must be exercised at least once. In the case of the tamper inputs, both external inputs and internal sources shall be included, except that the voltage monitor circuit need not be exercised.

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APPENDIX

TEST METHOD 5

CONTROL UNIT

ELECTROMAGNETIC SUSCEPTIBILITY TEST

1. TEST APPARATUS:

a. Test fixture (see figure 5).
b. EMI test set-up (see figure 6).
c. 20 ohm, 50 watt, 5 percent resistor.
d. Power supply, Hewlett-Packard 6202B or equivalent.
e. 6-channel chart recorder, Brush 260 or equivalent.
f. Equipment as specified in MIL-STD-462, Test Methods CS01, CS02, CS06 and RS06.

2. TEST PREPARATION:

a. Drill or punch two 7/8 inch conduit holes in top left (inside marked rectangle) surface of CU enclosure. Use care to insure metal shavings do not fall into power supply.
b. Place CU inside EMI chamber in location where it will be tested and place test apparatus (a., c., and d. above) outside EMI chamber.
c. Connect 1/2 inch electrical metallic tubing (EMT) between CU and the test fixture (figure 5), using ones of the holes drilled or punched in step a.
d. Pull 10 wires through EMT and connect to test fixture as shown in figure 6. Connect other test apparatus as shown in figure 6.
e. Disconnect red and black battery leads from CU TB2-1 and TB2-4 and connect the 20 ohm resistor between these 2 terminals.
f. Adjust potentiometers A10R8 (entrance time delay) and A9R6 (exit time delay) to maximum clockwise position in order to minimize time delay on alarm processing to the delayed alarm output.
g. Set Latch/Nonlatch switch to LATCH position.

3. TEST PROCEDURE:

a. The CU is now ready to be connected to the EMI test equipment as specified in MIL-STD-462, test method CS01. Connect the ac power source and test equipment to CU TB1 using the second conduit hole drilled or punched in step 2 a. Turn the CU, external 5 V power supply, and chart recorder on. Close and lock the CU door.
b. Insure S1 and S2 are closed (non-alarm condition). Rotate the mode switch to TEST/RESET and back to SECURE. Wait 12 seconds and then verify output conditions on the chart recorder are correct. All outputs should be low except for the Access/Secure output and the AA output which should be high (see table VI).
c. Conduct test method CS01. The chart recorder shall be continuously monitored. No erroneous outputs shall occur as a result of the test.
d. Repeat the test, but initiate momentary intrusion alarms (S1 open) at various frequencies throughout the test frequency band to insure that alarms are processed normally during CU operation under adverse EMI conditions. Outputs shall be as specified in table VI. Always reset CU

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(mode switch to TEST/RESET and back to SECURE then wait 12 seconds) after instantaneous alarm terminates.

e. Turn the mode switch to ACCESS. All outputs shall be low except for the AA which shall be high. Repeat step c.
f. Repeat step d., except use S2 (instead of S1) to generate tamper alarms.
g. Test method CS01 is now complete. Turn off all test equipment.
h. Repeat steps a. through f. for test methods CS02 and CS06.
i. When test method CS06 is complete, disconnect CS06 test equipment. Connect ac source to CU. Turn the CU, 5 V power supply, and chart recorder on. Close and lock CU.
j. Set up EMI test equipment for MIL-STD-462, test method RS03. Perform steps b. through f. above.
k. EMI testing is now complete.

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APPENDIX

TABLE VI. Output conditions.

CONDITION	INST	DELAYED	AA	S/A	AC FAIL
SECURE MODE					
INTRUSION	X[1]	X[2]	X[2]	HIGH	LOW
TAMPER	X[1]	X[3]	X[3]	HIGH	LOW
NORMAL (NON-ALARM)	LOW	LOW	HIGH	HIGH	LOW
ACCESS MODE					
INTRUSION	LOW	LOW	HIGH	LOW	LOW
TAMPER	X[1]	X[3]	X[3]	LOW	LOW
NORMAL (NON-ALARM)	LOW	LOW	HIGH	LOW	LOW
TEST/RESET MODE					
INTRUSION	X[4]	X[1]	HIGH	LOW	LOW
TAMPER	X[4]	X[1]	HIGH	LOW	LOW
NORMAL (NON-ALARM)	LOW	LOW	HIGH	LOW	LOW

NOTES

X INDICATES ALARM CONDITION, HIGH (>4.5 V) FOR INST AND DELAYED, LOW (<0.5 V) FOR AA OUTPUT.

-1 NO DELAY, OUTPUT ALARM PRESENT UNTIL 10 ± 2 SECONDS AFTER CESSATION OF ALARM INPUT.
-2 APPEARS AFTER 10 ± 2 SECONDS DELAY, LATCHED UNTIL RESET BY TURNING MODE SWITCH TO TEST/RESET.
-3 NO DELAY, LATCHED UNTIL RESET BY TURNING MODE SWITCH TO TEST/RESET.
-4 SAME AS -1 EXCEPT INTERNAL ALARM (SONALERT) ALSO SOUNDS WHILE OUTPUT ALARM IS PRESENT.

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APPENDIX

TEST METHOD 6

CONTROL UNIT

RELIABILITY TEST

1. TEST APPARATUS:

   a. CU reliability test fixture (see figure 7).
   b. CU reliability test set-up (see figure 8).
   c. Power supplies, Hewlett-Packard 6202B or equivalent (2 required).
   d. Chart recorder, 6 channel, Brush 260 or equivalent or other means of continuous monitoring.
   e. Digital multimeter, Hewlett Packard 3466A or equivalent.

2. TEST PREPARATION:

   a. Open CU door and set tamper switch to maintenance position (all the way out).
   b. Connect test fixture to CU as shown in figure 8. Set power supply 1 to 5 ±0.2 Vdc and power supply 2 to 24.5 ±0.5 Vdc. AC power input (115 Vac nominal) should be switched off.
   c. Set switches S1 through S14 closed, S16 open, and S15 to position 2.
   d. Set potentiometers on CU A9 and A10 cards to maximum clockwise position to obtain minimum entrance and exit time delays.
   e. Set CU Latch/Non-Latch switch to NON-LATCH.
   f. Set CU mode switch to SECURE.
   g. Close ac power input switch.
   h. Turn on CU power switch.
   i. If an alarm occurs, turn mode switch to TEST/RESET, then return to SECURE position. After 12 seconds, outputs 1, 2, and 4 should be low (less than 1 Vdc) and outputs 3 and 5 high (5 ±1 Vdc). If not, take corrective action before proceeding with test.

3. TEST PROCEDURE:

   a. Reliability test begins after step h. of test preparation. During the reliability test, outputs 1 through 5 shall be continuously monitored. Nine times each 24 hours (except where noted herein), but not more than once per hour, the tests of steps b. through aa. shall be performed. Except where specified during these test periods, 115 Vac (nominal) power shall be continuously applied to the CU, the CU power switch shall be on, and the CU door shall be closed. Any alarms that occur during door closing after performing tests may be neglected. Such alarms shall be reset by performing step i. of test preparation. Test periods are considered as part of the total elapsed time of the reliability test.
   b. Set Latch/Non-latch switch to LATCH position.
   c. Set mode switch to ACCESS position. Observe outputs after 12 seconds. Outputs 1, 2, 3, and 4 shall be low. Output 5 shall be high. All A12 PWB LED's shall be unlit.
   d. Momentarily (for 150 to 500 ms) open switch S2. Within 1 second,

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outputs 1 and 2 shall go high and output 5 low. Output 1 shall go low after 10 ±2 seconds. After 12 seconds, outputs 2 and 5 shall remain high and low respectively. All LED's on A12 PWB shall remain unlit.

e. Set mode switch to TEST/RESET position and observe outputs. Immediately, output 2 shall go low and output 5 high.

f. Return mode switch to ACCESS position and repeat steps d. and e. for switches S4, S6, S8, S10, S12 and S14 and the CU tamper switch.

g. With mode switch still set to TEST/RESET, open S3 and observe outputs. Within 1 second, the sonalert shall begin sounding, outputs 1 and 2 shall go high, and LED A in the CU shall light. Output 5 shall be unchanged (high).

h. Set mode switch to SECURE position and observe outputs for the next 25 seconds. Immediately, the sonalert shall cease sounding, LED A shall extinguish, output 2 shall go low and outputs 1 and 5 shall remain high. After 4.5 to 6 seconds output 3 shall go high. Output 2 shall go high and output 5 low 20 ±4 seconds after the mode switch is placed in SECURE. LED A shall light 10 ±2 seconds after the mode switch is placed in SECURE.

i. Close switch S3 and observe outputs for at least 14 seconds. Output 1 shall go low after 10 ±2 seconds. Nothing else shall change.

j. Set mode switch to TEST/RESET position and immediately return to SECURE. Output 2 shall go low and output 5 high immediately. Output 3 shall go low for 4.5 to 6.5 seconds. LED A shall extinguish.

k. Wait 15 seconds and momentarily open S5. Observe outputs. Within 1 second, output 1 shall go high and LED B shall light. Output 2 goes shall go high and output 5 low after 10 ±2 seconds.

l. Set Latch/Non-Latch switch to NON-LATCH.

m. Rotate mode switch to TEST/RESET then return to SECURE position to reset latched alarms. Wait 15 seconds. Disregard output changes.

n. Momentarily open switch S7 and observe outputs. Output 1 shall go high immediately and LED C shall light. After a delay of 10 ±2 seconds output 2 shall go high and output 5 low. In an additional 10 ±2 seconds, output 2 shall go low and output 5 shall remain low. Output 1 shall go low 10 ±2 seconds after it went high.

o. Rotate mode switch to TEST/RESET, then return to SECURE position and observe outputs. Output 5 shall go high at TEST/RESET position and LED C shall immediately extinguish at SECURE position. Output 3 shall go low at TEST/RESET position and remain low for 4.5 to 6 seconds after arriving at the SECURE position, then shall go high.

p. At least 12 seconds after placing mode switch in SECURE position in step o., momentarily activate switches S9 and S11 simultaneously and observe outputs. Output 1 shall go high immediately and LED's D and E shall light. After 10 ±2 seconds, output 2 shall go high and output 5 low. In an additional 10 ±2 seconds, output 2 shall go low and output 5 shall remain low.

q. Momentarily open switch S1 and observe outputs. Outputs 1 and 2 shall immediately go high, then go low after 10 ±2 seconds. Output 3 shall remain high. LED's D and E shall remain lit and all other LED's unlit. Output 5 shall remain low.

r. Rotate mode switch to TEST/RESET, then return to SECURE position. Output 5 shall go high. LED's D and E shall extinguish at return to SECURE. Output 3 shall go low at TEST/RESET and go high 4.5 to 6

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seconds after return to SECURE.

s. Momentarily open switch S1 and observe outputs. Outputs 1 and 2 shall immediately go high, then go low after 10 ±2 seconds. Outputs 3 and 5 shall remain high. LED F shall light.

t. Set mode switch to TEST/RESET, then to SECURE position. Disregard output changes.

NOTE: Steps u. through aa. below need be performed only once per 24 hour period.

u. In sequence, set switch S15 to position 1, close switch S16 and open switch providing ac power to CU. Observe outputs. Output 4 shall go high 3 ±2 seconds after removal of ac power. No other output changes shall occur.

v. Open switch S16, place S15 in position 2. Do not restore ac input power.

w. Remove fuse F2 from CU power supply.

x. Set voltage on power supply #2 to 20 ±1 Vdc, then close switch S16.

y. Slowly reduce power supply #2 voltage while monitoring output 1 (disregard other outputs). Output 1 shall be low above 19.0 Vdc and shall go high at no less than 18.3 Vdc.

z. Increase power supply #2 voltage to 20 ±1 Vdc. After output 1 goes low, slowly increase voltage while monitoring output 1. Output 1 shall go high at a voltage not less than 21.0 Vdc and not greater than 21.7 Vdc.

aa. Place S15 in position 1 and return power supply #2 to 24.5 ±.5 Vdc. Replace fuse F2 and restore ac power to the CU.

bb. At all times when tests are not being performed, the CU mode switch shall be in SECURE position, Latch/Nonlatch switch in NONLATCH position, and outputs 1 through 5 shall be continuously monitored. Outputs 1, 2 and 4 shall be low and outputs 3 and 5 high at all times.

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APPENDIX

TEST METHOD 7

POWER SUPPLY ASSEMBLY

1. TEST APPARATUS:

a. Oscilloscope, Tektronix Model T922, or equivalent.
b. DVM, Hewlett-Packard Model 5314A, or equivalent.
c. Autotransformer (VR1), GenRad Model W10-MT3A, or equivalent.
d. Power supply, dc, Hewlett-Packard Model 6200, or equivalent.
e. Sound level meter, B&K Model 2209, or equivalent.
f. Ammeter, -5, 0, +5 A dc (M1), GE Model 250-LSLS, or equivalent.
g. Milliammeter, 0 - 300 mA dc (M2), GE Model 250-KGKG, or equivalent.
h. Milliammeter, 0 - 800 mA dc (M3), GE Model 250-KWKW, or equivalent.
i. Milliammeter, 0 - 100 mA dc (M4), GE Model 250-JRJR, or equivalent.
j. Resistor, 15 ohm adjustable, 100 W, Clarostat Model VK-100-NA-15, orequivalent.
k. Resistor, 5 ohm, ±5%, 100 W, Ohmite Model 0600A, or equivalent.
l. Resistor, 150 ohm adjustable, 12 W, Ohmite Model 1013, or equivalent.
m. Resistor, 50 ohm adjustable, 25 W, Ohmite Model 0366, or equivalent.
n. Resistor, 100 ohm, ±5% 1/2 W.
o. Switches (S1-S8), SPST, Leviton Model 19916, or equivalent.
p. Lamp, Chicago Miniature Model 1815, or equivalent.
q. Test Fixture (see figure 9).
r. Diode, 1N4004 or equivalent.

2. TEST PROCEDURE (ROOM TEMPERATURE):

2.1 General. Unless otherwise specified, all alignment, burn-in, and testing procedures will be conducted at prevailing room conditions (68 °F to 78 °F).

2.2 Alignment procedure. The following alignment procedure will be performed on each power supply prior to its subjection to burn-in and functional tests:

a. Connect the power supply to the test fixture as shown in figure 9.
b. Set autotransformer (VRI) for output voltage of 115 Vac to power supply (TB1-1 to TB1-2).
c. Close S1, S2, S3, and S4 and open S5, S6, S7, and S8. Place the power supply switch in the ON position. Allow unit to operate for 5 minutes prior to making the following adjustments:

1. Adjust R3 on card A1 to attain 27.5 ±0.1 Vdc at output #1B. M2 shall indicate 0.250 ±0.035 A.
2. Adjust R6 on card A1 to attain 20 ±0.1 Vdc at output #2. M3 will indicate 0.600 ±0.025 A.
3. Measure 5 ±0.35 Vdc at output #3. M4 shall indicate 50 ±7 mA.

d. The No. 1815 lamp (TB1-6 to TB1-5) will light when ac is present on TB1-1 and TB1-2.

2.3 Burn-in. Burn-in will be 24 hours minimum and will be conducted under the following conditions:

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a. Connect the power supply to the test fixture as shown in figure 9.
b. Set VR1 for input voltage of 115 ±1 Vac at power supply input (TB1-1 to TB1-2).
c. Close S1, S2, S3, and S4 and open S5, S6, S7, and S8. Place the power supply switch in the ON position. The No. 1815 lamp shall light.
d. Verify output #1B is 27.5 ±1.0 Vdc. M2 shall indicate 0.250 ±0.035 A.
e. Verify output #2 is 20 ±1.0 Vdc. M3 shall indicate 0.600 ±0.045 A.
f. Verify output #3 is 5 ±0.35 Vdc. M4 shall indicate 50 ±7.0 mA.
g. Some time within the last 8 hours of burn-in, open S2 and close S5 to simulate a fast charge condition. M1 shall indicate 1.7 ±0.15 A. After 4 hours in this condition, repeat steps d., e., and f.; then open S5 and close S2. M2 shall indicate 0.250 ±0.035 A.
h. Failure of any test shall constitute failure of the burn-in requirement and will require failure analysis, repair, and repeat of the entire 24-hour burn-in.
i. The burn-in is to be completed prior to the remaining test procedures. After successful completion of burn-in, repeat alignment procedure of 2.2 above.

2.4 Functional test.

2.4.1 Output voltage levels.

a. Connect the test apparatus as shown in figure 9. Open S1 through S8. Set the power supply assembly power switch in the OFF position and apply 115 Vac, 60 Hz line power to VR1.
b. Adjust the output of VR1 to 115 ±1 Vac.
c. Close S1.
d. Place the power supply assembly power switch in the ON position. The No. 1815 lamp shall light.
e. Observe and record the output voltage levels for outputs #1B, 2, and 3 in column E of Appendix table V.
f. The output voltages must be within the levels stated in column D of Appendix table V to be acceptable.

2.4.2 Load regulation.

a. Close S1, S3, S4, and S5 and open S2, S6, S7, and S8.
b. Maintain the output on VR1 at 115 ±1 Vac.
c. M1 shall indicate 1.7 ±0.15 A, M2 shall indicate 0.0 A, M3 shall indicate 0.6 ±0.1 A, and M4 shall indicate 50 ±6 mA.
d. Observe the voltages on outputs #1A, 2, 3, and 4, and record in column F of Appendix table V.
e. Open S5 and close S2. M1 shall now indicate 0.0 A and M2 shall indicate 250 ±35 mA.
f. Record the voltage on output #1B in column F of Appendix table V.
g. For acceptable load regulation, the output voltage measurements taken in column F shall not differ from those taken in column E by more than that stated in column G. The voltages taken in column F shall also meet the requirements given in column D.

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2.4.3 Ripple and noise (periodic and random deviations).

a. Close S1, S3, S4, and S5 and open S2, S6, S7, and S8.
b. Maintain the output of VR1 at 115 ±1 Vac.
c. With an oscilloscope, measure the ac p-p ripple voltages on outputs #1A, 2, and 3. Record this data in column H of Appendix table V.
d. Open S5 and close S2. Measure and record the ac p-p ripple voltage on output #1B in column H of Appendix table V.
e. For acceptable ripple content, the ac p-p voltages shall be less than stated in column I of Appendix table V.

2.4.4 Line regulation.

a. Close S1, S3, S4, and S5 and open S2, S6, S7, and S8.
b. Adjust the output of VR1 to 110 ±1 Vac.
c. Measure and record, in column J of Appendix table V, the voltages on outputs #1A, 2, 3, and 4.
d. Open S5, close S2, and record the voltage on output #1B.
e. Open S2, close S5, and increase the output of VR1 to 125 ±1 Vac.
f. Measure and record, in column K of Appendix table V, the voltages on outputs #1A, 2, 3, and 4.
g. Open S5, close S2, and record the voltage on output #1B in column K.
h. For acceptable line regulation, the output voltage measurements recorded in columns J and K shall not differ from those recorded in column F by more than that stated in column L. The voltages recorded in columns J and K shall also meet the requirements given in column D.

2.4.5 Current limit.

a. Close S1, S3, S4, and S5 and open S2, S6, S7, and S8.
b. Adjust the output of VR1 to 115 ±1 Vac.
c. Record in Appendix Table V the current indicated on M1. It shall be 1.7 ±0.15 A.
d. Close S6. Record in Appendix table V the current indicated on M1.
e. The current indicated on M1 shall not exceed 2.4 A.
f. Open S6.
g. For acceptance, the current limit characteristics shall be within the above specified limits.

2.4.6 Standby power.

a. Close S1, S3, S4, and S7 and open S2, S5, S6, and S8.
b. Place the power supply (PS2) switch in the ON position.
c. Open S1. The lamp across TB1-5 and TB1-6 shall extinguish. M3 shall continue to indicate 0.6 ±0.1 A and M4 shall continue to indicate 50 ±6 mA. M1 shall indicate current is being supplied from the power supply. M1 shall indicate no more than 50 ma higher than the sum of M3 and M4.
d. Close S1. The lamp across TB1-5 and TB1-6 shall illuminate. M3 shall continue to indicate 0.6 ±0.1 A and M4 shall continue to indicate 50 ± 6 mA. M1 will indicate less than 10 ma.
e. For acceptance, the standby power characteristics shall be within the above specified limits.

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2.4.7 Output #5.

a. Close S1, S2, S3, and S4 and open S5, S6, S7, and S8.
b. Place the power supply power switch in the ON position and adjust the output of VR1 to 125 ±1 Vac.
c. The voltage across TB5-1 and TB5-2 shall be 125 ±1 Vac.
d. For acceptance, the output #5 characteristics shall be within the above specified limits.

2.4.8 Output #6, 5 Vdc.

a. Close S1, S2, S3, and S4 and open S5, S6, S7, and S8.
b. Place the power supply power switch in the ON position and adjust the output of VR1 to 115 ±1 Vac.
c. The voltage across TB4-4 and TB4-2 shall be 5 ±0.35 Vdc.
d. For acceptance, the 5 Vdc output characteristics shall be within the above specified limits.

2.4.9 Output #7, ac presence.

a. Close S1, S3, S4, and S7 and open S2, S5, S6, and S8.
b. Place the power switch in the ON position and adjust the output of VR1 to 115 Vac.
c. The resistance across TB3-1 and TB3-2 shall be less than 10 ohms.
d. Open S1.
e. The resistance across TB3-1 and TB3-2 shall be 200K ±20K ohms within 5 seconds after opening S1.
f. For acceptance, the ac presence output characteristics shall be within the above specified limits.

2.4.10 Sonalert output.

a. Open S1, S2, S3, S4, S5, S6, and S7. Close S8.
b. Adjust power supply 1 for 20 ±1 Vdc.
c. The sonalert shall produce a minimum audible sound of 60 dB at a distance of 10 feet as measured with the sound level meter. NOTE: During environmental tests, verification that the sonalert is producing an audible output is the only test requirement.
d. Open S8. The audible sound shall stop.
e. For acceptance, the sonalert input characteristics shall be within the above specified limits.

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APPENDIX

TEST METHOD 8

TAMPER/ALARM PWB ASSEMBLY (13220E4213)

A1 THROUGH A6 CARDS

FUNCTIONAL TEST

1. TEST APPARATUS:

   a. Power supply, Hewlett-Packard 6202B or equivalent.
   b. DVM, Fairchild 7000A or equivalent.
   c. Test fixture (see figure 10).
   d. Test fixture schematic (see figure 11).
   e. A1-A6 and A7 card test set-up (see figure 12).

2. TEST PROCEDURE:

   a. Connect test apparatus as shown in figure 12.

      NOTE: Test fixture shall be kept at ambient room temperature, with no more than a ±5 °F swing during tests, without recalibrating the fixture. When performing functional test as a part of the low or high temperature test, a connecting cable must be run from the test fixture (outside the temperature chamber) to the circuit card (inside the chamber). Cable length shall be less than two feet, if possible.

   b. Turn off power supply.
   c. Install card to be tested in test fixture and connect test point adapter.
   d. Turn on power supply, and adjust output to 19.9 ±0.1 Vdc using the DVM.
   e. Place switch S1 in position 1 (intrusion).
   f. Momentarily activate switch S2 (50 ms timer) and observe lights. The amber and green lights shall appear simultaneously for approximately 2 seconds. The red light shall not appear.
   g. Momentarily activate switch S3 (150 ms timer) and observe lights. Response shall be same as step f.

      NOTE: If a failure occurs during step g., repeat step g. two more times. Occurrence of two failures out of three attempts constitutes a test failure and the card shall be rejected.

   h. Place switch S1 in position 2 (tamper).
   i. Repeat step f.
   j. Repeat step g.
   k. Turn off power supply.
   l. Remove test point adapter from card, and remove card from test fixture.

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APPENDIX

TEST METHOD 9

VOLTAGE MONITOR PWB ASSEMBLY (13220E4214)

A7 CARD

FUNCTIONAL TEST

1. TEST APPARATUS:

   a. Power supply, Hewlett-packard 6202B or equivalent.
   b. DVM, Fairchild 7000A or equivalent.
   c. Test fixture (see figure 10).
   d. Test fixture schematic (see figure 11).
   e. A1-A6 and A7 card test set-up (see figure 12).

2. TEST PROCEDURE:

   a. Connect test apparatus as shown in figure 12.

      NOTE: Test fixture shall be kept at ambient room temperature, with no more than a ±5 °F swing during tests, without recalibrating the test fixture. When performing functional test as a part of the low or high temperature test, a connecting cable must be run from the test fixture (outside the temperature chamber) to the circuit card (inside the chamber). Cable length shall be less than two feet, if possible.

   b. Turn off power supply.
   c. Install A7 card to be tested in test fixture, and connect test point adapter.
   d. Turn on power supply, adjust voltage to 19.0 Vdc using the DVM, and observe test fixture lights. The white light shall be continuously ON with no intermittent flashing.
   e. Slowly decrease power supply voltage using the DVM and observe test fixture lights. The white light shall become continuously OFF at a voltage below 19.0 but not less than 18.3 Vdc.
   f. Adjust power supply voltage to 21.0 Vdc using the DVM, and observe test fixture lights. The white light shall be continuously ON.
   g. Slowly increase power supply to 21.7 Vdc using the DVM and observe test fixture lights. The white light shall become continuously OFF at a voltage greater than 21.0 Vdc but not greater than 21.7 Vdc.
   h. Adjust power supply output to 19.9 ±0.1 Vdc using the DVM.
   i. Place switch S1 in position 1 (door tamper).
   j. Momentarily activate switch S2 (50 ms timer) and observe lights. The amber and green lights shall be on for approximately 2 seconds. Red light shall not illuminate, even briefly.

      NOTE: Disregard condition of white light during steps j. through n.

   k. Momentarily activate switch S3 (150 ms timer) and observe lights. Responses shall be as in step j.

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NOTE: If a failure occurs during step k., repeat step k. two more times. Two failures out of three attempts constitutes a test failure and the card shall be rejected.

l. Place switch S1 in position 2 (telephone dialer tamper).
m. Repeat step j.
n. Repeat step k.
o. Place switch S4 in position 1 (alarm), momentarily activate S5, and observe test fixture lights. The white light shall turn OFF, and green light shall turn ON while S5 is activated. When S5 is released, white light shall turn ON and green light shall turn OFF. The red light shall never be activated.
p. Place switch S4 in position 2 (nonalarm), momentarily activate S6 and observe test fixture lights. The white light shall be ON and the green light shall turn ON while S6 is activated. When S6 is released, white light shall be ON, and green light shall turn OFF. The red light shall never be activated.
q. Turn off power supply.
r. Remove test point adapter from A7 card, and remove card from test fixture.

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APPENDIX

TEST METHOD 10

AUDIBLE ALARM PWB ASSEMBLY (13220E4215)

A8 CARD

FUNCTIONAL TEST

1. TEST APPARATUS:

a. Power supply, Hewlett-Packard 6202B or equivalent (two required).
b. DVM, Fairchild 7000A or equivalent (two required).
c. Test fixture (see figure 13).
d. Test fixture schematic (see figure 14).
e. A8 card test set-up (see figure 15).

2. TEST PROCEDURE:

a. Connect test apparatus as shown in figure 15.

NOTE: Test fixture shall be kept at ambient room temperature, with no more than a ±5 °F swing during tests, without recalibrating the fixture. When performing functional test as a part of the low or high temperature test, a connecting cable must be run from the test fixture (outside the temperature chamber) to the circuit card (inside the chamber). Cable length shall be less than two feet, if possible.

b. Turn off both power supplies.
c. Install A8 card in test set.
d. Turn on both the 20 Vdc and 5 Vdc power supplies, and adjust to 19.9 ±0.1 Vdc and 4.9 ±0.1 Vdc, respectively, using the DVM's.
e. Place S1, S4, S5, and S6 in position 1, and momentarily depress reset switch S3. Observe test set indicators. All test set indicators shall be off.

NOTE: Throughout this test, if the yellow test-in-progress indicator is ON, wait until it turns OFF before depressing S3.

f. Wait to allow the A8 card circuitry to stabilize - approximately 30 seconds.
g. Verify that the yellow test-in-progress indicator is OFF.
h. Momentarily depress S2, and observe test set low and high limit indicators. The yellow test-in-progress indicator shall light and before it extinguishes, both the low and high limit green indicators shall light and shall remain lit when the test-in-progress indicator turns off. Neither red indicator shall light.
i. Wait until the test-in-progress indicator goes OFF. Place S4 in position 2 and momentarily depress reset switch S3.
j. Repeat steps g. and h.
k. Place S1 and S5 in position 2, and momentarily depress reset switch S3.
l. Repeat steps g. and h.
m. Wait until the test-in-progress indicator goes OFF; then depress S7,

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and observe the AA LED. The AA LED shall be ON when and only when the switch S7 is depressed.

n. Place S6 in position 2. Depress S8, and observe the AA LED. Response shall be as in step m.
o. Depress S9, and observe the ACC LED. Response shall be as in step m.
p. Depress S10, and observe the SOD LED. Response shall be as in step m.
q. Turn off both power supplies.
r. Remove the A8 card from the test set.

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APPENDIX

TEST METHOD 11

INSTANTANEOUS ALARM PWB ASSEMBLY (13220E4216)

A9 CARD

FUNCTIONAL TEST

1. TEST APPARATUS:

   a. Power supply, Hewlett-Packard 6202B or equivalent (two required).
   b. DVM, Fairchild 7000A or equivalent (two required).
   c. Test fixture (see figure 16).
   d. Test fixture schematic A (see figure 17).
   e. Test fixture schematic B (see figure 18).
   f. A9 card test set-up (see figure 19).

2. TEST PROCEDURE:

   a. Connect test apparatus as shown in figure 19.

NOTE: Test fixture shall be kept at ambient room temperature, with no more than a ±5 °F swing during tests without recalibrating the fixture. When performing functional test as a part of the low or high temperature test, a connecting cable must be run from the test fixture (outside the temperature chamber) to the circuit card (inside the chamber). Cable length should be less than two feet, if possible.

   b. Turn off both power supplies.
   c. Install A9 card in test set.
   d. Turn on both the 20 Vdc and 5 Vdc power supplies, and adjust to 19.9 ±0.1 Vdc and 4.9 ±0.1 Vdc, respectively, using the DVM's.
   e. Adjust the potentiometer on the A9 card to the minimum exit delay setting (full clockwise).
   f. Place S1, S4, S5, and S6 in position 1, and momentarily depress reset switch S3. All test indicators except the ACC LED shall be off, and the ACC LED shall be turned on.

NOTE: If the yellow test-in-progress indicator is on, wait until it turns off before depressing S3.

   g. Wait to allow the A9 card circuitry to stabilize - approximately 30 seconds.
   h. Verify that the yellow test-in-progress indicator is off.
   i. Momentarily depress S2, and observe test set low and high limit indicators and NL LED. The Yellow test-in-progress indicator and the NL LED shall light, and before the test-in-progress indicator extinguishes, both the low and high limit green indicators shall light and shall remain lit when the test-in-progress indicator turns off. The NL LED shall extinguish when the high limit indicator lights.
   j. Wait until the test-in-progress indicator goes off; place S5 in position 2; and momentarily depress reset switch S3.

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k. Repeat steps h. and i.
l. Wait until the test-in-progress indicator goes off; place S5 in position 3; and momentarily depress reset switch S3.
m. Repeat steps h. and i.
n. Wait until the test-in-progress indicator goes off; then depress S7, and observe the IA1 LED. The LED shall be off before depressing S7; the LED shall turn on while S7 is depressed; and the LED shall turn off when S7 is released.
o. Momentarily depress reset switch S3.
p. Verify that the yellow test-in-progress indicator is off; momentarily depress S2; then immediately hold S8 depressed and observe the test-in-progress, the green high limit, and NL LED's. The yellow test-in-progress indicator and the NL LED shall illuminate. The IA2 LED shall illuminate when S8 is depressed. The green low limit and high limit indicators shall illuminate before the yellow test-in-progress indicator extinguishes. The NL and IA2 LED's shall both extinguish at the time the high limit indicator illuminates.
q. Wait until the test-in-progress indicator goes off; depress S9, and observe the NL LED. The LED shall be off before S9 is depressed; the LED shall go on when S9 is depressed; and the LED shall go off when S9 is released.
r. Place S4 in position 2 and momentarily depress reset switch S3. The responses shall be as in step f.
s. Place S6 in position 2, and observe the test-in-progress, low and high limit and SEC LED's. Both the yellow test-in-progress and green SEC LED's shall light when S6 is placed in position 2. Before the test-in-progress indicator goes off, both low and high limit green indicators shall turn on.
t. Wait until the test-in-progress indicator goes off; place S6 in position 1 and S1 in position 2.
u. Verify that the test-in-progress indicator is off, and adjust the potentiometer on the A9 card to the maximum exit delay setting (full counterclockwise).
v. Momentarily depress reset switch S3. The responses shall be as in step f.
w. Repeat step s.

NOTE: This test step will take approximately 2-1/2 minutes. Wait until the yellow test-in-progress indicator goes off before proceeding to step x.

x. Depress S10 and observe the EI LED. The LED shall be off before depressing S10; the LED shall be on when S10 is depressed; and the LED shall be off when S10 is released.
y. Turn off both power supplies.
z. Remove the A9 card from the test set.

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APPENDIX

TEST METHOD 12

LATCHED ALARM PWB ASSEMBLY (13220E4217)

A10 CARD

FUNCTIONAL TEST

1. TEST APPARATUS:

   a. Power supply, Hewlett-Packard 6202B or equivalent (two required).
   b. DVM, Fairchild 7000A or equivalent (two required).
   c. Test fixture (see figure 20).
   d. Test fixture schematic (see figure 21).
   e. Test fixture parts list (see figure 22).
   f. A10 card test set-up (see figure 23).

2. TEST PROCEDURE:

   a. Connect test apparatus as shown in figure 23.

NOTE: Test fixture shall be kept at ambient room temperature, with no more than a ±5 °F swing during tests without recalibrating the fixture. When performing functional test as a part of the low or high temperature test, a connecting cable must be run from the test fixture (outside the temperature chamber) to the circuit card (inside the chamber). Cable length shall be less than two feet, if possible.

   b. Turn off both power supplies.
   c. Install A10 card in test set, and plug in test point adapter.
   d. Turn on both the 20 Vdc and 5 Vdc power supplies, and adjust to 19.9 ±0.1 Vdc and 4.9 ±0.1 Vdc, respectively, using the DVM's.
   e. Adjust the potentiometer on the A10 card to the minimum entrance delay setting (full clockwise).
   f. Place both S1 and S4 in position 1, and momentarily depress both reset switches S3 and S6. All test set indicators shall be off.

NOTE: If the yellow test-in-progress indicator is on, wait until it turns off before depressing S3 and S6.

   g. Wait to allow the A10 card circuitry to stabilize - approximately 30 seconds.
   h. Verify that the yellow test-in-progress indicator is off.
   i. Momentarily depress S2, and observe test set low and high limit indicators and the AA and DIT LED's. The yellow test-in-progress indicator shall light, and before it extinguishes both the low and high limit green indicators and the AA and DIT LED's will light and will remain lit when the test-in-progress indicator turns off.
   j. Wait until the test-in-progress indicator goes off, adjust the potentiometer on the A10 card to the maximum entrance delay setting (full counterclockwise).

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k. Place S1 in position 2, and momentarily depress S3 and S6. All low limit and high limit indicators and the AA and DIT LED's shall turn off.
l. Verify that the yellow test-in-progress indicator is off.
m. Momentarily depress S2, and observe test set low and high limit indicators. Responses shall be as in step i.

NOTE: This test step will take approximately 2-1/2 minutes. Wait until the yellow test-in-progress indicator goes off before proceeding to step n.

n. Momentarily depress S6. Both the AA and DIT LED's shall turn off.
o. Momentarily depress S5. Both the AA and DIT LED's shall turn on and remain on when S5 is released.
p. Place S4 in position 2. AA LED shall turn off.
q. Momentarily depress S6. DIT LED shall turn off.
r. Momentarily depress S5. Only the DIT LED shall turn on and it shall remain on when S5 is released.
s. Momentarily depress S7 and observe MR1 LED. The LED shall be off before depressing the switch; the LED shall turn on while the switch is depressed; and the LED shall turn off when the switch is released.
t. Momentarily depress S8 and observe the MR2 LED. Responses shall be as in step s.
u. Turn off both power supplies.
v. Remove the test point adapter, and remove the A10 card from the test set.

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APPENDIX

TEST METHOD 13

ALARM I.D. PWB ASSEMBLY (13220E4218)

A11 CARD

AND

LED PWB ASSEMBLY (13220E4219)

A12 CARD

FUNCTIONAL TEST

1. TEST APPARATUS:

   a. Power supply, Hewlett-Packard 6202B or equivalent (two required).
   b. DVM, Fairchild 7000A or equivalent (two required).
   c. Test fixture (see figure 24).
   d. Test fixture schematic (see figure 25).
   e. A11/A12 card test set-up (see figure 26).

2. TEST PROCEDURE A12/A11 CARDS:

   a. Connect test apparatus as shown in figure 26.

   NOTE: Test fixture shall be kept at ambient room temperature, with no more than a ±5 °F swing during tests without recalibrating the fixture. When performing functional test as a part of the low or high temperature test, a connecting cable must be run from the test fixture (outside the temperature chamber) to the circuit card (inside the chamber). Cable length shall be less than two feet, if possible.

   b. Turn off both power supplies.
   c. Install A12 card to be tested in test fixture.
   d. Turn on the 20 Vdc supply. Adjust voltage to 19.9 ±0.1 Vdc using the DVM. Depress S5 and observe LED's. It shall be visually apparent that all LED's are lit with no angular misalignment.
   e. Turn off 20 Vdc power supply.
   f. Remove A12 card from test fixture, and install A12 card onto the A11 card to be tested.
   g. Install A11 card to be tested in test fixture.
   h. Turn on both the 20 Vdc and 5 Vdc power supplies, and adjust to 19.9 ±0.1 Vdc and 4.9 ±0.1 Vdc, respectively, using the DVM's.
   i. Place S1 in test 1 position.
   j. Momentarily depress reset switch S3 and observe amber test fixture light and A12 card lights. The amber test fixture light shall turn on for approximately 5 seconds. The A12 card LED's shall either turn off (if they were on) or remain off and shall continue to be off, with no blinking, after the amber test fixture light turns off.

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k. Momentarily depress S2, and observe A12 LED's. LED's F, A, B, C, D, and E shall light in sequence within 1 second and shall remain lit until reset.
l. Place S1 in test 2 position and S4 in F position.
m. Momentarily depress reset switch S3 and observe amber test fixture light and A12 card LED's. Response shall be as in step j.
n. Momentarily depress S2 and observe green test fixture light and A12 card LED's. Only the F LED shall light and remain lit. The green test fixture light will be on at the end of the test.
o. Successively place S4 in positions A, B, C, D, and E. Repeat steps m. and n. for each position. Only the single LED indicated by the setting of S4 shall light. Activation or momentary blinking of other LED's shall be cause for rejection.
p. Turn off both power supplies.
q. Remove the A11 card from the test fixture.

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APPENDIX

TEST METHOD 14

STATUS PROCESSOR INTERCONNECTING WIRING SUBASSEMBLY (13220E4225)

FUNCTIONAL TEST

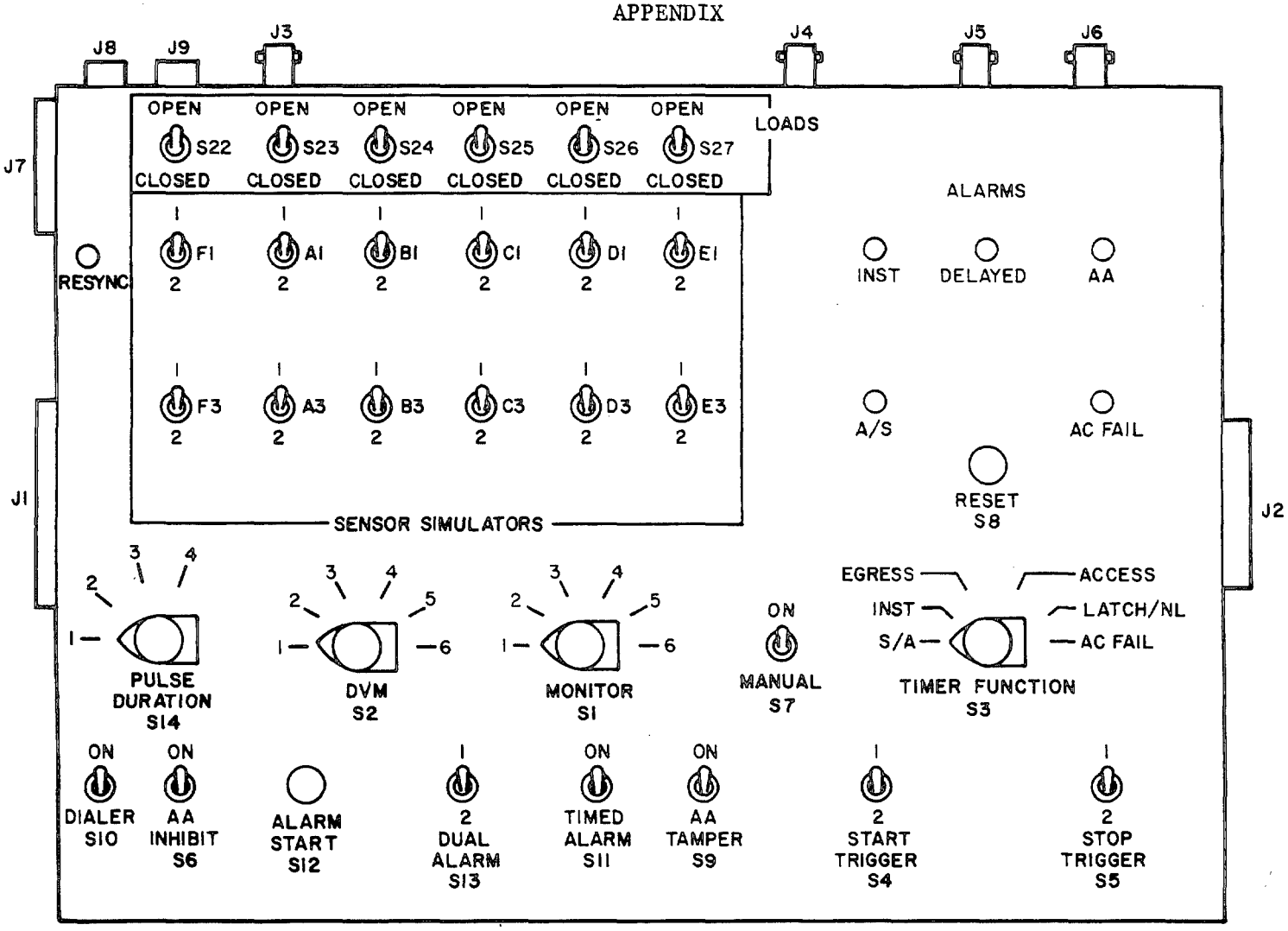
1. TEST APPARATUS:

   a. Ohmmeter, Triplett 630-NS or equivalent.

2. TEST PROCEDURE:

   a. Place switch S5 in LATCH position. Using ohmmeter, verify a resistance greater than 100K exists from J9-D to J12-2 and from J8-5 to J10-18. Verify a resistance less than 1 ohm exists from J10-L to S5-6.
   b. Place switch S5 in NONLATCH position. Verify a resistance greater than 100K from J10-L to S5-6 and less than 1 ohm from J9-D to J12-2 and from J8-5 to J10-18.
   c. Verify a resistance greater than 100K exists from J12-23 to J12-34. Depress switch S2 and verify the resistance from J12-23 to J12-34 becomes less than 1 ohm.
   d. Verify proper operation of diodes CR1 and CR2 by measuring forward and reverse resistances across each diode. Forward resistance is that measured with negative ohmmeter probe on S5-6 and positive probe on J12-25 and J12-16 for CR1 and CR2, respectively. Reverse resistance is measured by reversing the meter probes. Reverse resistance shall be at least 10 times the forward resistance.
   e. Remaining wiring shall be checked by inspection or continuity checks. Interconnecting wiring shall be in accordance with the drawing.
   f. Failure to conform to one or more of the above requirements shall constitute failure of this test.

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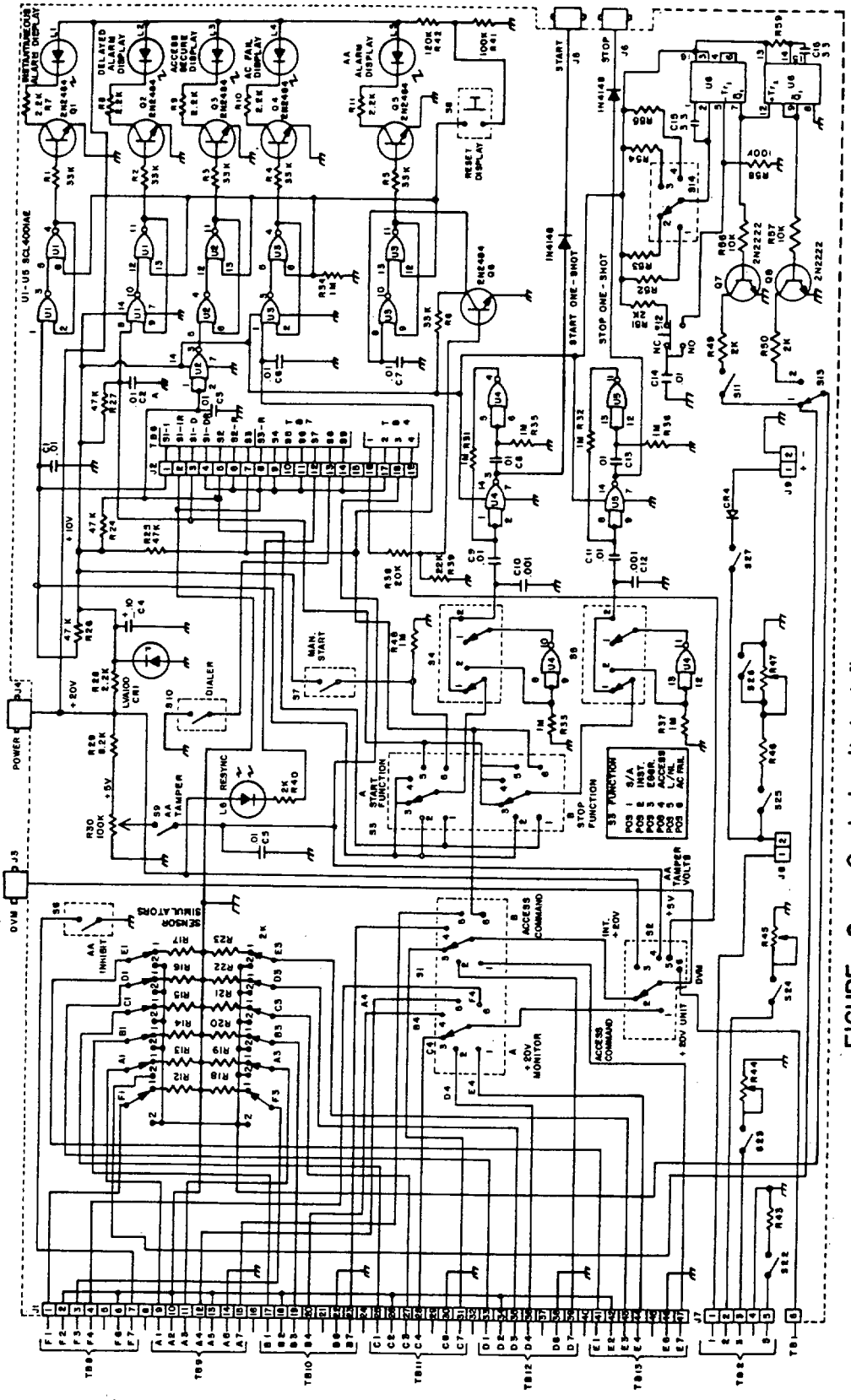


NOTE:
SEE FIGURE 2 FOR TEST FIXTURE SCHEMATIC DIAGRAM.

FIGURE 1. Control unit test fixture.

X-2978E

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X-2979F

FIGURE 2. Control unit test fixture schematic.

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QTY	ITEM	DESCRIPTION	REF SYMBOLS
2	CAPACITOR	0.001 uF 50 V	C10, 12
12	CAPACITOR	0.01 uF 50 V	C1-9, 11, 13, 14
2	CAPACITOR	3.3 uF 10% 15 WVDC MINIMUM	C15, 16
3	DIODE	1N4148 OR EQUIV	CR2, 3, 4
1	DIODE, ZENER (10.0 V)	LVA100	CR1
6	DIODE, LIGHT EMITTING	MONSANTO MV5053 OR EQUIV	L1-6
5	INTEGRATED CIRCUIT	CD4001/MC14001 OR EQUIV	U1-5
1	INTEGRATED CIRCUIT	CD4098/MC14098 OR EQUIV	U6
16	RESISTOR	2,000 OHMS 5% 1/4W	R40, 49-51, R12-23
1	RESISTOR	3,000 OHMS 5% 1/4W	R60
6	RESISTOR	2,200 OHMS 5% 1/4W	R7-11, 28
1	RESISTOR	8,200 OHMS 5% 1/4W	R29
2	RESISTOR	10,000 OHMS 5% 1/4W	R56, 57
1	RESISTOR	20,000 OHMS 5% 1/4W	38
1	RESISTOR	22,000 OHMS 5% 1/4W	R39
6	RESISTOR	33,000 OHMS 5% 1/4W	R1-6
4	RESISTOR	47,000 OHMS 5% 1/4W	R24-27
3	RESISTOR	100,000 OHMS 5% 1/4W	R41, 58, 59
1	RESISTOR	120,000 OHMS 5% 1/4W	R42
8	RESISTOR	1 MEG OHMS 5% 1/4W	R31-37, 48
1	RESISTOR, VARIABLE	100,000 OHMS	R30
1	RESISTOR	100 OHMS 2% 1/2W	R43
1	RESISTOR, VARIABLE	50 OHMS 25W	R44
1	RESISTOR, VARIABLE	150 OHMS 12W	R45
1	RESISTOR	5 OHMS 5% 100W	R46
1	RESISTOR, VARIABLE	15 OHMS 100W	R47
4	RESISTOR	SEE NOTE 3	R52-55
2	SWITCH, DP6T ROTARY	CTS T206 OR EQUIV	S1, 3
2	SWITCH, DPDT TOGGLE	ALCO MTA206N OR EQUIV	S4, 5
2	SWITCH, SP6T ROTARY	CTS T205 OR EQUIV	S2, 14
21	SWITCH, SPST TOGGLE	CUTLER-HAMMER 7580K7 OR EQUIV	S6, 7, 9-11, 22-27
13	SWITCH, SPDT TOGGLE	CUTLER-HAMMER A123S1YZQ OR EQUIV	S13, A1, B1, C1, D1, E1, F1, A3, B3, C3, D3, E3, F3
1	SWITCH, PUSHBUTTON	CUTLER-HAMMER 8411K12 OR EQUIV	S8
1	SWITCH, PUSHBUTTON	CUTLER-HAMMER PB12631ZQ OR EDUIV	S12
7	SWITCH (NOT USED)		S15-21
6	TRANSISTOR	2N2484A OR EQUIV	Q1-6
2	TRANSISTOR	2N2222A OR EQUIV	Q7, 8

FIGURE 3. Control unit test fixture parts list.

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NOTES: 1. CALIBRATE RESISTORS R44, R45, AND R47 AS FOLLOWS (SWITCHES S22-S27 OPEN UNLESS OTHERWISE SPECIFIED):

A. WITH AMMETER PLUGGED INTO J8, CLOSE S25 AND APPLY 27.5 VDC AT J7-1 (REFERENCE J7-4). ADJUST R47 FOR READING OF 1.70 ± .02 AMPERES.
B. OPEN S25, CLOSE S24 AND APPLY 27.5 VDC AT J7-2. ADJUST R45 FOR READING OF 250 ± 5 MA.
C. OPEN S24, CLOSE S23 AND APPLY 20.0 VDC AT J7-3. ADJUST R44 FOR READING OF 600 ± 10 MA.

2. PLACE R60 ACROSS J2-17 AND J2-18. ADJUST R30 FOR 4.5 ± 0.2 VDC ACROSS J2-17 (GROUND) AND J2-18 (POSITIVE). REMOVE R49 AFTER ADJUSTMENT.
3. RESISTORS R52, 53, 54, AND 55 ARE SELECTED (ADJUSTED IF POTS ARE USED) TO PROVIDE OUTPUT PULSE TIMES AT U6 PIN 7 AS FOLLOWS WHEN U6 IS TRIGGERED BY DEPRESSING S12:

R52:	240 ± 15 MS	(NOMINAL VALUE 145K OHMS)
R53:	55 ± 5 MS	(NOMINAL VALUE 33K OHMS)
R54:	775 ± 25 MS	(NOMINAL VALUE 470K OHMS)
R55:	1.55 ± .05 S	(NOMINAL VALUE 940K OHMS)

FIGURE 3. Control unit test fixture parts list. (Cont.)

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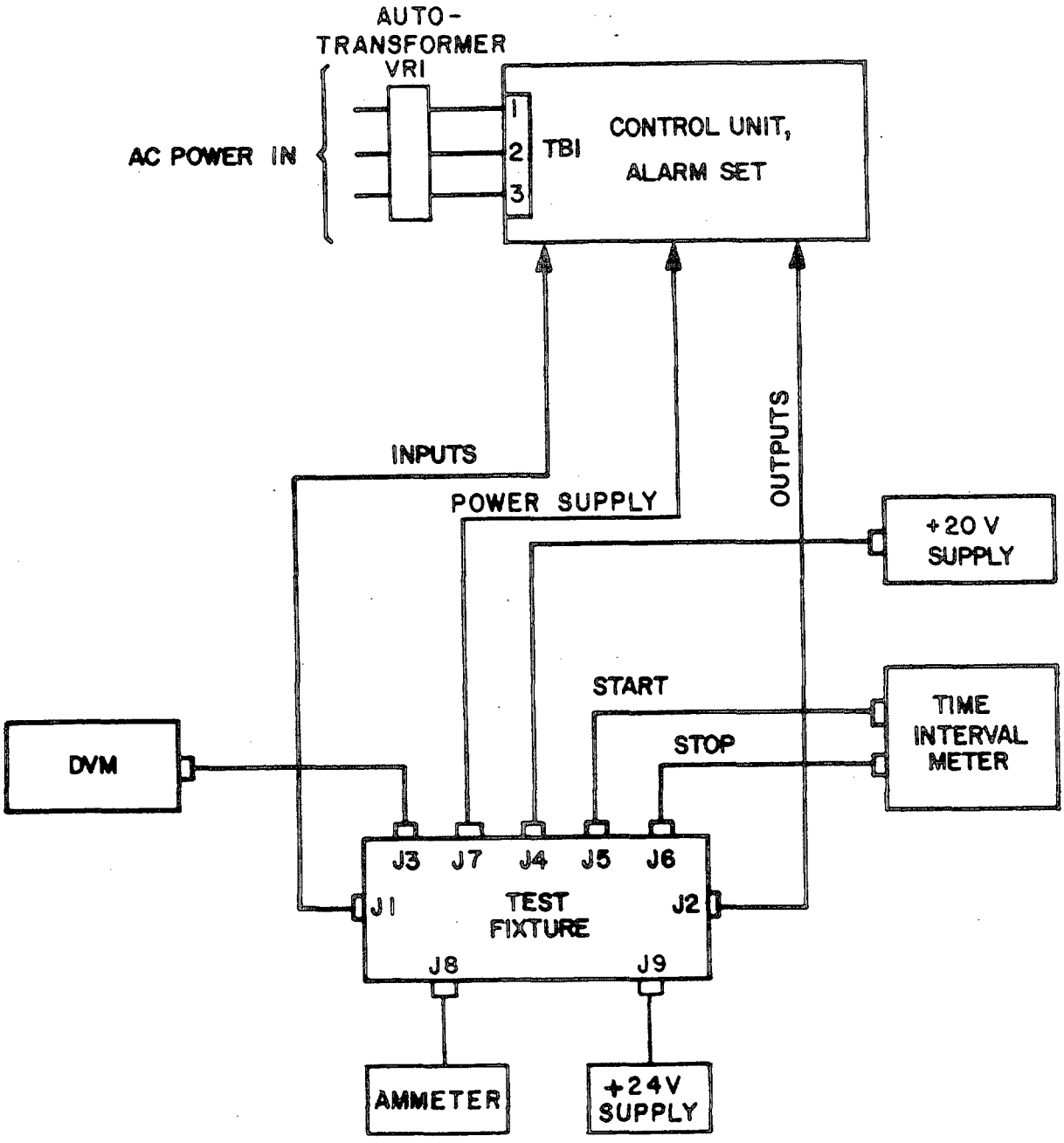
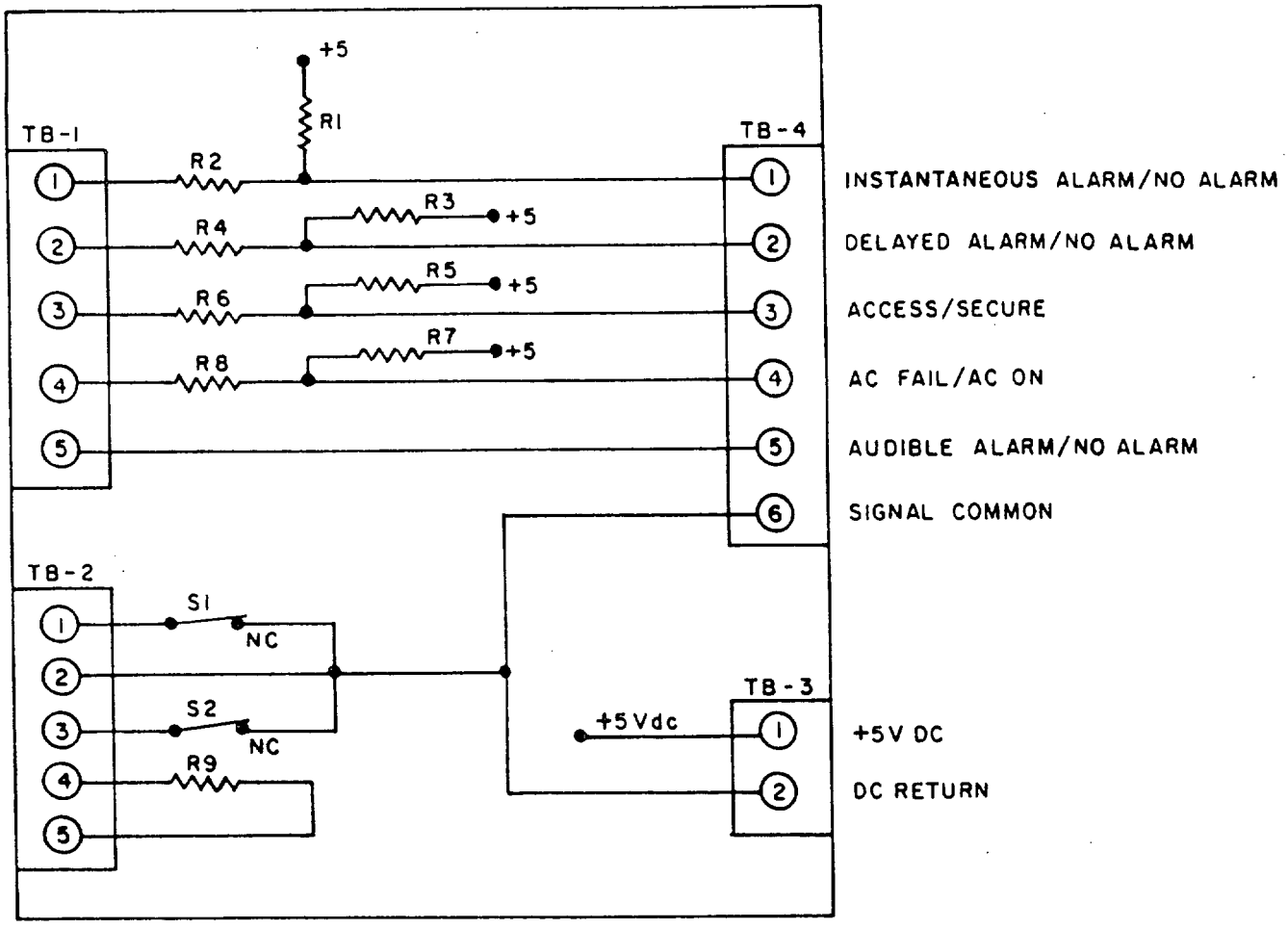


FIGURE 4. Control unit test setup. X-2980A

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PARTS:
R1, R3, R5, R7 ARE 10K ±5% OHMS, 1/4 WATT
R2, R4, R6, R8 ARE 470 ±5% OHMS, 1/4 WATT
R9 IS 200 ±5% OHMS, 2 WATT
S1, S2 ARE SPST NORMALLY CLOSED SWITCHES

FIGURE 5. Control unit EMI test fixture.

X-4286

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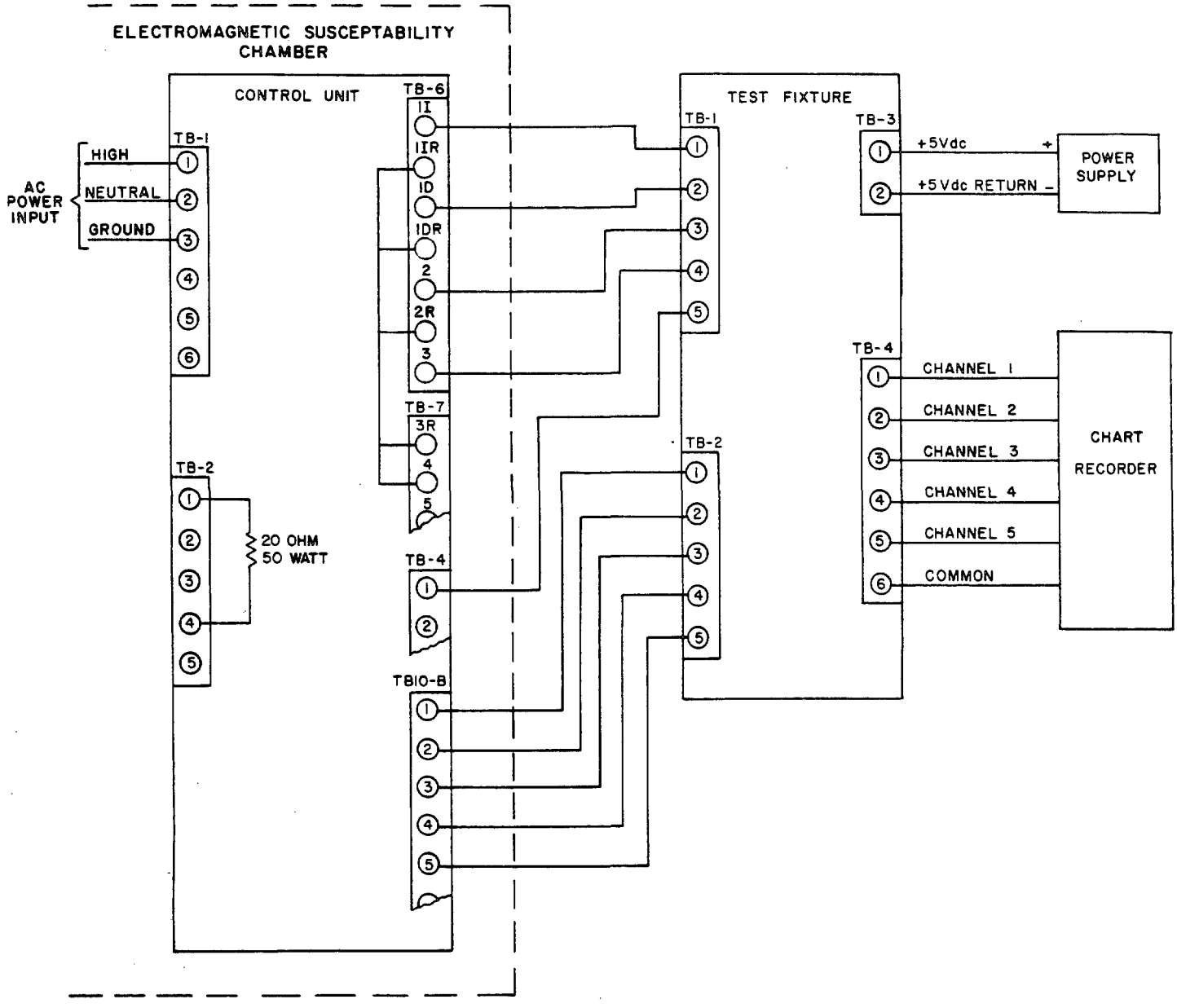
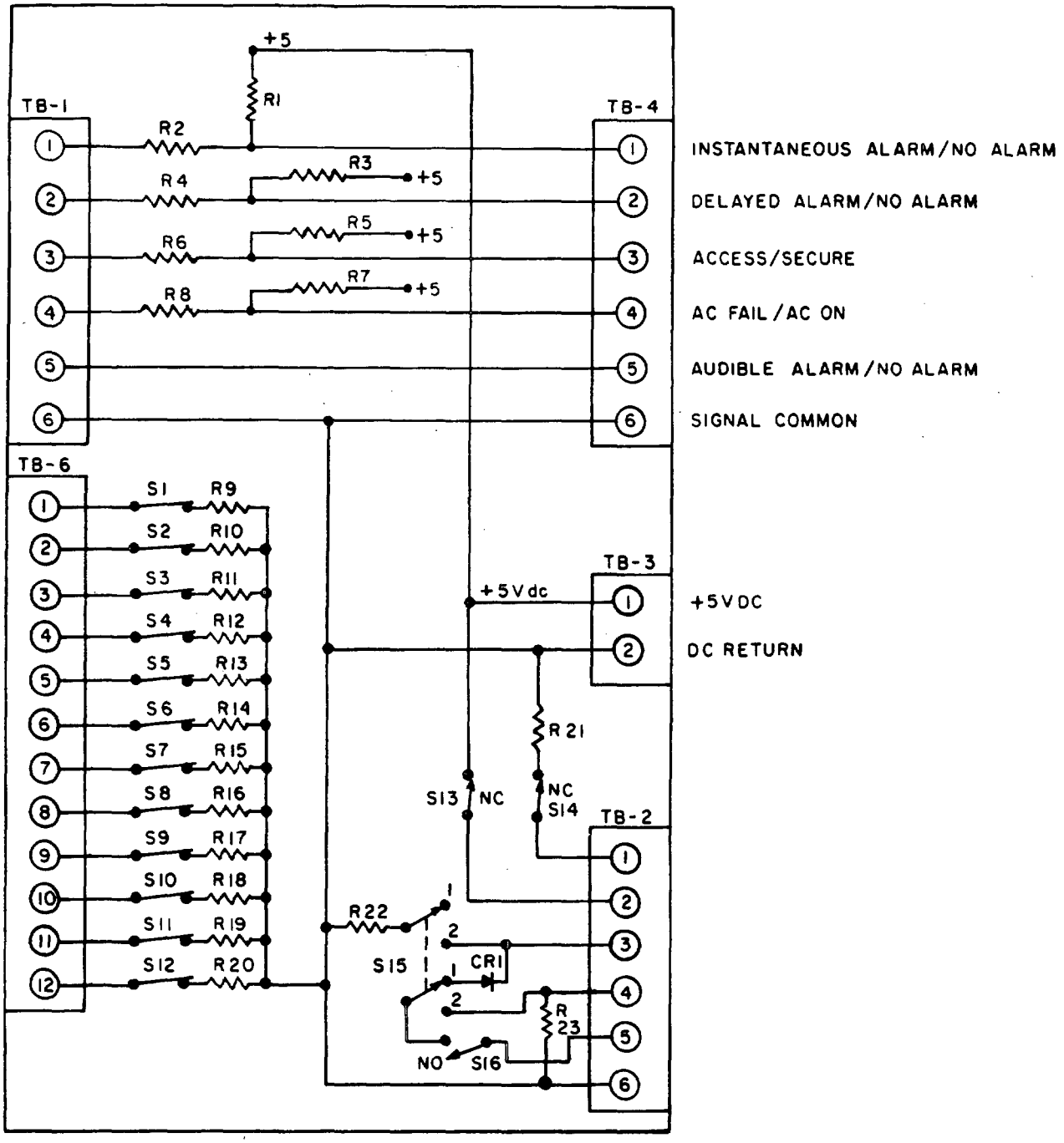


FIGURE 6. Control unit EMI test setup.

X-4287A

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PARTS:
R1, R3, R5, R7 - 10K ±5% OHMS, 1/4 WATT
R2, R4, R6, R8 - 470 ±5% OHMS, 1/4 WATT
R9 - R21 - 20K ±5% OHMS, 1/4 WATT
R22 - 20 OHMS, 50 WATTS
R23 - 50 OHMS, 10 WATTS
CR1 - IN4004 OR EQUIV.
S1 - S14, S16 SPST SWITCH
S15 - DPDT SWITCH

FIGURE 7. Control unit reliability test fixture.

X-4285A

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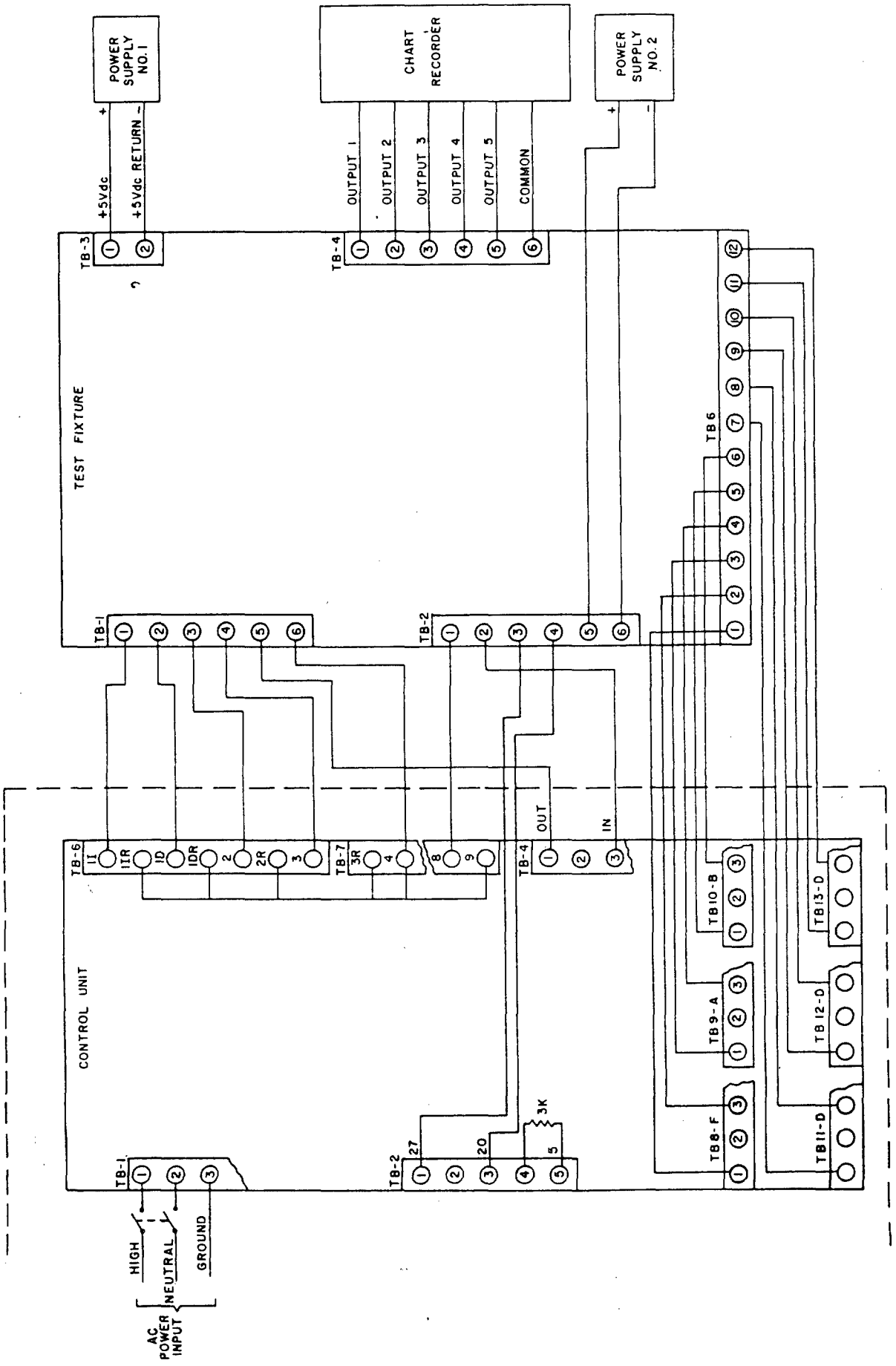
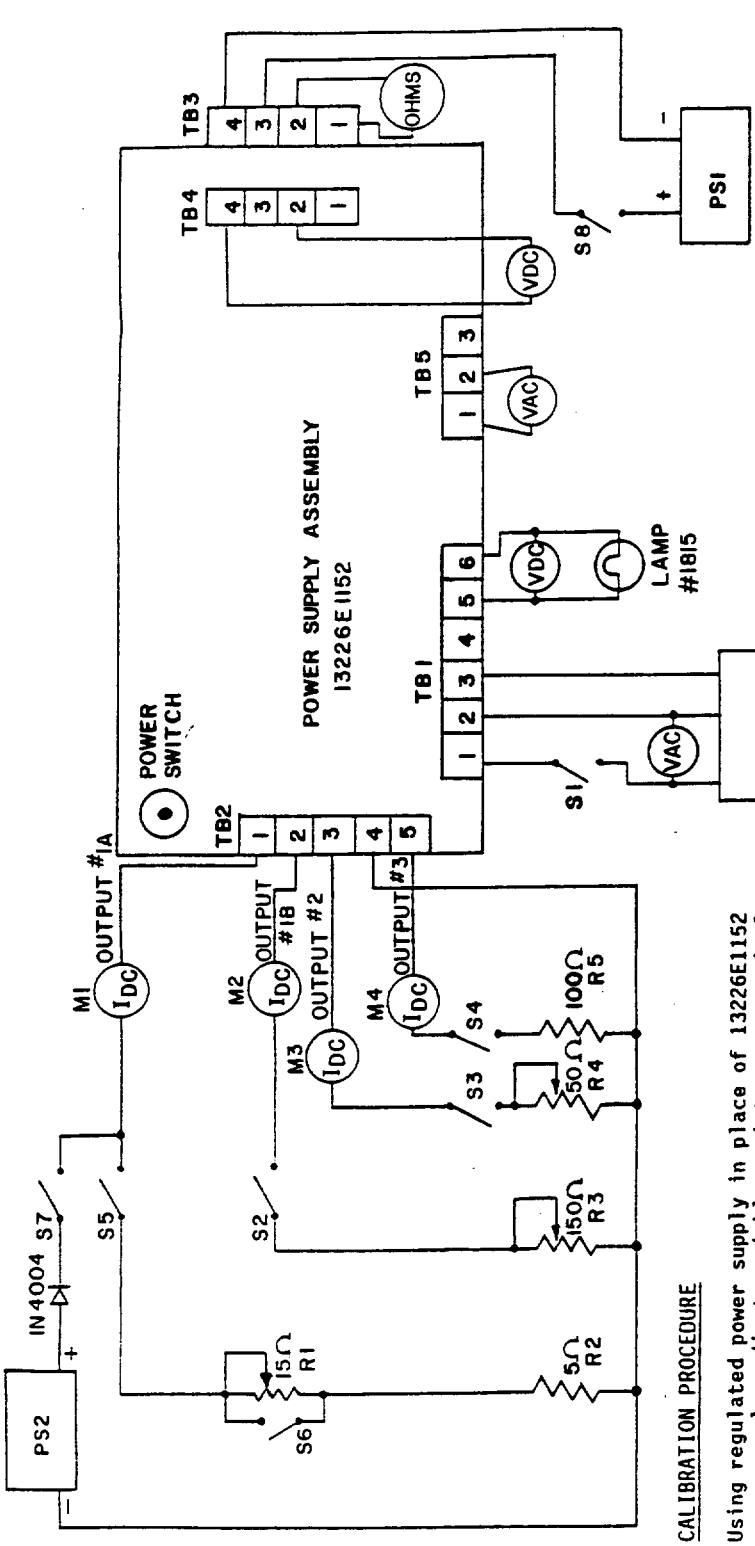


FIGURE 8. Control unit reliability test setup.

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X-3938D

FIGURE 9. Power supply assembly test fixture.

CALIBRATION PROCEDURE

Using regulated power supply in place of 13226E1152 power supply, adjust variable resistors successively to obtain the following (all switches open unless specified)

1. With 27.5 Vdc applied at TB2-1 and S5 closed, adjust R1 for reading on M1 of 1.70 + 0.2 A.
2. With 27.5 Vdc applied at TB2-2 and S2 closed adjust R3 for reading on M2 of 250 + 5 ma.
3. With 20.0 Vdc applied at TB2-3, close S3 and adjust R4 for reading on M3 of 600 + 10 ma.
4. With 5.0 Vdc applied at TB2-5, close S4 and verify reading on M4 is 50 ± 3 ma. If not re-select R5.

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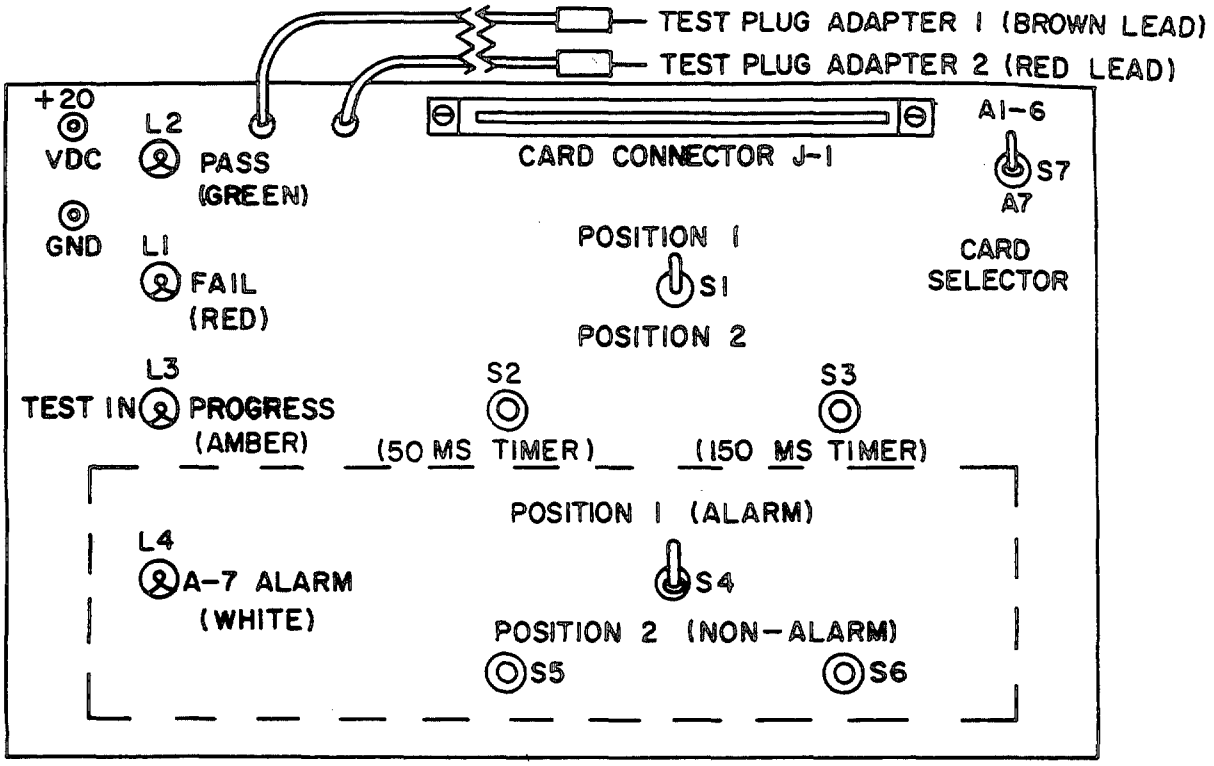
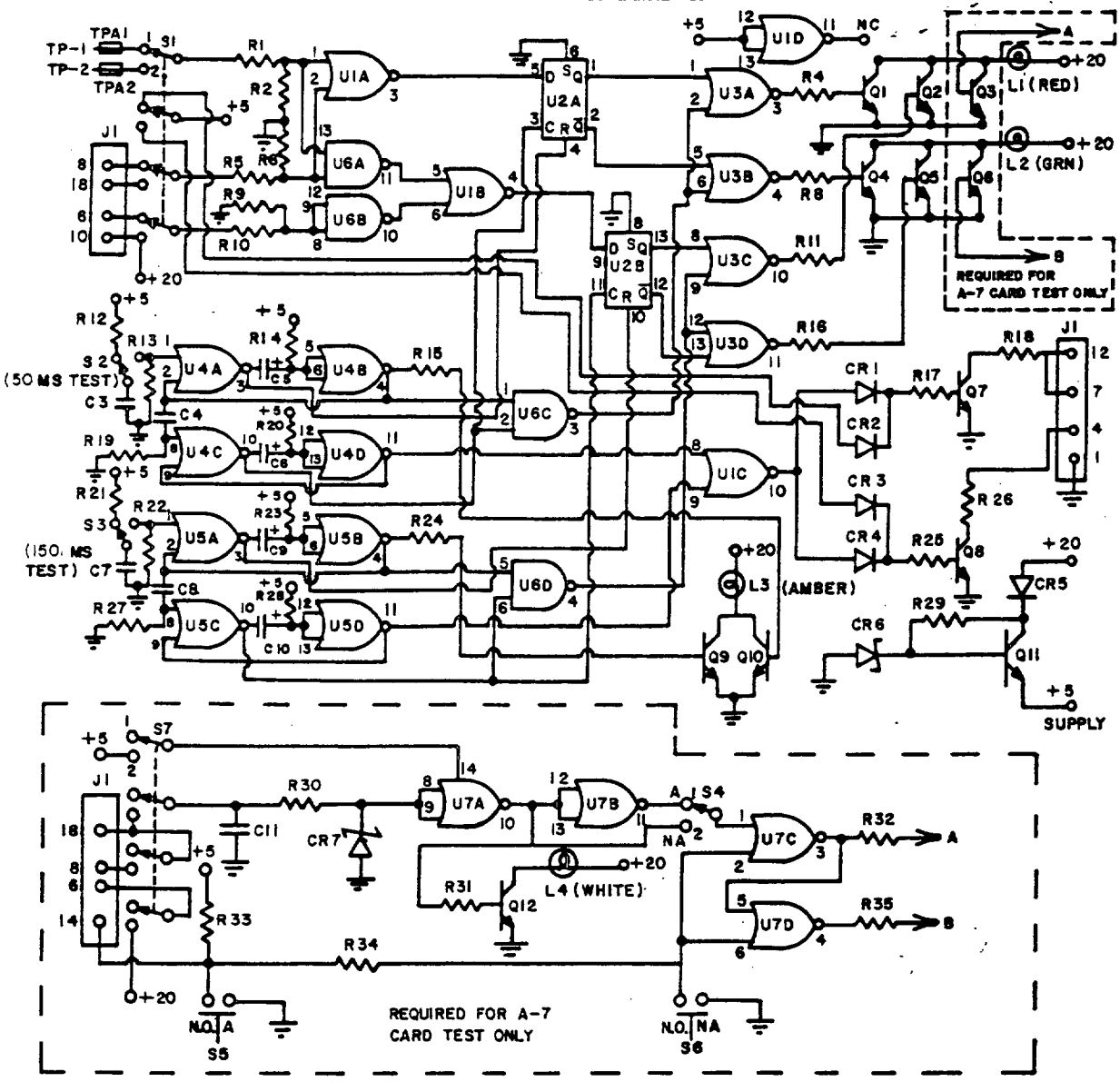


FIGURE 10. A1-A6 and A7 cards test fixture.

X-3911B

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R#	VALUE (OHMS)
1,5,10,30	15K
2,6,9	4.7K
3,7	NOT USED
4,8,11,15,16,17,24,25,31,32,35	10K
12,13,19,21,22,27,34	47K
14,23	NOTE 3
18,26	2K
20	NOTE 1
28	NOTE 2
29	1K (1/2W)
33	680

ALL RESISTORS 1/4W, ±10% TOLERANCE UNLESS NOTED.

C#	VALUE	VOLTAGE
1,2	NOT USED	
3,4,7,8	.01μF	12V
5,9	43μF	12V
6,10	2.2μF	12V
11	.001μF	25V

ALL CAPACITORS ARE LOW LEAKAGE MYLAR, POLYCARBONATE OR TANTALUM (±10% TOLERANCE).

U#	TYPE
1,3,4,5,7	CD4001 QUAD 2 INPUT NOR GATE
2	CD4013 DUAL "D" FLIP-FLOP W/SET/RESET
6	CD4011 QUAD 2 INPUT NAND GATE

S#	TYPE
1,7	4PDT TOGGLE, CUTLER-HAMMER #SF4SCY191 OR EQ.
2,3	MOM. SPDT (BBM), GRAYHILL #46-101 OR EQ.
4	SPDT TOGGLE, CUTLER-HAMMER #8282K14 OR EQ.
5,6	MOM. SPST-N.O., GRAYHILL #39-1 OR EQ.

CR#	TYPE
1,2,3,4,5	1N456 OR EQ.
6	1N753 - 6.2V ZENER
7	1N751 - 5.1V ZENER

J1 - CARD CONNECTOR, 18 PIN, ELCO 6007 OR EQ.

Q1 THRU Q12 - 2N2222A OR EQ.

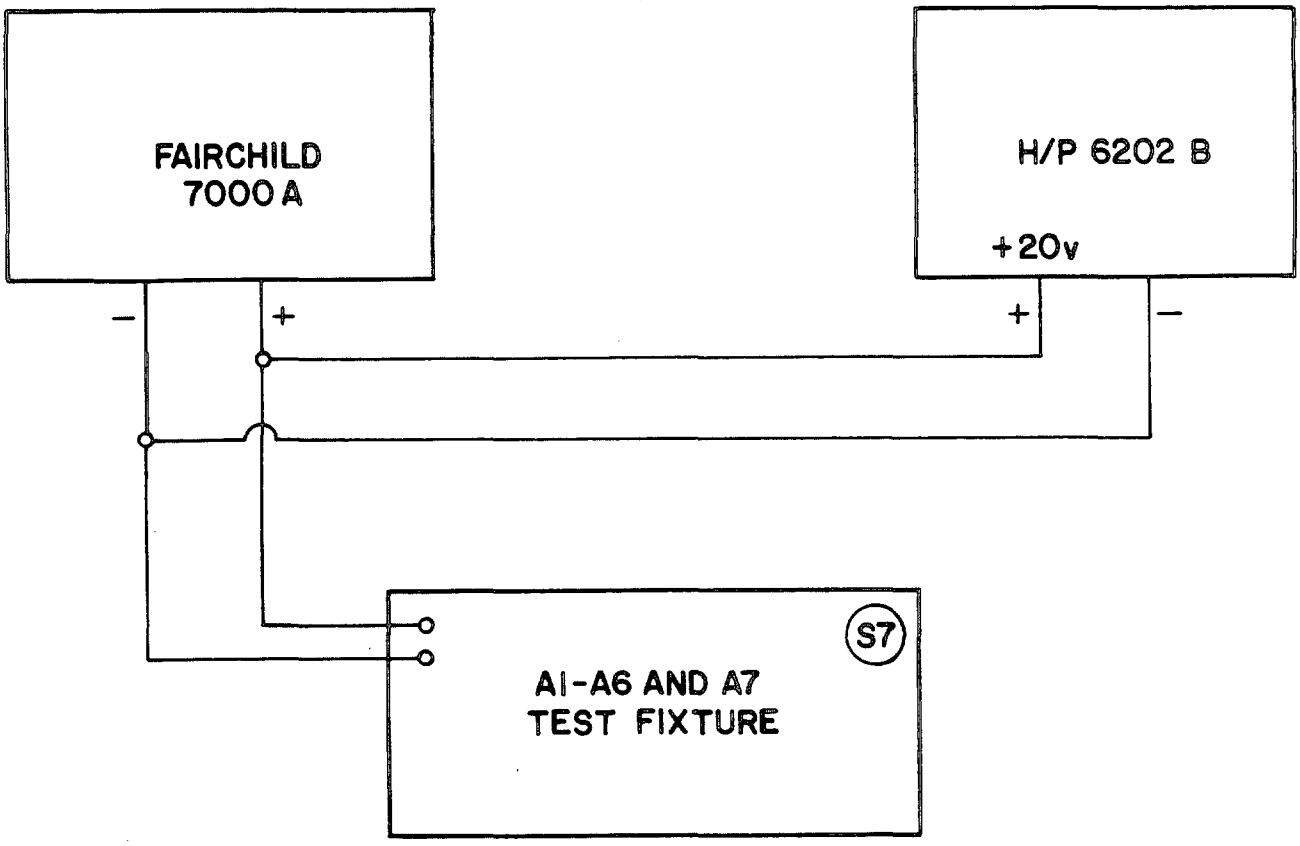
L1 THRU L4 - #327, SYLV. #33309-0 OR EQ.

TPA 1, 2 - TIP PLUG, SMITH #492 OR EQ.

NOTE 1: R20 APPROXIMATELY 60K Ω; SELECT FOR 50 (+0, -2) MS PULSE DURATION AT U4D OUTPUT.
NOTE 2: R28 APPROXIMATELY 150K Ω; SELECT FOR 150 (+4, 0) MS PULSE DURATION AT U5D OUTPUT.
NOTE 3: R14, R23 APPROXIMATELY 100K Ω; SELECT FOR TIMER OUTPUT = 2 SECONDS.
NOTE 4: CIRCUIT TRACE LENGTH SHOULD BE MINIMUM PRACTICABLE TO AVOID CIRCUIT OSCILLATION.

FIGURE 11. A1-A6 AND A7 CARDS TEST FIXTURE SCHEMATIC. X-3912C

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NOTE:
SWITCH S7 ON TEST SET MUST BE IN POSITION 1 FOR TEST OF A1-A6 CARDS, AND IN POSITION 2 FOR TEST OF THE A7 CARD.

FIGURE 12. A1-A6 and A7 card test set-up.

X-3913B

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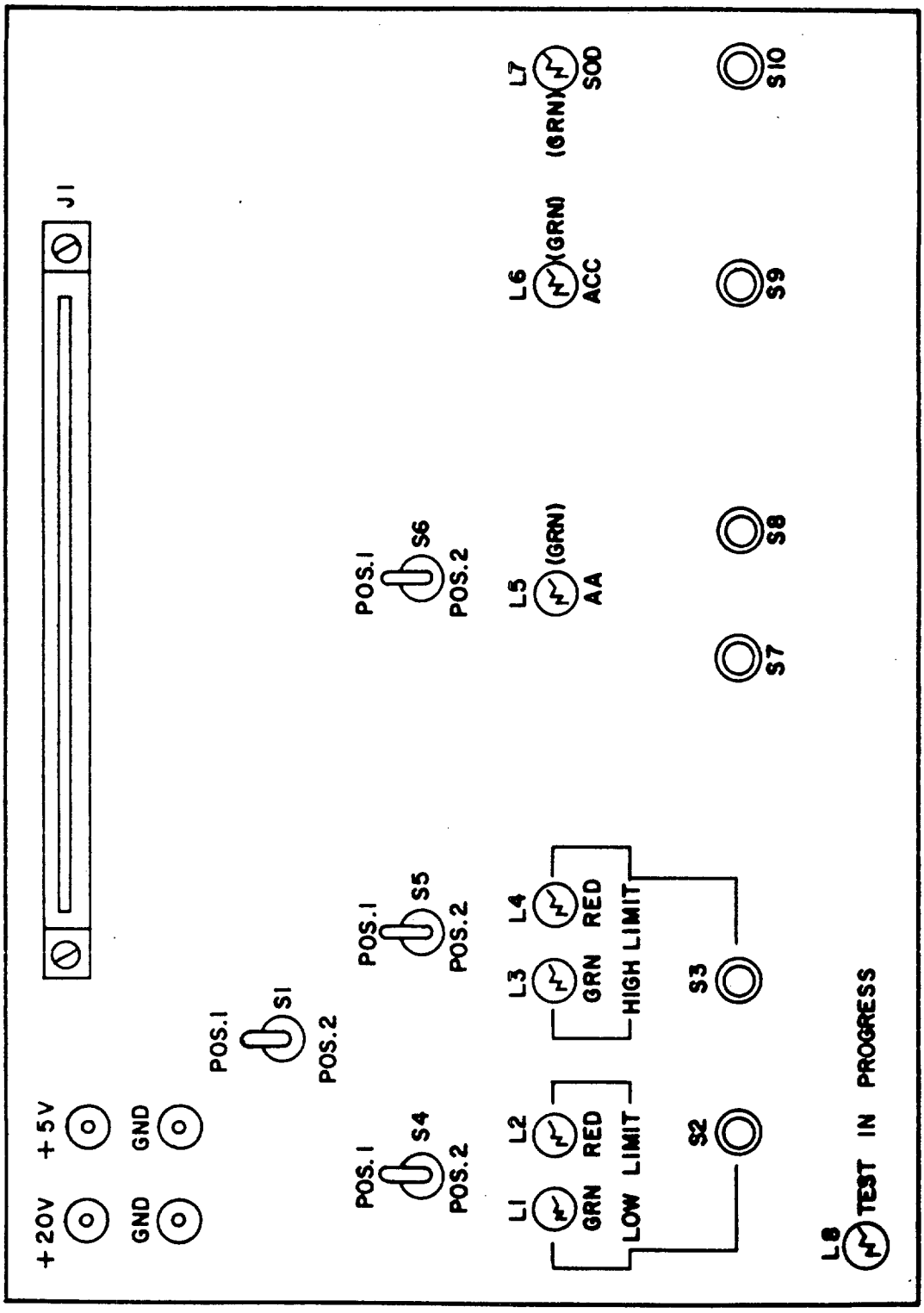
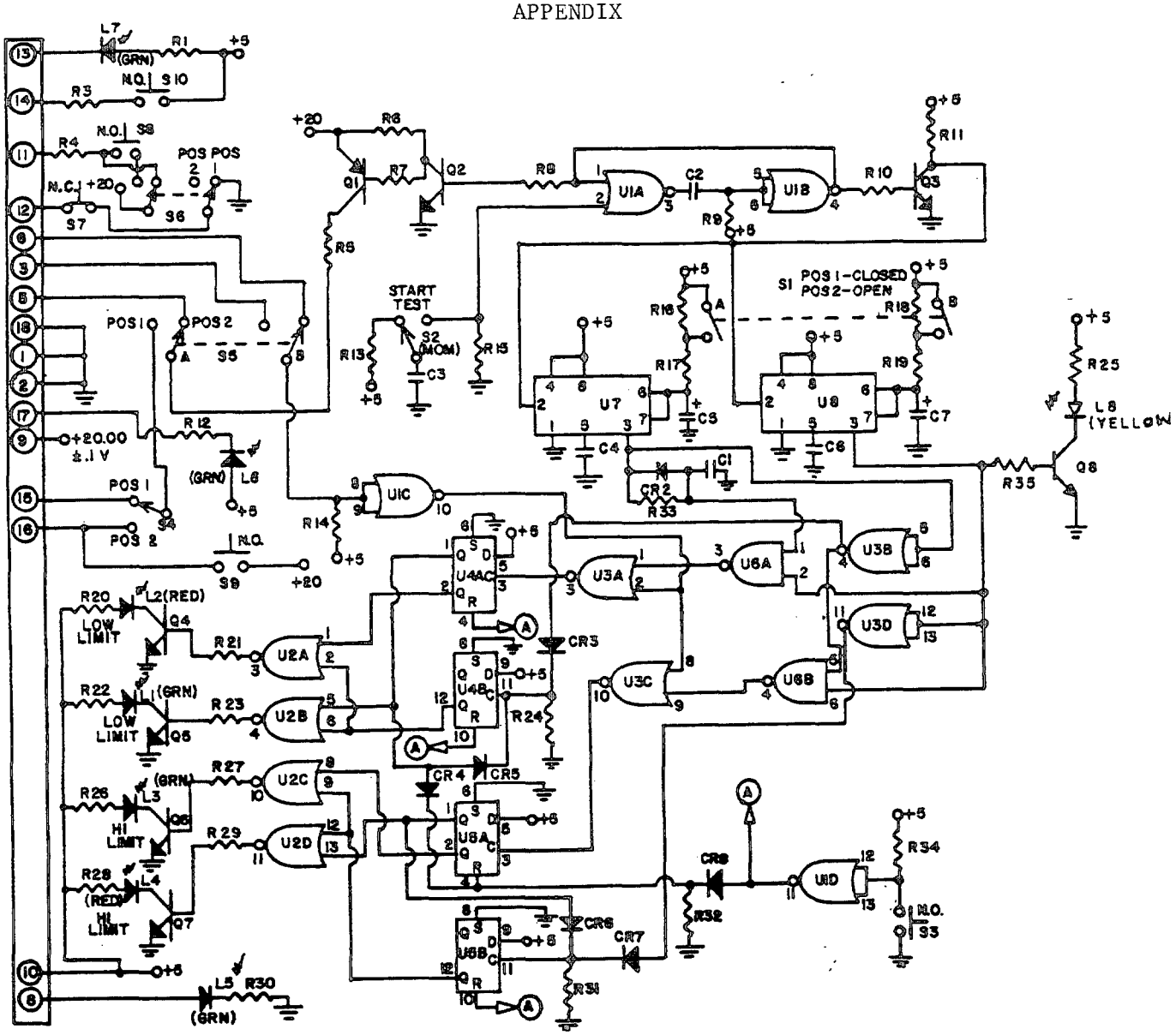


FIGURE 13. A8 card test fixture.

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U#	TYPE
1,2,3	CD4001 QUAD 2-INPUT NOR GATE
4,5	CD4013 DUAL "D" FLIP-FLOP W/SET/RESET
6	CD4011 QUAD 2-INPUT NAND GATE
7,8	555 TIMER

NOTE: TIE ALL UNUSED GATE INPUTS TO +5 VOLTS.

C#	VALUE
1	2.2μF
2-8	0.1μF
3,4,6	0.01μF
5,7	45μF

ALL CAPACITORS ARE 12 WVDC LOW LEAKAGE MYLAR, POLYCARBONATE OR TANTALUM (±10% TOLERANCE).

CR#	TYPE
1	NOT USED
2-8	1N456 OR EQ.

Q#	TYPE
1	2N2907A OR EQ.
2-8	2N2222A OR EQ.

R#	VALUE (OHMS)
1,12,20,22,25,26,28	470
2	NOT USED
3,4,5,30	1K
6,7,11,14,21,23,27,29,35	10K
8,10,13,15,24,31,32,33,34	47K
9	NOTE 1
16,17	NOTE 2
18,19	NOTE 3

ALL RESISTORS 1/4W, ±10% TOLERANCE UNLESS NOTED.

S#	TYPE
1	DPST, TOGGLE, ALCO SWITCH #MCT-21ON OR EQ.
2	SPDT MOM. GRAYHILL #46-101 OR EQ.
4	SPDT TOGGLE
3,8,9,10	SPST MOM. N.O., GRAYHILL #39-1 OR EQ.
5,6	DPDT TOGGLE, CUTLER-HAMMER #SF2SCY191 OR EQ.
7	SPST MOM. N.C., GRAYHILL #39-2 OR EQ.

L#	TYPE
1,3,5,6,7	MV5253 (GREEN) OR EQ.
2,4	MV5053 (RED) OR EQ.
8	MV5353 (YELLOW) OR EQ.

J1-CARD CONNECTOR, 18 PIN, ELCO #6007 OR EQ.

NOTES:
1. R9=390K, SELECTED FOR U1B OUTPUT OF 20.0 ± 1.0 MS.
2. R17=20K, SELECTED FOR U7 OUTPUT OF 4.0 ± 0.1 SECONDS; R16=75K, SELECTED FOR U7 OUTPUT OF 8.0 ± 0.1 SEC.
3. R19=125K, SELECTED FOR U8 OUTPUT OF 7.0 ± 0.1 SECONDS; R18=100K, SELECTED FOR U8 OUTPUT OF 12.0 ± 0.1 SEC.
4. CIRCUIT TRACE LENGTH SHOULD BE THE MINIMUM PRACTICABLE TO AVOID CIRCUIT OSCILLATION.

FIGURE 14. A8 CARD TEST FIXTURE SCHEMATIC

X-3915B

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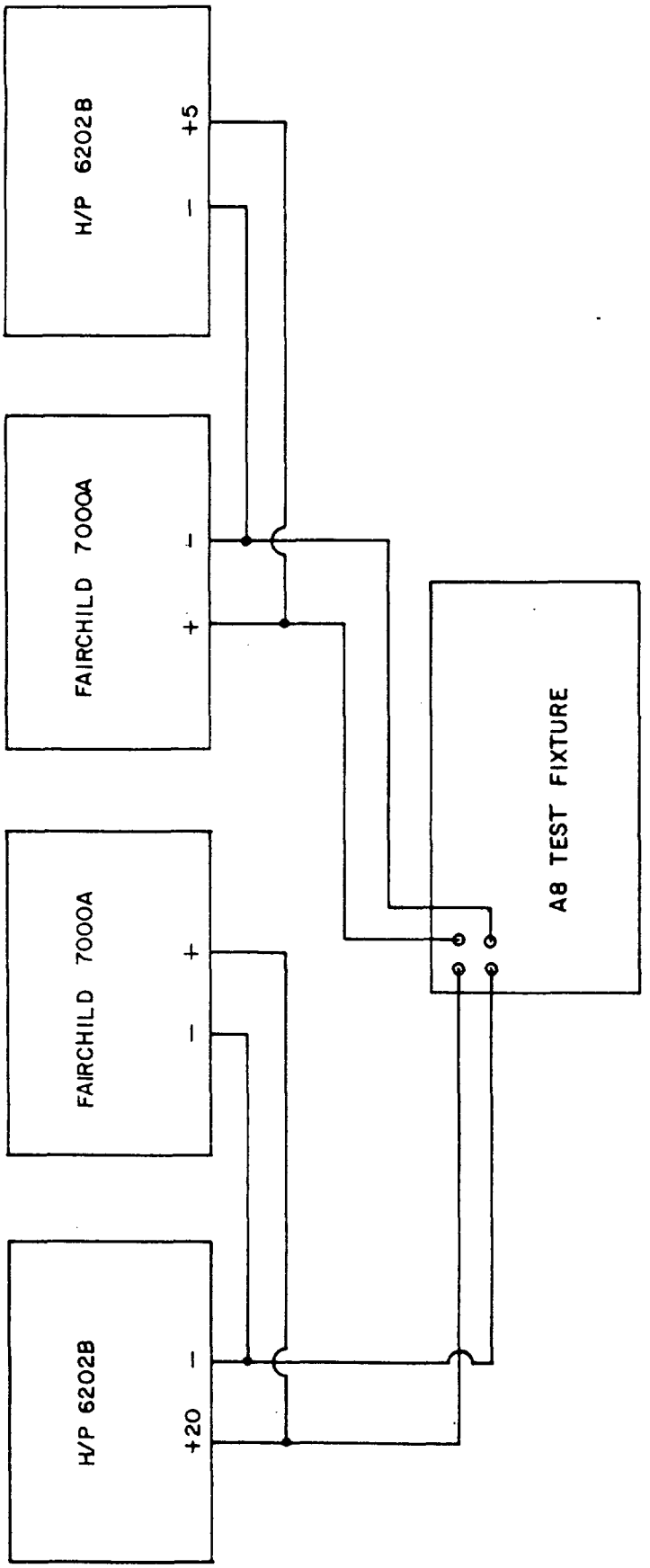


FIGURE 15. A8 card test set-up.

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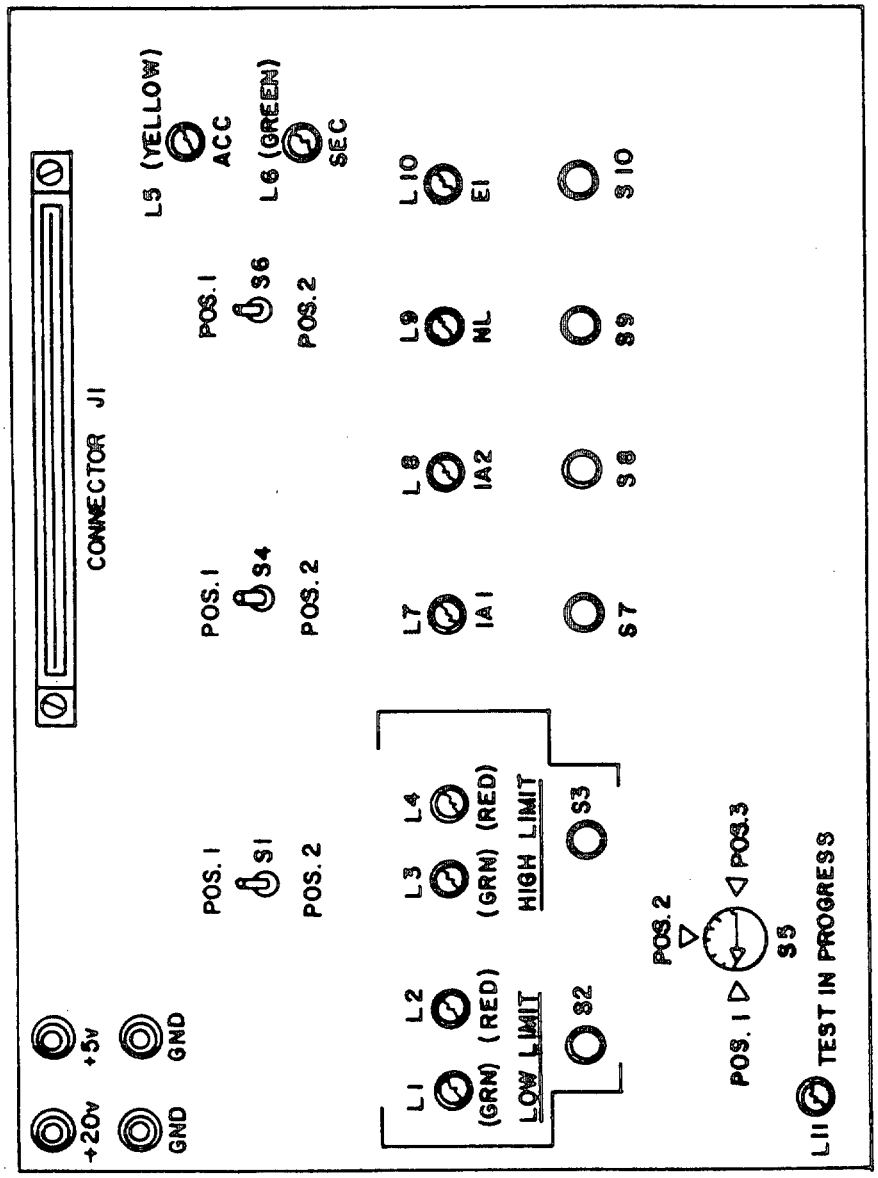
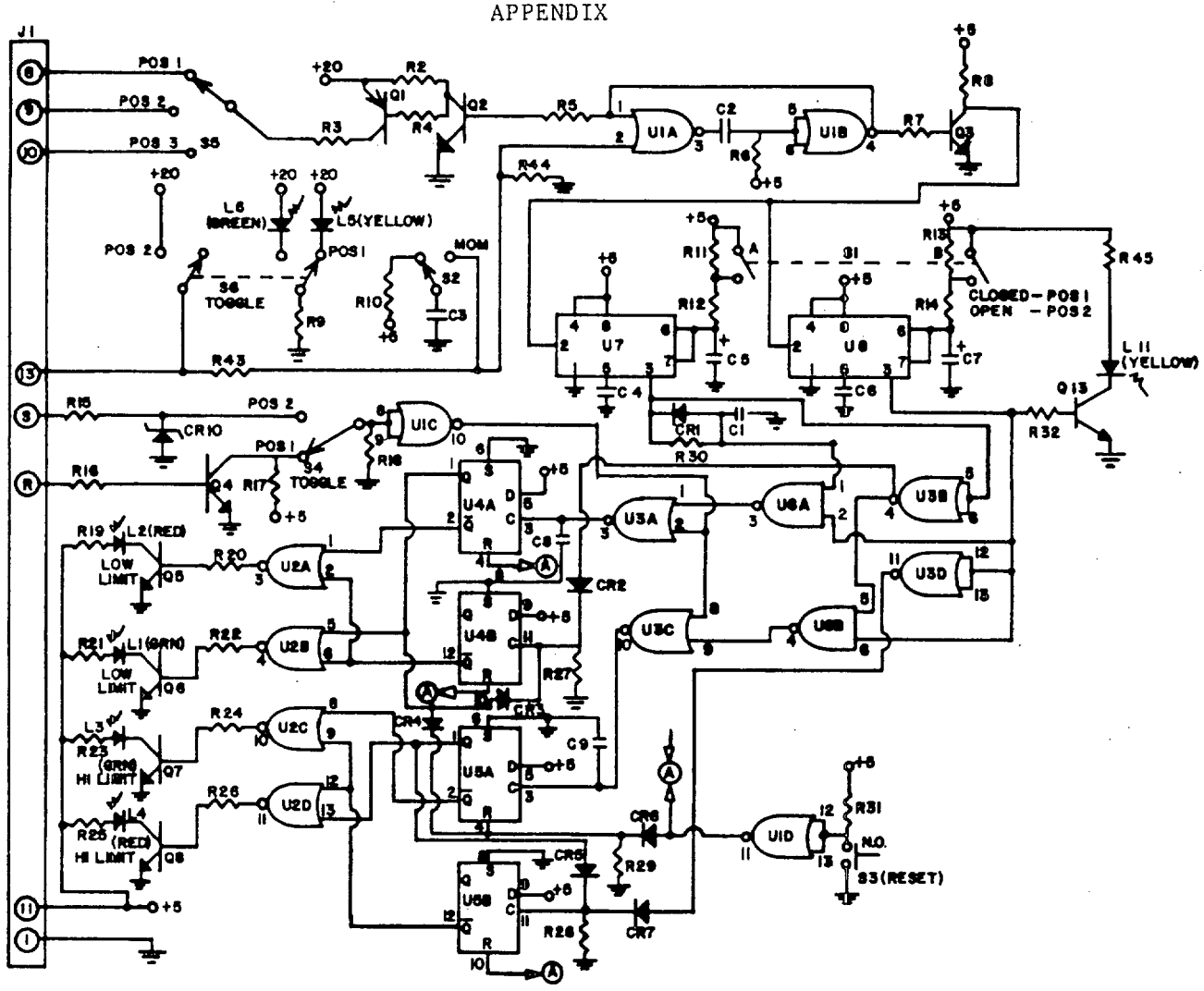


FIGURE 16. A9 card test fixture. X-3917A

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U#	TYPE
1,2,3,9	#4001 QUAD 2-INPUT NOR GATE
4, 5	#4013 DUAL "D" FLIP-FLOP W/SET/RESET
6	#4011 QUAD 2-INPUT NAND GATE
7, 8	#555 TIMER

NOTE: TIE ALL UNUSED GATE INPUTS TO +5 VOLTS. TIE PIN 7 OF EACH CMOS IC TO CIRCUIT GROUND. TIE PIN 14 OF EACH CMOS IC TO +5 VOLTS.

L#	TYPE
1, 3, 6	MV5253 (GREEN) OR EQ.
2, 4, 7, 8, 9, 10	MV5053 (RED) OR EQ.
5, 11	MV353 (YELLOW) OR EQ.

S#	TYPE
1	DPST TOGGLE, ALCOSWITCH #MCT-21ON OR EQ.
2, 4	SPDT MOM. GRAYHILL #46-101 OR EQ.
3, 7, 8, 9, 10	SPST MOM. N.O., GRAYHILL #39-1 OR EQ.
5	3 POSITION ROTARY, GRAYHILL SERIES 50 OR EQ.
6	DPDT TOGGLE, CUTLER-HAMMER #SF2SCY191 OR EQ.

Q#	TYPE
1	2N2907 OR EQ.
2 THRU 13	2N2222A OR EQ.

CR#	TYPE
1 THRU 7	IN456 OR EQ.
10	IN751 ZENER OR EQ.

R#	VALUE
2, 4, 8, 20, 22, 24, 26, 32, 33, 37, 39	10K
3, 9, 17, 41	1K
5, 7, 10, 16, 18, 27, 28, 29, 30, 31, 35, 36, 44	47K
6	NOTE 1
11, 12	NOTE 2
13, 14	NOTE 3
15	4.7K
19, 21, 23, 25, 34, 38, 40, 42	470Ω
43	20K
45	170

ALL RESISTORS 1/4W, ±10% TOLERANCE UNLESS NOTED.

C#	VALUE
1	2.2μF
2	0.1μF
3, 4, 6	0.01μF
5, 7	45μF
8, 9	1000pF

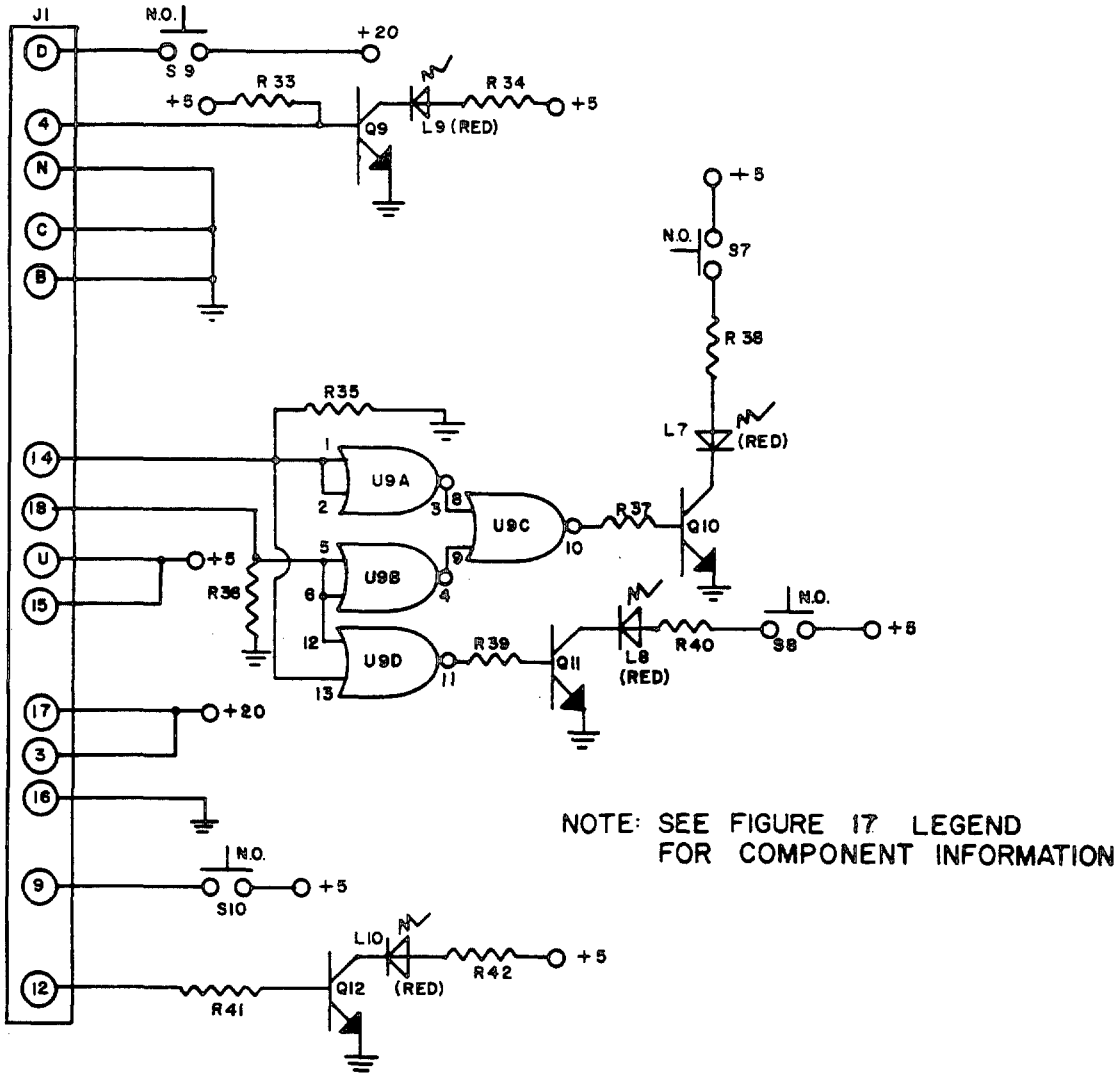
ALL CAPACITORS ARE 12 WVDC LOW LEAKAGE MYLOR, POLY-CARBONATE OR TANTALUM (±10% TOLERANCE)

J1 CARD CONNECTOR, 36 PIN, AMPHENOL #225-27821-101 OR EQ.

NOTE:
1. R6 ≅ 1.5M, SELECTED FOR U1B OUTPUT OF 75.0 +1.0 MS.
2. R11 ≅ 1.4M, SELECTED FOR U7 OUTPUT OF 80±2 SECONDS. R12 ≅ 163K, SELECTED FOR U7 OUTPUT 8+0.0, -0.2 SECONDS.
3. R13 ≅ 2.55M, SELECTED FOR U8 OUTPUT OF 153±3 SECONDS. R14 ≅ 230K SELECTED FOR U8 OUTPUT 12.0 +0.2, -0.0 SECONDS.
4. CIRCUIT TRACE LENGTH SHOULD BE THE MINIMUM PRACTICABLE TO AVOID CIRCUIT OSCILLATION.

FIGURE 17. A9 card test fixture schematic A. X-3918 A

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J-SIIDS A-9 CARD TEST FIXTURE
SCHEMATIC B – NON-TIMER FUNCTIONS

FIGURE 18. A9 card test fixture schematic B. X-3919B

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X-3920A

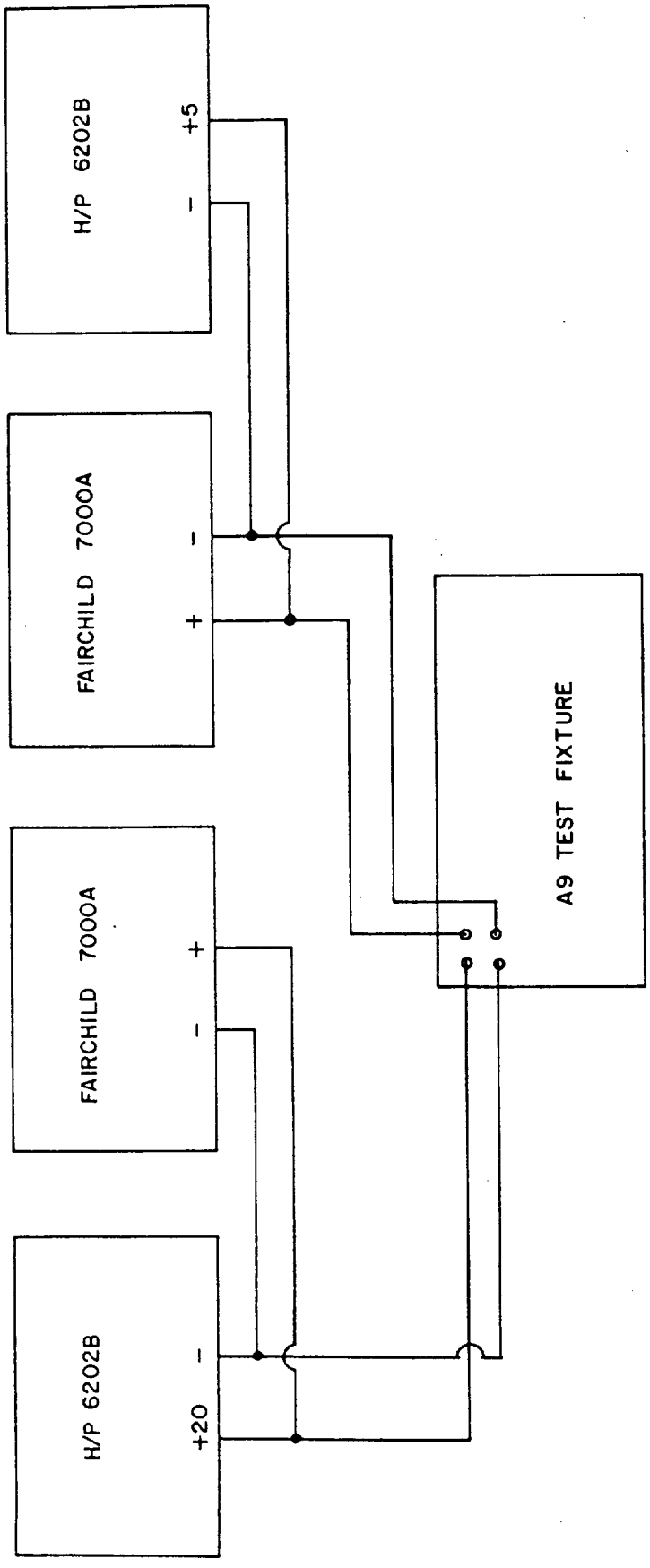
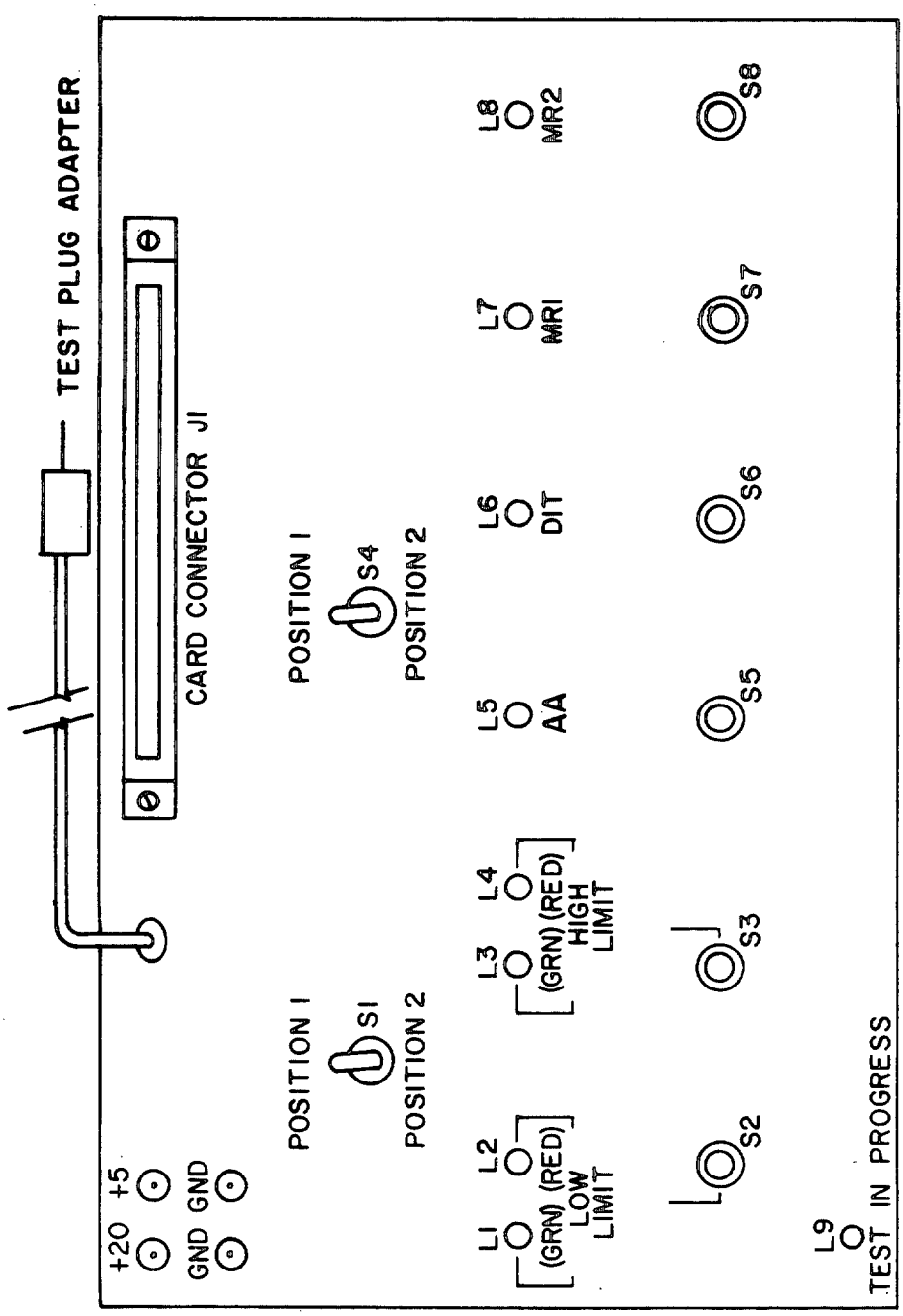


FIGURE 19. A9 card test set-up.

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X-39218

FIGURE 20. A10 card test fixture.

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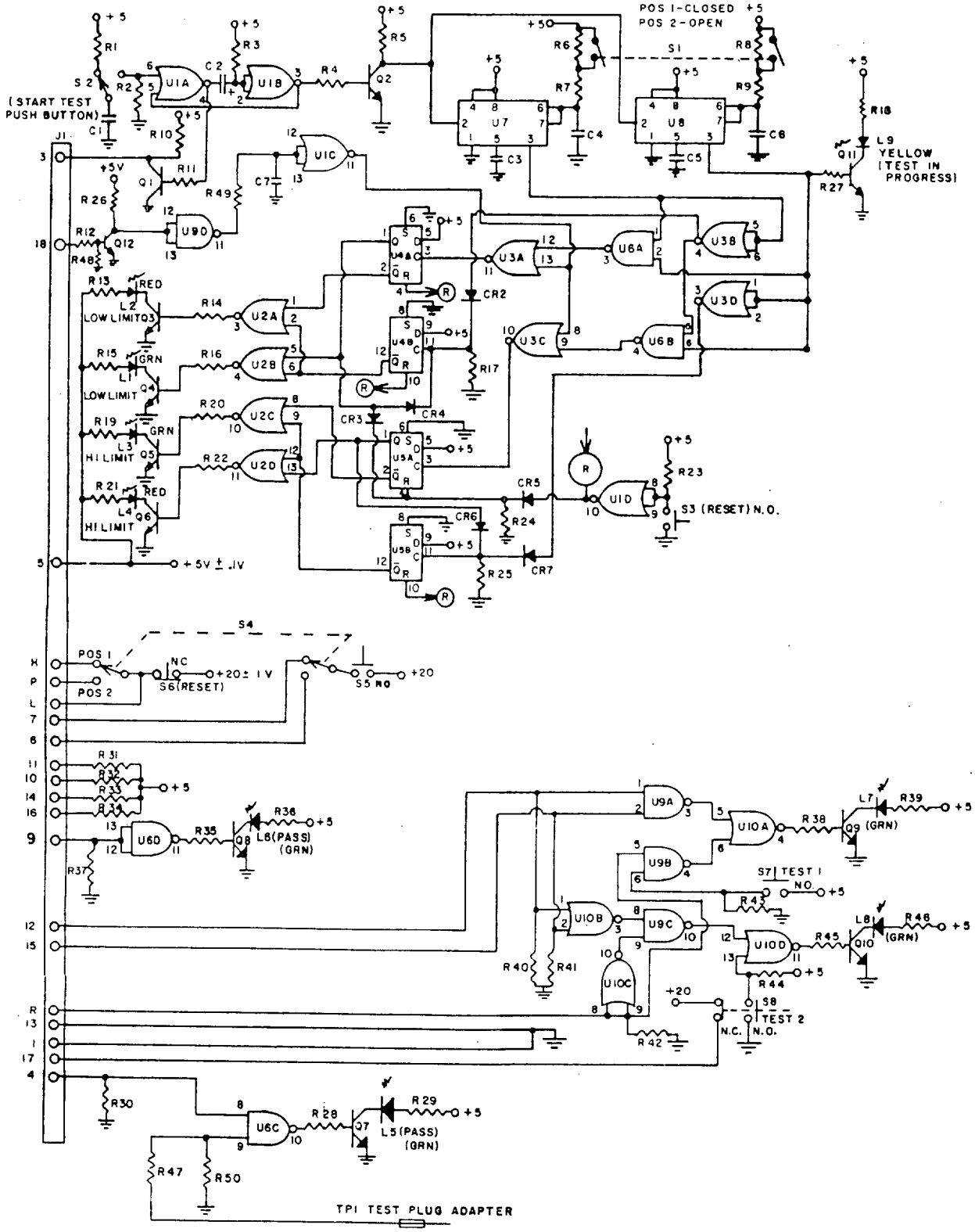


FIGURE 21. A10 card test fixture schematic.

X-3922C

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:10Z
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U#	TYPE
1,2,3,10	#001 QUAD 2-INPUT NOR GATE
4,5	#4013 DUAL "D" FLIP-FLOP W/SET/RESET
6,9	#4011 QUAD 2-INPUT NAND GATE
7,8	#555 TIMER

C#		VALUE
1,3,5	.01 µF	ALL CAPACITORS LOW LEAKAGE MYLAR, POLYCARBONATE OR TANTALUM, ± 10% TOLERANCE, 12 WVDC MINIMUM
2,7	.1 µF	
4,6	45 µF	TANTALUM, ± 10% TOLERANCE, 12 WVDC MINIMUM

S#	TYPE
1	TOG. DPST, ALCO SWITCH #MCT-210N
2	MOM. SPDT, GRAYHILL #46-101 OR EQ
3,5,7	MOM. SPDT, GRAYHILL #39-1 OR EQ
4	TOG. DPDT, CUTLER-HAMMER #SF2SCY191 OR EQ
6	MOM. SPST, N.C., GRAYHILL #39-2 OR EQ
8	MOM. DPDT, ALCO SWITCH MTA-206R OR EQ

CR#	TYPE
2-7	IN456 OR EQ

L#	TYPE
1,3,5,6,7,8	MONSANTO MV5253 OR EQ
2,4	MONSANTO MV5053 OR EQ
9	MONSANTO MV5353 OR EQ

Q#	TYPE
1-12	2N2222A or EQ

R#	VALUE
1,2,4,11,17,23,24,25,30,37,40,41,42,43	47K ohms
3	TBD NOTE 1
5,10,14,16,20,22,26,27,28,35,38,44,45	10K ohms
6	TBD NOTE 2
7	TBD NOTE 2
8	TBD NOTE 3
9	TBD NOTE 3
12,48,50,	4.7K ohms
13,15,18,19,21,29,36,39,46	330 ohms
31	270 ohms
32,33,34	1K ohm
47	15K ohms
49	TBD NOTE 4

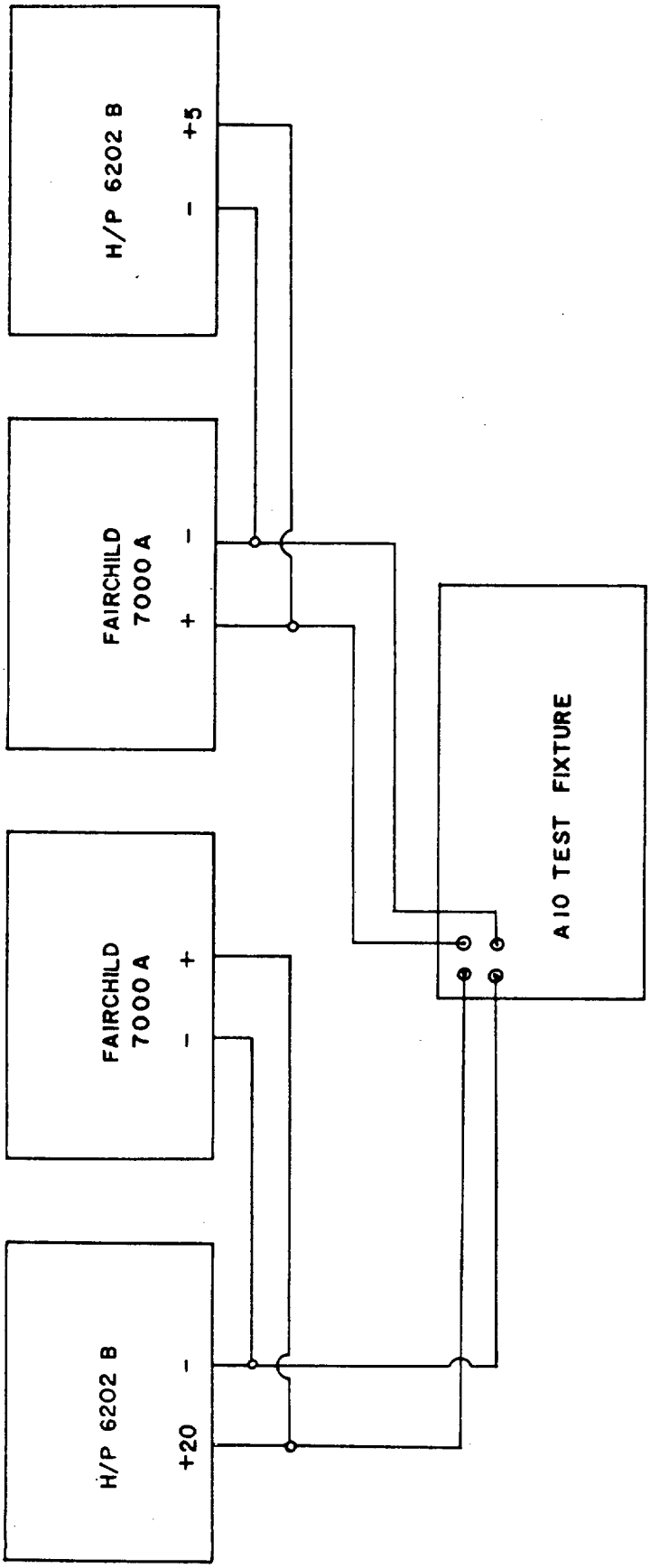
ALL RESISTORS 1/4 WATT, ± 10% EXCEPT WHERE NOTED

J1 CARD CONNECTOR, AMPHENOL #225-27821-101 OR EQ
TP1 TEST PLUG ADAPTER, TIP PLUG, SMITH #492 OR EQ

NOTE 1: R3≅390K. SELECTED FOR U1B OUTPUT AT 20.0±0.5 MS.
NOTE 2: R6≅1.4M. SELECTED FOR U7 OUTPUT AT 80±2 SECONDS. R7≅163k, SELECTED FOR U7 OUTPUT AT 8.0+0.0, -0.2 SECONDS.
NOTE 3: R8≅2.55 M, SELECTED FOR U8 OUTPUT AT 153±3 SECONDS. R9≅230K, SELECTED FOR U8 OUTPUT AT 12.0 +0.2, -0.0 SECONDS.
NOTE 4: R49≅374K, SELECTED FOR UIC OUTPUT AT 24±1MS.
NOTE 5: CIRCUIT TRACE LENGTH SHOULD BE THE MINIMUM PRACTICABLE TO AVOID CIRCUIT OSCILLATION.
NOTE 6: CR1 NOT USED.
NOTE 7: PIN 14 OF EACH CMOS IC CONNECTED TO +5 V DC. PIN 7 OF EACH CMOS IC CONNECTED TO GROUND.

FIGURE 22. AIO card test fixture parts list. X-4289A

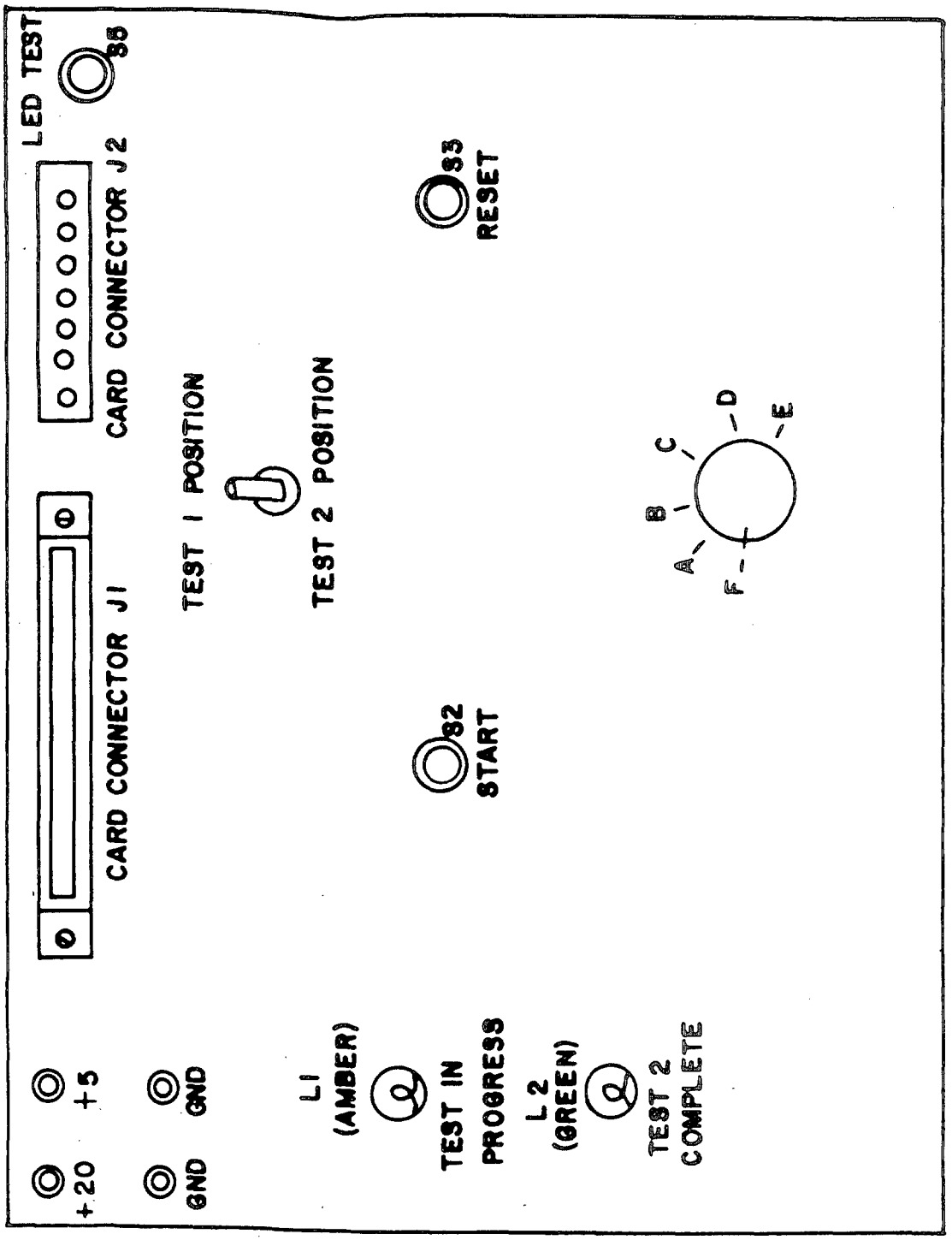
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X-3924 A

FIGURE 23. A10 card test set-up.

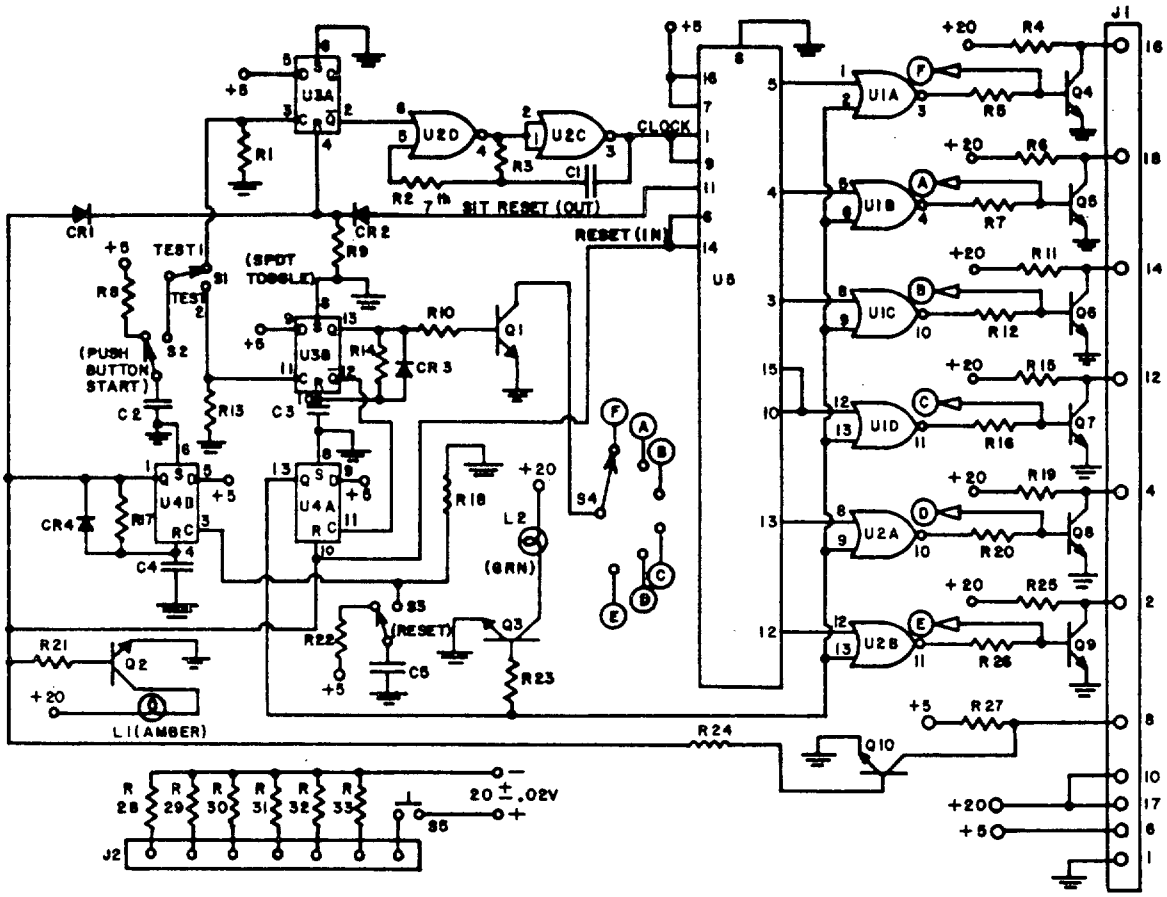
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X-3908 A

FIGURE 24. A11/A12 cards test fixture.

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U#	TYPE
1,2	#4001 QUAD 2 - INPUT NOR GATE
3,4	#4013 DUAL "D" FLIP-FLOP W/SET/RESET
5	#4015 STATIC SHIFT REGISTER

ALL CAPACITORS ARE LOW LEAKAGE MYLAR, POLYCARBONATE OR TANTALUM (±10% TOL.)

C#	VALUE	VOLTAGE
1	.1μF	12V
2,5	.01μF	12V
3	2μF	12V
4	10μF	12V

S#	TYPE
1	SPDT TOGGLE CUTLER-HAMMER #8282K14 OR EQ.
2	SPDT MOM. GRAYHILL #46-101 OR EQ.
3	MOM, SPDT GRAYHILL #46-101 OR EQ.
4	6 POSITION ROTARY, GRAYHILL SERIES 50 OR EQ.
5	N.O. SPST MOM. GRAYHILL #39-1 OR EQ.

ALL RESISTORS 1/4W, ±10% (UNLESS NOTED)

R#	VALUE
1,8,9,13,18,21,22,23,24	47K
2	NOTE 1
3	NOTE 2
4, 6, 11, 15, 19, 25	39K
5, 7, 10, 12, 16, 20, 26, 27	10K
14	NOTE 3
17	NOTE 4
28, 29, 30, 31, 32, 33	2.2K

J1 - CARD CONNECTOR, 18 PIN, ELCO 6007 OR EQ

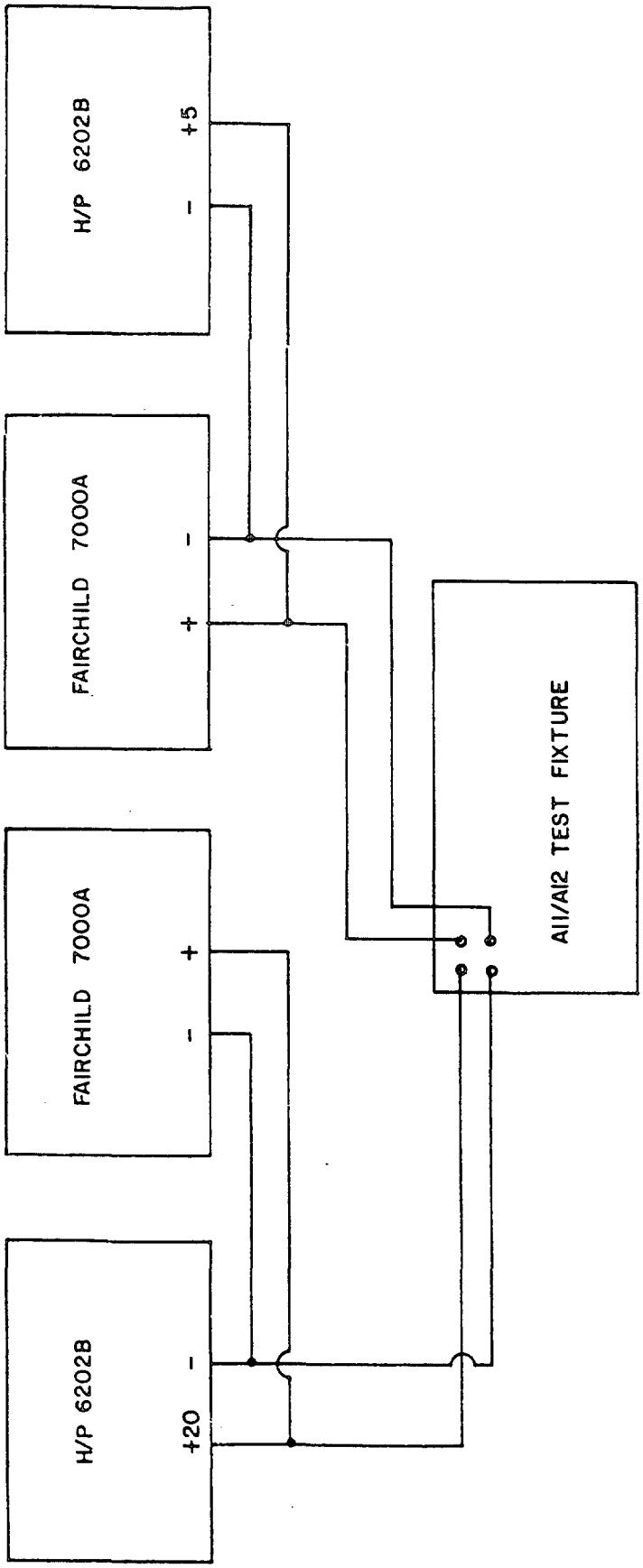
Q1 THRU Q10, 2N2222A OR EQ.

L1, L2, - #327SYLV, #33309-0 OR EQ.
CR1 THRU CR4 - IN456 OR EQ.

NOTE 1. APPROX. 4 TIMES R3
NOTE 2. TBD, APPROX. 560K, FOR U2C OUTPUT OF 6.25 Hz ±.2Hz.
NOTE 3. TBD, APPROX. 1M, FOR 1.5 SEC ±.1 SEC OUTPUT AT U3B-Q.
NOTE 4. TBD, APPROX. 820 K, FOR 5.0 SEC ± .5 SEC OUTPUT AT U4B-Q.
NOTE 5. ALL CIRCUIT TRACES SHOULD BE THE MINIMUM PRACTICAL TO AVOID CIRCUIT OSCILLATIONS

FIGURE 25. A11/A12 cards test fixture schematic. X-3909A

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X-3910A

FIGURE 26. A11/A12 cards test set-up.

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